Latest Syllabus

CHEMISTRY (52)
CLASS 10

There will be one paper of ***two hours*** *duration of 80 marks and Internal Assessment of practical work carrying 20 marks.*

The paper will be divided into ***two*** *sections, Section I (40 marks) and Section II (40 marks).*

Note: *All chemical processes / reactions should be studied with reference to the reactants, products, conditions, observation and the (balanced) equations and diagrams.*

1. Periodic Properties and variations of Properties – Physical and Chemical

(i) Periodic properties and their variations in groups and periods.

Definitions and trends of the following periodic properties in groups and periods should be studied:

- *atomic size*
- *metallic character*
- *non-metallic character*
- *ionisation potential*
- *electron affinity*
- *electronegativity*

(ii) Periodicity on the basis of atomic number for elements.

- *The study of modern periodic table up to period 3 (students to be exposed to the complete modern periodic table but no questions will be asked on elements beyond period 3 – Argon);*
- *Periodicity and other related properties to be explained on the basis of nuclear charge and shells (not orbitals).*

(Special reference to the alkali metals and halogen groups).

2. Chemical Bonding

Electrovalent, covalent and co-ordinate bonding, structures of various compounds, Electron dot structure.

(a) Electrovalent bonding:

- *Electron dot structure of Electrovalent compounds NaCl, $MgCl_2$, CaO.*
- *Characteristic properties of electrovalent compounds – state of existence, melting and boiling points, conductivity (heat and electricity), dissociation in solution and in molten state to be linked with electrolysis.*

(b) Covalent Bonding:

- *Electron dot structure of covalent molecules on the basis of duplet and octet of electrons (example: hydrogen, chlorine, nitrogen, ammonia, carbon tetrachloride, methane.*
- *Polar Covalent compounds – based on difference in electronegativity:*

  Examples – HCl, and H_2O including structures.
- *Characteristic properties of Covalent compounds – state of existence, melting and boiling points, conductivity (heat and electricity), ionisation in solution.*

  Comparison of Electrovalent and Covalent compounds.

(c) Coordinate Bonding:

- *Definition*
- *The lone pair effect of the oxygen atom of the water molecule and the nitrogen atom of the ammonia molecule to explain the formation of H_3O^+ and OH^- ions in water and NH_4^+ ion.*

  The meaning of lone pair; the formation of hydronium ion and ammonium ion must be explained with the help of electron dot diagrams.

3. Study of Acids, Bases and Salts

(i) Simple definitions, in terms of the molecules and their characteristic properties.

(ii) Ions present in mineral acids, alkalis and salts and their solutions; use of litmus and pH paper to test for acidity and alkalinity.

- *Examples with equation for the ionisation/ dissociation of ions of acids, bases and salts.*
- *Acids form hydronium ions (only positive ions) which turn blue litmus red, alkalis form hydroxyl ions (only negative ions) with water which turns red litmus blue.*
- *Salts are formed by partial or complete replacement of the hydrogen ion of an acid by a metal. (To be explained with suitable examples).*
- *Introduction to pH scale to test for acidity, neutrality and alkalinity by using pH paper or Universal indicator.*

(iii) Definition of salt; types of salts.

Types of salts: normal salts, acid salt, basic salt, definition and examples.

Contd...

(iv) Action of dilute acids on salts.

Decomposition of hydrogen carbonates, carbonates, sulphites and sulphides by appropriate acids with heating if necessary. (Relevant laboratory work must be done).

(v) Methods of preparation of Normal salts with **relevant equations.** (Details of apparatus or procedures not required).

Methods included are:

- *Direct combination*
- *Displacement*
- *Precipitation (double decomposition)*
- *Neutralization of insoluble base*
- *Neutralisation of an alkali (titration)*
- *Action of dilute acids on carbonates and bi-carbonates.*

4. Analytical Chemistry

(i) Action of Ammonium Hydroxide and Sodium Hydroxide on solution of salts: colour of salt and its solution; formation and colour of hydroxide precipitated for solutions of salts of Ca, Fe, Cu, Zn and Pb; special action of ammonium hydroxide on solutions of copper salt and sodium hydroxide on ammonium salts.

On solution of salts:

- *Colour of salt and its solution.*
- *Action on addition of Sodium Hydroxide to solution of Ca, Fe, Cu, Zn, and Pb salts drop by drop in excess. Formation and colour of hydroxide precipitated to be highlighted with the help of equations.*
- *Action on addition of Ammonium Hydroxide to solution of Ca, Fe, Cu, Zn, and Pb salts drop by drop in excess. Formation and colour of hydroxide precipitated to be highlighted with the help of equations.*
- *Special action of Ammonium Hydroxide on solutions of copper salts and sodium hydroxide on ammonium salts.*

(ii) Action of alkalis *(NaOH, KOH)* on certain metals, their oxides and hydroxides.

The metals must include aluminium, zinc and lead, their oxides and hydroxides, which react with caustic alkalis (NaOH, KOH), showing the amphoteric nature of these substances.

5. Mole Concept and Stoichiometry

(i) Gay Lussac's Law of Combining Volumes; Avogadro's Law.

- *Idea of mole – a number just as a dozen, a gross (Avogadro's number).*
- *Avogadro's Law - statement and explanation.*
- *Gay Lussac's Law of Combining Volumes. – Statement and explanation.*
- *Understanding molar volume- "the mass of 22.4 litres of any gas at S.T.P. is equal to its molar mass". (Questions will not be set on formal proof but may be taught for clear understanding).*
- *Simple calculations based on the molar volume and Gay Lussac's law.*

(ii) Refer to the atomicity of hydrogen, oxygen, nitrogen and chlorine (proof not required).

The explanation can be given using equations for the formation of HCl, NH_3, and NO.

(iii) Vapour Density and its relation to relative molecular mass:

- *Molecular mass = 2×vapour density (formal proof not required)*
- *Deduction of simple (empirical) and molecular formula from:*

(a) the percentage composition of a compound.

(b) the masses of combining elements.

(iv) Mole and its relation to mass.

- *Relating mole and atomic mass; arriving at gram atomic mass and then gram atom; atomic mass is a number dealing with one atom; gram atomic mass is the mass of one mole of atoms.*
- *Relating mole and molecular mass arriving at gram molecular mass and gram molecule – molecular mass is a number dealing with a molecule, gram molecular mass is the mass of one mole of molecules.*
- *Simple calculations based on relation of mole to mass, volume and Avogadro's number.*

(v) Simple calculations based on chemical equations

Related to weight and/or volumes of both reactants and products.

6. Electrolysis

(i) Electrolytes and non-electrolytes.

Definitions and examples.

(ii) Substances containing molecules only, ions only, both molecules and ions.

- *Substances containing molecules only ions only, both molecules and ions.*
- *Examples: relating their composition with their behaviour as **strong and weak electrolytes as well as non-electrolytes**.*

(iii) Definition and explanation of electrolysis, electrolyte, electrode, anode, cathode, anion, cation, oxidation and reduction (on the basis of loss and gain of electrons).

(iv) An elementary study of the migration of ions, with reference to the factors influencing selective discharge of ions (reference should be made to the activity series as indicating the tendency of metals, e.g., Na, Mg, Fe, Cu, to form ions) illustrated by the electrolysis of:

- *Molten lead bromide*

Contd...

- *acidified water with platinum electrodes*
- *Aqueous copper (II) sulphate with copper electrodes; electron transfer at the electrodes.*

The above electrolytic processes can be studied in terms of electrolyte used, electrodes used, ionization reaction, anode reaction, cathode reaction, use of selective discharge theory, wherever applicable.

(v) Applications of electrolysis.

- *Electroplating with nickel and silver, choice of electrolyte for electroplating.*
- *Electro refining of copper.*

Reasons and conditions for electroplating; names of the electrolytes and the electrodes used should be given. Equations for the reactions at the electrodes should be given for electroplating, refining of copper.

7. Metallurgy

(i) Occurrence of metals in nature:

- *Mineral and ore - Meaning only.*
- *Common ores of iron, aluminium and zinc.*

(ii) Stages involved in the extraction of metals.

(a) Dressing of the ore – hydrolytic method, magnetic separation, froth flotation method.

(b) Conversion of concentrated ore to its oxide-roasting and calcination (definition, examples with equations).

(c) Reduction of metallic oxides- some can be reduced by hydrogen, carbon and carbon monoxide (e.g. copper oxide, lead (II) oxide, iron (III) oxide and zinc oxide) and some cannot (e.g. Al_2O_3, MgO) - refer to activity series). Active metals by electrolysis e.g. sodium, potassium and calcium. (reference only).

Equations with conditions should be given.

(d) Electro refining – reference only.

(ii) Extraction of Aluminium.

(a) Chemical method for purifying bauxite by using NaOH – Baeyer's Process.

(b) Electrolytic extraction – Hall Heroult's process:

Structure of electrolytic cell - the various components as part of the electrolyte, electrodes and electrode reactions.

Description of the changes occurring, purpose of the substances used and the main reactions with their equations.

(iii) Alloys – composition and uses.

Stainless steel, duralumin, magnalium, brass, bronze, fuse metal / solder.

8. Study of Compounds

A. Hydrogen Chloride

Hydrogen chloride: preparation of hydrogen chloride from sodium chloride; refer to the density and solubility of hydrogen chloride (fountain experiment); reaction with ammonia; acidic properties of its solution.

- *Preparation of hydrogen chloride from sodium chloride; the laboratory method of preparation can be learnt in terms of reactants, product, condition, equation, diagram or setting of the apparatus, procedure, observation, precaution, collection of the gas and identification.*
- *Simple experiment to show the density of the gas (Hydrogen Chloride) –heavier than air.*
- *Solubility of hydrogen chloride (fountain experiment); setting of the apparatus, procedure, observation, inference.*
- *Method of preparation of hydrochloric acid by dissolving the gas in water- the special funnel arrangement and the mechanism by which the back suction is avoided should be learnt.*
- *Reaction with ammonia*
- *Acidic properties of its solution - reaction with metals, their oxides, hydroxides and carbonates to give their chlorides; decomposition of carbonates, hydrogen carbonates, sulphides, sulphites.*
- *Precipitation reactions with silver nitrate solution and lead nitrate solution.*

B. Ammonia

Ammonia: its laboratory preparation from ammonium chloride and collection; ammonia from nitrides like Mg_3N_2 and AlN and ammonium salts. Manufacture by Haber's Process; density and solubility of ammonia (fountain experiment); aqueous solution of ammonia; its reactions with hydrogen chloride and with hot copper (II) oxide and chlorine; the burning of ammonia in oxygen; uses of ammonia.

- *Laboratory preparation from ammonium chloride and collection; (the preparation to be studied in terms of, setting of the apparatus and diagram, procedure, observation, collection and identification)*
- *Ammonia from nitrides like Mg_3N_2 and AlN using warm water.*

  Ammonia from ammonium salts using alkalies.

  The reactions to be studied in terms of reactants, products, conditions and equations.
- *Manufacture by Haber's Process.*
- *Density and solubility of ammonia (fountain experiment).*
- *The burning of ammonia in oxygen.*
- *The catalytic oxidation of ammonia (with conditions and reaction)*
- *Its reactions with hydrogen chloride and with hot copper (II) oxide, lead monoxide and chlorine (both chlorine in excess and ammonia in excess).*

Contd...

All these reactions may be studied in terms of reactants, products, conditions, equations and observations.

- *Aqueous solution of ammonia - reaction with sulphuric acid, nitric acid, hydrochloric acid and solutions of iron(III) chloride, iron(II) sulphate, lead nitrate, zinc nitrate and copper sulphate.*
- *Uses of ammonia - manufacture of fertilizers, explosives, nitric acid, refrigerant gas (Chlorofluro carbon – and its suitable alternatives which are nonozone depleting), and cleansing agents.*

C. Nitric Acid

Nitric Acid: one laboratory method of preparation of nitric acid from potassium nitrate or sodium nitrate. Large scale preparation. Nitric acid as an oxidizing agent.

- *Laboratory preparation of nitric acid from potassium nitrate or sodium nitrate; the laboratory method to be studied in terms of reactants, products, conditions, equations, setting up of apparatus, diagram, precautions, collection and identification/tests.*
- *Manufacture of Nitric acid by Ostwald's process (Only equations with conditions where applicable).*
- *As an oxidising agent: its reaction with copper, carbon, sulphur.*

D. Sulphuric Acid

Large scale preparation, its behaviour as an acid when dilute, as an oxidizing agent when concentrated - oxidation of carbon and sulphur as a dehydrating agent - dehydration of sugar and copper (II) sulphate crystals; its non-volatile nature.

- *Manufacture by Contact Process Equations with conditions where applicable).*
- *Its behaviour as an acid when dilute -reaction with metal, metal oxide, metal hydroxide, metal carbonate, metal bicarbonate, metal sulphite, metal sulphide.*
- *Concentrated sulphuric acid as an oxidizing agent - the oxidation of carbon sulphur and copper.*
- *Concentrated sulphuric acid as a dehydrating agent- (a) the dehydration of sugar (b) Copper (II) sulphate crystals.*
- *Non-volatile nature of sulphuric acid - reaction with sodium or potassium chloride and sodium or potassium nitrate.*

9. Organic Chemistry

(i) Introduction to Organic compounds.

- *Unique nature of Carbon atom – tetra valency, catenation.*
- *Formation of single, double and triple bonds, straight chain, branched chain, cyclic compounds (only benzene).*

(ii) Structure and Isomerism.

- *Structure of compounds with single, double and triple bonds.*
- *Structural formulae of hydrocarbons. Structural formula must be given for: alkanes, alkenes, alkynes up to 5 carbon atoms.*
- *Isomerism – structural (chain, position)*

(iii) Homologous series – characteristics with examples.

Alkane, alkene, alkyne series and their gradation in properties and the relationship with the molecular mass or molecular formula.

(iv) Simple nomenclature.

Simple nomenclature of the hydrocarbons with simple functional groups – (double bond, triple bond, alcoholic, aldehydic, carboxylic group) longest chain rule and smallest number for functional groups rule – trivial and IUPAC names (compounds with only one functional group).

(v) Hydrocarbons: alkanes, alkenes, alkynes.

- *Alkanes - general formula; methane (greenhouse gas) and ethane - methods of preparation from sodium ethanoate (sodium acetate), sodium propanoate (sodium propionate), from iodomethane (methyl iodide) and bromoethane (ethyl bromide). Complete combustion of methane and ethane, reaction of methane and ethane with chlorine through substitution.*
- *Alkenes – (unsaturated hydrocarbons with a double bond); ethene as an example. Methods of preparation of ethene by dehydro halogenation reaction and dehydration reactions.*
- *Alkynes - (unsaturated hydrocarbons with a triple bond); ethyne as an example of alkyne; Methods of preparation from calcium carbide and 1,2 dibromoethane ethylene dibromide).*

Only main properties, particularly addition products with hydrogen and halogen namely Cl_2, Br_2 and I_2 pertaining to alkenes and alkynes.

- *Uses of methane, ethane, ethene, ethyne.*

(vi) Alcohols: ethanol – preparation, properties and uses.

- *Preparation of ethanol by hydrolysis of alkyl halide.*
- *Properties – Physical: Nature, Solubility, Density, Boiling Points. Chemical: Combustion, action with sodium, ester formation with acetic acid, dehydration with conc. Sulphuric acid to prepare ethene.*
- *Denatured and spurious alcohol.*
- *Important uses of Ethanol.*

(vii) Carboxylic acids (aliphatic - mono carboxylic acid): Acetic acid – properties and uses of acetic acid.

Contd...

- *Structure of acetic acid.*
- *Properties of Acetic Acid: Physical properties – odour (vinegar), glacial acetic acid (effect of sufficient cooling to produce ice like crystals). Chemical properties – action with litmus, alkalis and alcohol (idea of esterification).*
- Uses of acetic acid.

INTERNAL ASSESSMENT OF PRACTICAL WORK

Candidates will be asked to observe the effect of reagents and/or of heat on substances supplied to them. The exercises will be simple and may include the recognition and identification of certain gases and ions listed below. The examiners will not, however, be restricted in their choice to substances containing the listed ions.

Gases: Hydrogen, Oxygen, Carbon dioxide, Chlorine, Hydrogen chloride, Sulphur dioxide, Hydrogen sulphide, Ammonia, Water vapour, Nitrogen dioxide.

Ions: Calcium, Copper, Iron, Lead, Zinc and Ammonium, Carbonate, Chloride, Nitrate, Sulphide, Sulphite and Sulphate.

Knowledge of a formal scheme of analysis is not required. Semi-micro techniques are acceptable but candidates using such techniques may need to adapt the instructions given to suit the size of the apparatus being used.

Candidates are expected to have completed the following minimum practical work:

1. Action of heat on the following substances:
   (a) Copper carbonate, zinc carbonate
   (b) zinc nitrate, copper nitrate, lead nitrate Make observations, identify the products and make deductions where possible (equations not required).
2. Make a solution of the unknown substance: add sodium hydroxide solution or ammonium hydroxide solution, make observations and give your deduction. Warming the mixture may be needed. Choose from substances containing Ca^{2+}, Cu^{2+}, Fe^{2+}, Fe^{3+}, Pb^{2+}, Zn^{2+}, NH_4^+.
3. Supply a solution of a dilute acid and alkali. Determine which is acidic and which is basic, giving two tests for each..
4. Add concentrated hydrochloric acid to each of the given substances, warm, make observations, identify any product and make deductions: (a) copper oxide (b) manganese dioxide.

EVALUATION

The assignments/project work are to be evaluated by the subject teacher and by an External Examiner. (The External Examiner may be a teacher nominated by the Head of the school, who could be from the faculty, but not teaching the subject in the section/class. For example, a teacher of Chemistry of Class VIII may be deputed to be an External Examiner for Class X Chemistry projects.)

The Internal Examiner and the External Examiner will assess the assignments independently.

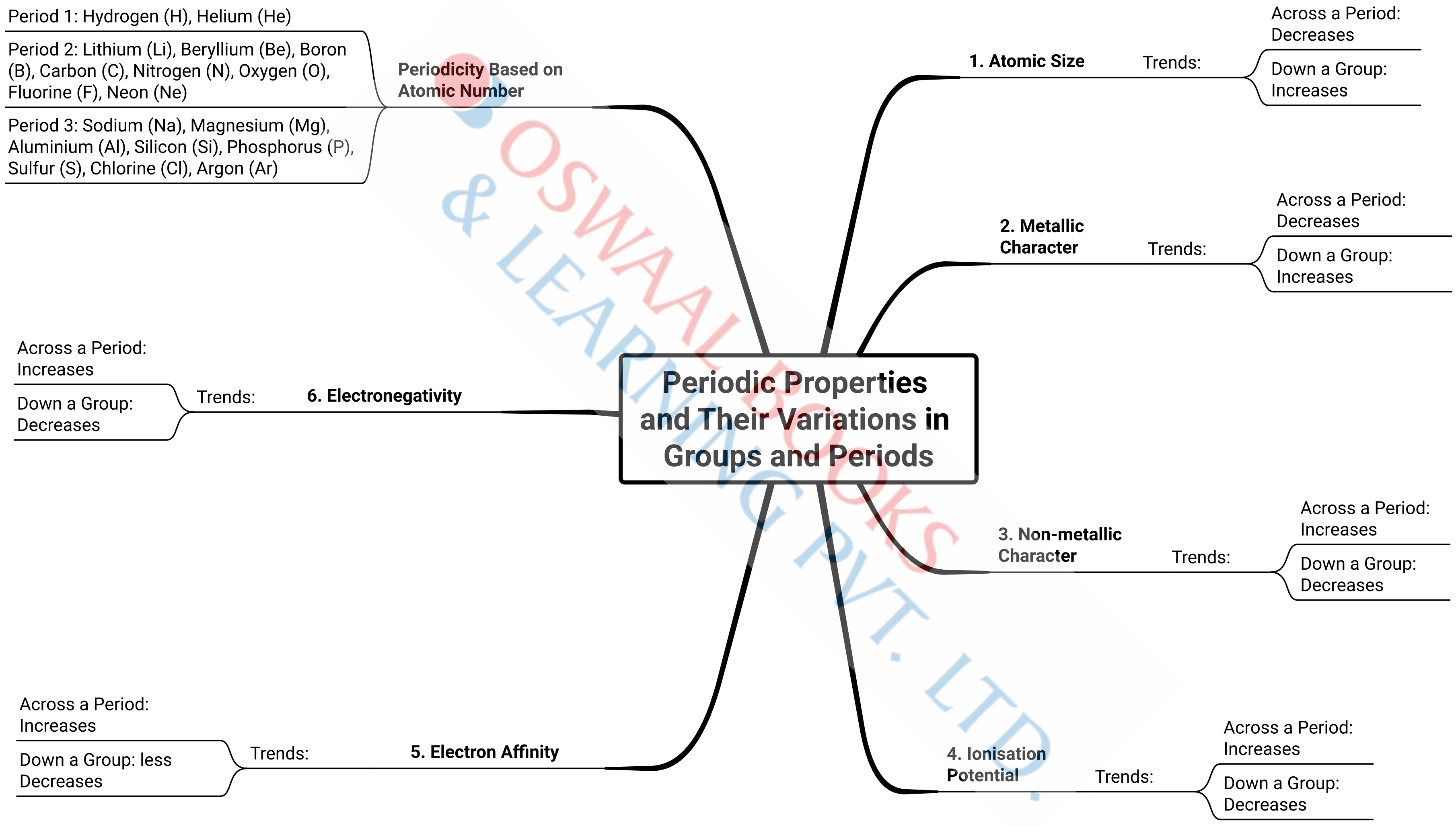
Award of Marks	**(20 Marks)**
Subject Teacher (Internal Examiner)	10 marks
External Examiner	10 marks

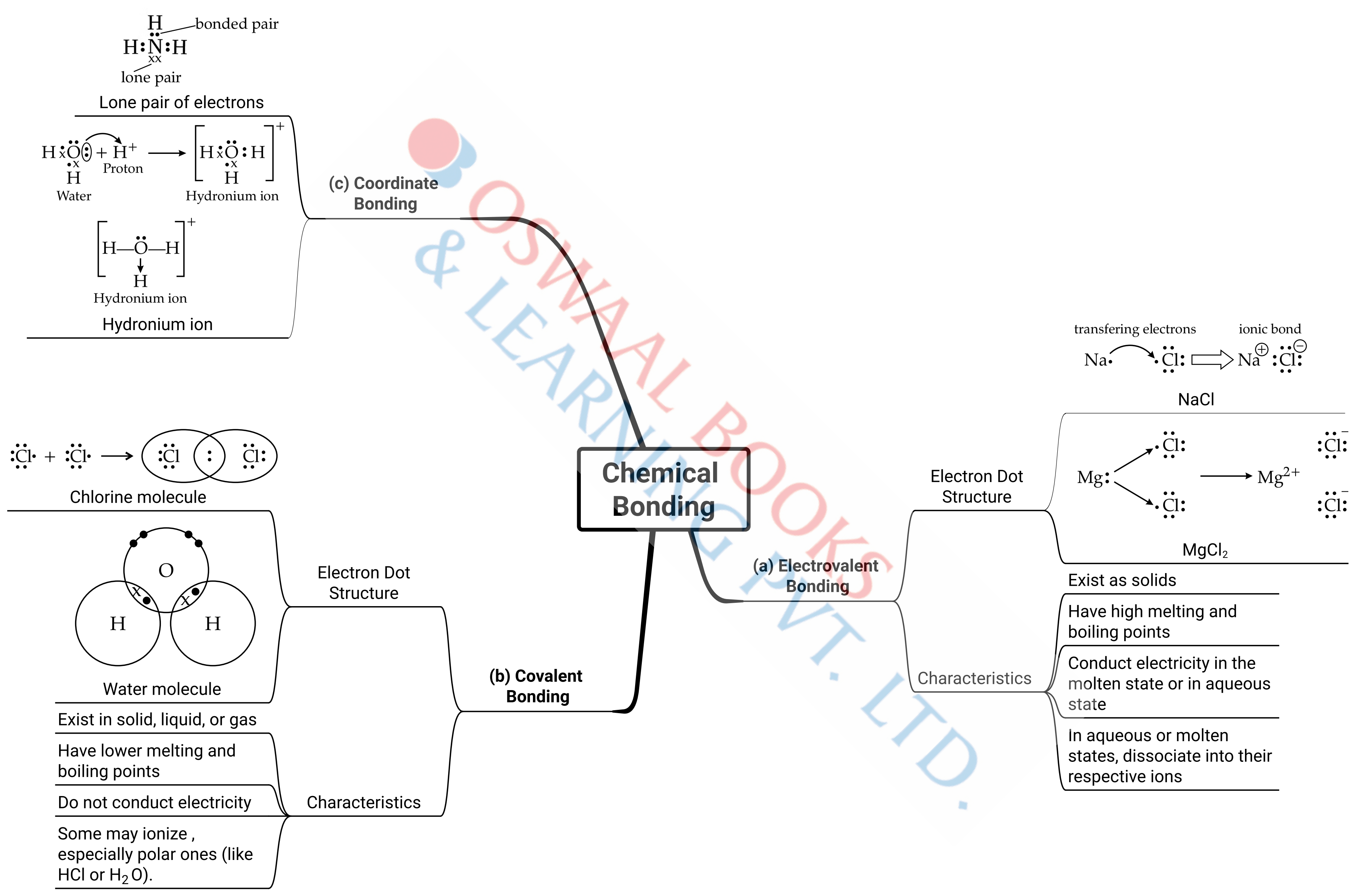
The total marks obtained out of 20 are to be sent to the Council by the Head of the school.

The Head of the school will be responsible for the online entry of marks on the Council's CAREERS portal by the due date.

*Note: According to the recommendation of International Union of Pure and Applied Chemistry (IUPAC), **the groups are numbered from 1 to 18 replacing the older notation of groups IA …. VIIA, VIII, IB …… VIIB and 0.** However, for the examination both notations will be accepted.*

Old notation	IA	IIA	IIIB	IVB	VB	VIB	VIIB	VIII			IB	IIB	IIIA	IVA	VA	VIA	VIIA	0
New notation	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18


Periodic Properties and Their Variations in Groups and Periods
1. Atomic Size
Trends:
Across a Period: Decreases
Down a Group: Increases
2. Metallic Character
Trends:
Across a Period: Decreases
Down a Group: Increases
3. Non-metallic Character
Trends:
Across a Period: Increases
Down a Group: Decreases
4. Ionisation Potential
Trends:
Across a Period: Increases
Down a Group: Decreases
5. Electron Affinity
Trends:
Across a Period: Increases
Down a Group: less Decreases
6. Electronegativity
Trends:
Across a Period: Increases
Down a Group: Decreases
Periodicity Based on Atomic Number
Period 1: Hydrogen (H), Helium (He)
Period 2: Lithium (Li), Beryllium (Be), Boron (B), Carbon (C), Nitrogen (N), Oxygen (O), Fluorine (F), Neon (Ne)
Period 3: Sodium (Na), Magnesium (Mg), Aluminium (Al), Silicon (Si), Phosphorus (P), Sulfur (S), Chlorine (Cl), Argon (Ar)


Chemical Bonding
(a) Electrovalent Bonding
Electron Dot Structure
transfering electrons
ionic bond
NaCl
MgCl2
Characteristics
Exist as solids
Have high melting and boiling points
Conduct electricity in the molten state or in aqueous state
In aqueous or molten states, dissociate into their respective ions
(b) Covalent Bonding
Electron Dot Structure
Chlorine molecule
Water molecule
Characteristics
Exist in solid, liquid, or gas
Have lower melting and boiling points
Do not conduct electricity
Some may ionize , especially polar ones (like HCl or H2O).
(c) Coordinate Bonding
bonded pair
lone pair
Lone pair of electrons
Water
Proton
Hydronium ion
Hydronium ion
Hydronium ion

Study of Acids, Bases, and Salts

- **1. Ionization**
  - $HCl \rightarrow H^{+} + Cl^{-}$
    Acids form hydronium ions (H_3O+) in water
  - $NaOH \rightarrow Na^{+} + OH^{-}$
    Bases (Alkalis) form hydroxide ions (OH-) in water
  - Litmus Test
    - Acids: Turn blue litmus paper red.
    - Alkalis: Turn red litmus paper blue.
    - Neutral Solutions: Do not change the color of litmus paper.
- **2. Salts**
  - Types
    - Normal Salts
    - Acid Salts
    - Basic Salts
- **3. Dilute Acids and Salts**
  - $NaHCO_3 + HCl \rightarrow NaCl + H_2O + CO_2$
    On bicarbonates
  - $CaCO_3 + 2HCl \rightarrow CaCl_2 + H_2O + CO_2$
    On carbonates
  - $Na_2SO_3 + 2HCl \rightarrow 2NaCl + H_2O + SO_2$
    On sulphites
  - $Na_2S + 2HCl \rightarrow 2NaCl + H_2S$
    On sulphides
- **4. Methods of Preparation**
  - $Zn + 2HCl \rightarrow ZnCl_2 + H_2$
    Direct Combination
  - $Zn + CuSO_4 \rightarrow ZnSO_4 + Cu$
    Displacement
  - $BaCl_2 + Na_2SO_4 \rightarrow BaSO_4 + 2NaCl$
    Precipitation
  - $CuO + 2HCl \rightarrow CuCl_2 + H_2O$
    Neutralization

Analytical Chemistry

1. Action on metal salt solutions — $NaOH$

- **Calcium (Ca^{2+})**
  - $Ca^{2+} + 2OH^- \rightarrow Ca(OH)_2$
  - White precipitate, does not dissolve in excess
- **Iron (II) Salts (Fe^{2+})**
  - $Fe^{2+} + 2OH^- \rightarrow Fe(OH)_2$
  - Light green precipitate
  - In excess, the precipitate does not dissolve.
- **Iron (III) Salts (Fe^{3+})**
  - $Fe^{3+}\ (aq) + 3OH^-\ (aq) \rightarrow Fe(OH)_3$
  - Brown precipitate
  - In excess, the precipitate does not dissolve.
- **Copper Salts (Cu^{2+})**
  - $Cu^{2+}\ (aq) + 2OH^-\ (aq) \rightarrow Cu(OH)_2$
  - Light blue precipitate when added in drops
  - Does not dissolve in excess sodium hydroxide
- **Zinc Salts (Zn^{2+})**
  - $Zn^{2+}\ (aq) + 2OH^-\ (aq) \rightarrow Zn(OH)_2$
  - White precipitate when added in drops
  - $Zn(OH)_2 + 2NaOH \rightarrow Na_2\ ZnO_2 + 2H_2\ O$
  - In excess, dissolves to form a colourless solution
- **Lead Salts (Pb^{2+})**
  - $Pb^{2+}\ (aq) + 2OH^-\ (aq) \rightarrow Pb(OH)_2$
  - When added drop by drop white precipitate
  - $Pb(OH)_2 + 2NaOH \rightarrow Na_2\ PbO_2 + 2H_2\ O$
  - In excess, dissolves to form a colourless solution:

2. Action on metal salt solutions — $NH_4\ OH$

- **Iron (II) Salts (Fe^{2+})**
  - $Fe^{2+} + 2OH^- \rightarrow Fe(OH)_2$
  - Light green precipitate
  - In excess, the precipitate does not dissolve.
- **Iron (III) Salts (Fe^{3+})**
  - $Fe^{3+}\ (aq) + 3OH^-\ (aq) \rightarrow Fe(OH)_3$
  - Brown precipitate
  - In excess, the precipitate does not dissolve.
- **Copper Salts (Cu^{2+})**
  - $Cu^{2+}\ (aq) + 2OH^-\ (aq) \rightarrow Cu(OH)_2$
  - Light blue precipitate when added in drops
  - $Cu(OH)_2\ (s) + 4NH_3\ (aq) \rightarrow [Cu(NH_3)_4]^{2+}$
  - Dissolves in excess forming deep blue solution
- **Zinc Salts (Zn^{2+})**
  - $Zn^{2+}\ (aq) + 2OH^-\ (aq) \rightarrow Zn(OH)_2$
  - (white precipitate)
  - $Zn(OH)_2\ (s) + 2OH^-\ (aq) \rightarrow [Zn(OH)_4]^2$
  - In excess dissolves to form a colourless solution
- **Lead Salts (Pb^{2+})**
  - $Pb^{2+}\ (aq) + 2OH^-\ (aq) \rightarrow Pb(OH)_2$
  - White precipitate when added in drops
  - In excess, does not dissolve

3. Amphoteric metals and Oxides

- **React with both acids and bases**
  - $2Al + 2NaOH + 6H_2\ O \rightarrow 2NaAlO_2 + 3H_2$
  - Aluminium
  - $Al_2O_3 + 2NaOH + 3H_2O \rightarrow 2NaAlO_2 + 3H_2O$
  - $Zn + 2NaOH + 2H_2\ O \rightarrow Na_2\ ZnO_2 + H_2$
  - $ZnO + 2NaOH \rightarrow Na_2\ ZnO_2 + H_2\ O$
  - $Pb + 2NaOH + 2H_2\ O \rightarrow Pb_2\ ZnO_2 + H_2$
  - $ZnO + 2NaOH \rightarrow Na_2\ ZnO_2 + H_2\ O$

Mole Concept and Stoichiometry

1. Gay-Lussac's Law

$H_2 + Cl_2 \rightarrow 2HCl$

The volumes of the reacting gases and of the products are in whole number ratios.

2. Avogadro's Law

Equal volumes of gases at the same temperature and pressure contain an equal number of molecules.

3. The Mole Concept

Is a unit used to measure number of particles.

-no.of particles = Avogadro's number $= 6.022 \times 10^{23}$

4. Mole and Atomic Mass:

$AtomicMass = \sum(\text{Isotopic Mass} \times \text{Abundance Fraction})$

Atomic mass is the mass of a single atom

$$\text{Number of Moles (n)} = \frac{\text{Given Mass (m)}}{\text{Molar Mass (M)}}$$

5. Percentage Composition of a Compound

$$\text{Percentage of Oxygen} = \left(\frac{\text{Mass of Oxygen}}{\text{Molar Mass of Water}}\right) \times 100$$

The percentage by mass of each element in a compound.

6. Vapour Density and Its Relation to Relative Molecular Mass

Molecular mass = 2 × Vapour density

$MolecularFormula = (\text{Empirical Formula}) \times n$

Electrolysis

1. Key Terms

- **Electrolysis**: Electrical energy drives chemical reaction.
- **Electrodes**: Materials through which electric current enters or leaves the electrolytic solution.
- **Anion**: A negatively charged ion attracted to the anode and get oxidised.
- **Cation**: A positively charged ion attracted to the cathode and get reduced.
- **Oxidation and Reduction**
  - Oxidation: The loss of electrons by a substance.
  - Reduction: The gain of electrons by a substance.

2. Electrolytes and Non-Electrolytes

- **Electrolytes**
  - $NaCl(solid) \rightarrow \mathrm{Na^{+}\,(aq) + Cl^{-}\,(aq)}$
  - Conduct electricity in water or in a molten state, contains only ions
- **Non-Electrolytes**
  - $Sugar(C_6\,H_{12}\,O_6)$
  - Do not conduct electricity in water,, contains molecules Only
- **Weak-Electrolytes**
  - $\mathrm{CH_3\,COOH\,(aq) \rightleftharpoons CH_3\,COO^{-}\,(aq) + H^{+}\,(aq)}$
  - Dissociate partially, containing both molecules and ions:

3. Selective Discharge of Ions

- **Factors**
  - The higher the metal in the activity series, the more readily it forms cations.
  - Higher concentration is deposited first

4. Electrolysis Process

- **a. Molten Lead Bromide**
  - Electrolyte : Molten lead bromide , carbon electrodes
  - $PbBr_2 \rightarrow \mathrm{Pb^{2+} + 2Br^{-}}$
    Ionization Reaction:
  - $\mathrm{2Br^{-} \rightarrow Br_2} + 2e^{-}$
    Anode (Oxidation):
  - $Pb^{2+} + 2e^{-} \rightarrow \mathrm{Pb}$
    Cathode (Reduction):
- **b. Acidified Water**
  - Electrolyte : Acidified water , Platinum electrodes
  - $H_2O \rightarrow \mathrm{H^{+} + OH^{-}}$
    Ionization Reaction:
  - $\mathrm{4OH^{-} \rightarrow O_2 + 2H_2O} + 4e^{-}$
    Anode Reaction :
  - $\mathrm{4H^{+}} + 4e^{-} \rightarrow \mathrm{2H_2}$
    Cathode Reaction
- **c. Aqueous Copper (II) Sulfate**
  - Electrolyte : Aqueous copper(II) sulfate , Copper electrodes.
  - $CuSO_4 \rightarrow \mathrm{Cu^{2+} + SO_4^{2-}}$
    Ionization Reaction:
  - $Cu \rightarrow \mathrm{Cu^{2+}} + 2e^{-}$
    Anode Reaction :
  - $Cu^{2+} + 2e^{-} \rightarrow \mathrm{Cu}$
    Cathode Reaction :

5. Applications

- a. Electroplating: depositing a layer of metal onto a surface
- b. Electro Refining: to purify metals

Metallurgy

- **1. Occurrence**
  - Iron
    - Fe_2O_3 Hematite
    - Fe_3O_4 Magnetite
  - Aluminium
    - $Al_2O_3 \cdot 2H_2O$ Bauxite
  - Zinc
    - ZnS) Zinc Blende
    - $ZnCO_3$ Calamine
- **2. Stages**
  - 1. Dressing of the Ore:
    - Hydraulic Method:
    - Magnetic Separation:
    - Froth Flotation:
  - 2. concentrated Ore to Its Oxide:
    - $2ZnS + 3O_2 \rightarrow 2ZnO + 2SO_2 \uparrow$ Roasting:
    - $CaCO_3 \rightarrow CaO + CO_2 \uparrow$ Calcination
  - Reduction of Metallic Oxides:
    - $Fe_2O_3 + 3C \rightarrow 2Fe + 3CO$ Reduction by Carbon:
    - $ZnO + CO \rightarrow Zn + CO_2$ Reduction by Carbon Monoxide:
- **3. Extraction of Aluminium**
  - 1. Chemical Method for Purifying Bauxite – Bayer's Process
    - $Al_2O_3 + 2NaOH + 3H_2O \rightarrow 2Na[Al(OH)_4]$ The bauxite reacts with NaOH to form soluble sodium aluminate
    - $2Na[Al(OH)_4] \rightarrow Al_2O_3 + 2NaOH + 3H_2O$ Alumina is obtained from the solution
  - 2. Electrolytic Extraction of Aluminium – Hall-Héroult Process
    - (Al_2O_3) and (Na_3AlF_6) As electrolyte
    - $Al^{3+} + 3e^- \rightarrow Al$ At the cathode:
    - $2O^{2-} \rightarrow O_2 + 4e^-$ At the anode:
- **4. Alloys**
  - 1. Stainless Steel
    - Composition: Fe, Cr, Ni
    - Uses: Kitchenware, surgical instruments
  - 2. Duralumin
    - Composition: Al, Cu, Mn, Mg
    - Uses: Aircraft, aerospace engineering.
  - 3. Brass
    - Composition: Cu, Zn
    - Uses: Musical instruments, plumbing fittings, coins.
  - 4. Bronze
    - Composition: Cu, Sn
    - Uses: Statues, coins, bearings.
  - 5. Solder
    - Composition: Sn, Pb
    - Uses: Electrical soldering, joining metals.

Study of Compounds

A. Hydrogen Chloride

- **Preparation**
  - $NaCl + H_2SO_4 \rightarrow NaHSO_4 + HCl$
- **Acidic Properties:**
  - $Zn + 2HCl \rightarrow ZnCl_2 + H_2$
    - Reaction with Metals:
  - $CuO + 2HCl \rightarrow CuCl_2 + H_2O$
    - Reaction with Metal Oxides:
  - $CaCO_3 + 2HCl \rightarrow CaCl_2 + CO_2 + H_2O$
    - Reaction with Metal Carbonates:
- **Precipitation Reactions**
  - $HCl + AgNO_3 \rightarrow AgCl(s) + HNO_3$
  - $2HCl + Pb(NO_3)_2 \rightarrow PbCl_2\ (s) + 2HNO_3$

B. Ammonia

- **Laboratory Preparation**
  - $NH_4Cl + NaOH \rightarrow NH_3 + NaCl + H_2O$
  - $Mg_3N_2 + 6H_2O \rightarrow 3Mg(OH)_2 + 2NH_3$
  - $N_2 + 3H_2 \rightleftharpoons 2NH_3$
    - By Haber's Process:
- **Reactions :**
  - $4NH_3 + 3O_2 \rightarrow 2N_2 + 6H_2O$
    - (In the presence of a platinum catalyst at 900°C)
  - $NH_3 + HCl \rightarrow NH_4Cl$
  - $2NH_3 + 3CuO \rightarrow 3Cu + N_2 + 3H_2O$
  - $2NH_3 + 3Cl_2 \rightarrow N_2 + 6HCl$
    - Excess Chlorine:
  - $NH_3 + 3Cl_2 \rightarrow NCl_3 + 3HCl$
    - Limited Chlorine:

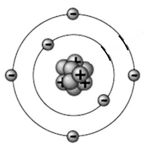
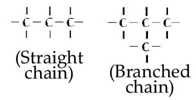
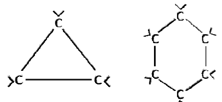
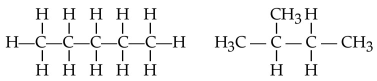
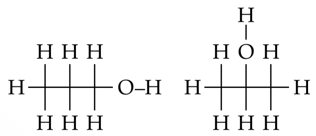
C. Nitric Acid

- **Large-Scale method**
  - Ostwald's Process:
- **Laboratory method**
  - $KNO_3 + H_2SO_4 \rightarrow KHSO_4 + HNO_3$
- **As Oxidizing Agent:**
  - $C + 4HNO_3 \rightarrow CO_2 + 2H_2O + 2NO_2$
  - $S + 2HNO_3 \rightarrow H_2SO_4 + 2NO_2$

D. Sulphuric Acid

- **Large-Scale methods**
  - By Contact Process:
- **- As an Acid:**
  - $Zn + H_2SO_4 \rightarrow ZnSO_4 + H_2$
    - Reaction with Metals:
  - $CuO + H_2SO_4 \rightarrow CuSO_4 + H_2O$
    - Reaction with Metal Oxides:
  - $Na_2CO_3 + H_2SO_4 \rightarrow Na_2SO_4 + CO_2 + H_2O$
    - Reaction with Metal Carbonates/ bicarbonates:
  - $Na2SO_3 + H_2SO_4 \rightarrow Na2SO_4 + H_2O$
    - Reaction with Metal Sulphites:
  - $FeS + H_2SO_4 \rightarrow FeSO_4 + H_2S$
    - Reaction with Metal Sulphides:
- **Concentrated Sulfuric Acid:**
  - $C + 2H_2SO_4 \rightarrow CO_2 + 2SO_2 + 2H_2O$
    - As an Oxidizing Agent:
  - $C_6H_{12}O_6 \rightarrow 6C + 6H_2O$
    - As a Dehydrating Agent:

Organic Chemistry

- **1. Unique Nature of Carbon Atom:**
  - 
  - Carbon atom
  - Tetra valency:
    - 
    - (Straight chain) (Branched chain)
  - Catenation:
    - 
    - (Cyclic or closed chain)
  - Homologous series:
    - CH_3CH_3 Ethane
    - $CH_3CH_2CH_3$ Propane
    - $CH_3CH_2CH_2CH_3$ Butane
    - $CH_3CH_2CH_2CH_2CH_3$ Pentane
- **2. Isomerism**
  - Chain Isomerism:
    - 
  - Position Isomerism:
    - 
- **3. Nomenclature**
  - Alcoholic Group: -ol
  - Aldehydic Group: -al
  - Carboxylic Group:-oic acid
  - Longest Chain Rule:
  - Smallest Number for Functional Groups
- **4. Alkanes, Alkenes, Alkynes**
  - Alkanes:
    - Preparation:
      - $CH_3COONa + NaOH \rightarrow CH_4 + Na_2CO_3$
      - Decarboxylation
    - Properties:
      - $CH_4 + 2O_2 \rightarrow CO_2 + 2H_2O$
      - Combustion
      - $CH_4 + Cl_2 \rightarrow CH_3Cl + HCl$
      - Substitution Reaction
  - Alkenes
    - Preparation:
      - $CH3 - CH_2 - X + OH^- \rightarrow CH_2 = CH_2 + HX$
      - Dehydrohalogenation
      - $C_2H_5OH \rightarrow C_2H_4 + H_2O$
      - Dehydration:
    - Properties:
      - $C_2H_4 + HCl \rightarrow C_2H_5Cl$
      - Addition Reactions:
  - Alkynes
    - Preparation:
      - $CH3 - CH_2 - X + OH^- \rightarrow CH_2 = CH_2 + HX$
      - Dehydrohalogenation
      - $CaC_2 + 2H_2O \rightarrow C_2H_2 + Ca(OH)_2$
      - from calcium carbide:
    - Properties:
      - $CH_2Br - CH_2Br + 2KOH(alc) \rightarrow C_2H_2 + 2KBr + H_2O$
      - Addition Reactions:
- **5. Ethanol**
  - Preparation :
    - $C_2H_5Cl + H_2O \rightarrow C_2H_5OH + HCl$
    - From Hydrolysis of Alkyl Halides:
  - Properties:
    - $2C_2H_5OH + 2Na \rightarrow 2C_2H_5ONa + H_2$
    - Reaction with Sodium:
    - $C_2H_5OH \xrightarrow{170°C} C_2H_4 + H_2O$
    - Dehydration with Conc. Sulfuric Acid:
- **6. Acetic Acid**
  - Preparation
    - $C_2H_5OH + 2[O] \rightarrow CH_3COOH + H_2O$
    - Oxidatio of alcohols:
    - $CH_3CHO + [O] \rightarrow CH_3COOH$
    - Oxidation of aldehydes:
  - Reactions
    - $CH_3COOH + NaOH \rightarrow CH_3COONa + H_2O$
    - Reaction with Alkalis
    - $CH_3COOH + C_2H_5OH \rightarrow CH_3COOC_2H_5 + H_2O$
    - Esterification:

Course of Action
Maximum Time = 3 Hrs
Maximum Questions = 70

1 PERIODIC PROPERTIES AND VARIATIONS OF PROPERTIES-PHYSICAL AND CHEMICAL

Learning Objectives

The students will learn:

- *Learn variations of periodic properties in groups and periods like atomic size, metallic properties, non metallic properties, ionisation potential, electron affinity and electronegativity.*
- *Brief idea of periodicity based on atomic number and the nuclear charge and shell.*
- *Modern periodic table up to 3rd period.*
- *Study of periodicity on the basis of nuclear charge and shells.*
- *Study of alkali metals, halogen groups.*

TOPIC-1: PERIODIC PROPERTIES AND THEIR VARIATIONS

Concepts covered: • Features of modern periodic table • Trends of the periodic properties in periodic table.

Revision Notes

➢ **General Introduction of Periodic Table:**

- According to modern periodic table, "The physical and chemical properties of elements are the periodic function of their atomic number".
- Niels Bohr gave the extended form of the table, known as long form of the modern **periodic table**.
- A tabular arrangement of elements in groups (vertical columns) and periods (horizontal rows) highlighting the regular trends in properties of elements is known as periodic table.

Locating Elements in Modern Periodic Table
Scan Me!

➢ **Salient features of the modern periodic table:**

Groups:

There are 1 to 18 vertical columns arranged from left to right in the modern periodic table known as groups.

It is assigned depending upon number of the valence electrons.

Modern Periodic Table – Groups
Scan Me!

- Group 1 elements are called alkali metals. They form strong alkalis with water.
- Group 2 elements are called alkaline earth metals. They form weaker alkalis as compared to group 1.
- Group 13 elements are known as boron family.
- Group 14 elements are known as carbon family.
- Group 15 elements are known as nitrogen family (Pnictogen).
- Group 16 elements are known as oxygen family or chalcogen family.
- Group 17 elements are called halogen family. They form salts with metallic elements.
- Group 18 (zero group) elements are called noble gases or inert gases. Due to stable electronic configuration, they hardly react with other elements. **[Board, 2023]**

The elements of groups 1, 2, 13, 14, 15 , 16 and 17 are known as the main group elements or representative elements. The outermost shell of all the elements of these groups are incomplete. The elements of groups 3, 4, 5, 6, 7, 8, 9, 10, 11 and 12 are known as transition elements. They have their two outermost shells incomplete.

Periods:

There are seven horizontal rows in the modern periodic table, called periods.

"The number of shells present in an atom determines its period".

- First period is the shortest period, containing only two elements, i.e., hydrogen and helium.
- Second period is a short period containing eight elements (Li, Be, B, C, N, O, F and Ne).

- Third period is also a short period containing eight elements (Na, Mg, Al, Si, P, S, Cl and Ar).
- Fourth and fifth periods are long periods, each containing eighteen elements (atomic numbers 19 to 36 in the fourth period and atomic numbers 37 to 54 in the fifth period).
- Sixth and seventh periods are longest periods, each containing 32 elements (atomic numbers 55 to 86 in the sixth period and atomic numbers 87 to 118 in the seventh period).

Periods	Number of elements
1	2
2	8
3	8
4	18
5	18
6	32
7	32

- In group 3 and sixth period, the elements from atomic numbers 57 to 71 are present, which are called Lanthanides (**rare earth metals**).
- In group 3 and seventh period, the elements from atomic numbers 89 to 103 are present, which are called Actinides (**radioactive elements**).

➢ The recurrence of elements with similar properties after certain regular intervals when these are arranged in the increasing order of their atomic number is called periodicity.

➢ The recurrence of similar electronic configuration, i.e., having same number of electrons in the outermost orbit, is the cause of periodicity.

➢ The chemical properties of elements depend upon the number of electrons in their outermost shell, so elements of the same group have similar properties.

Example: Characteristics of Halogens:

- They are coloured non-metals.
- They form monovalent anions, e.g., Cl^-.
- They are very reactive and so are found in combined state.
- Their melting and boiling points increase on moving down the group.
- They are good oxidising agents.
- They form hydracids.

➢ **Atomic Size (Atomic Radius):**

Atomic size is the most probable distance from the centre of the nucleus to the outermost shell containing electrons.

Atomic size depends upon the number of shells and nuclear charge.

(i) Number of shells: An increase in the number of shells increases the size of an atom because the distance between the outermost shell electrons and the nucleus increases.

(ii) Nuclear charge: The positive charge present on the nucleus of an atom is called nuclear charge. An increase in nuclear charge decreases the size of the atom because nucleus attracts the electrons more closely towards itself, leading to the decrease in atomic size.

Trends in Atomic Size: **[Board, 2024]**

(i) Down a group: Atomic size increases on moving down a group, i.e., from top to bottom, due to increase in the number of shells which outweighs the increased nuclear charge.

(ii) Across a period: Atomic size decreases from left to right across a period due to increase in the effective nuclear charge (atomic number).

➢ **Metallic Character:**

It refers to the tendency of a metal to **lose** their valence electrons to form a cation (positive ion). Thus, larger the tendency of a metal to lose their valence electrons, greater their metallic character.

Trends in Metallic Character:

(i) Down a group: Metallic character increases down a group due to increase in atomic size and nuclear charge.

(ii) Across a period: Metallic character decreases across a period, i.e., on moving from left to right.

Metallic and Non-metallic Properties

Scan Me!

➢ **Non-Metallic Character:**

Non-metallic character refers to the tendency of a non-metal to gain electrons to form an anion (negative ion).

➢ **Trends in Non-metallic Character:**

(i) Down a group: Non-metallic character decreases down a group due to increase in atomic size and nuclear charge.

(ii) Across a period: Non-metallic character increases across a period (left to right) due to decrease in atomic size.

MNEMONICS

Mnemonics for first 20 elements: (1 to 20)

First 10 elements.	Second 10 elements.
Mnemonics:	**Mnemonics:**
Hi, **He**llo **Li**sten **BBC** **N**ews **O**n **F**riday **N**ight	**Na** **M**an**g**o **A**man **S**e **P**epsi **S**oda **C**o**l**a **A**u**r** **K**eemti **Ca**r
Interpretation:	**Interpretation:**
Hi – Hydrogen	**Na** - Sodium
Hello – Helium	**M**an**g**o – Magnesium
Listen – Lithium	**A**man – Aluminium
B – Beryllium	**S**e – Silicon
B – Boron	**P**epsi – Phosphorus
C – Carbon	**S**oda – Sulphur
News – Nitrogen	**C**o**l**a – Chlorine
On – Oxygen	**A**u**r** – Argon
Friday - Fluorine	**K**eemti – Potassium
Night - Neon	**Ca**r – Calcium

Periodic Table of Elements

Atomic number
Symbol
Name
Atomic Mass

C Solid
Hg Liquid
H Gas
Rf Unknown

Metals: Alkali metal, Alkaline earth metal, Lanthanides, Actinides, Transition, Poor metals
Non-metals: Other non-metals, Noble gases

GROUP NUMBER

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18
1	1 H Hydrogen 1.0079																	2 He Helium 4.0026
2	3 Li Lithium 6.941	4 Be Beryllium 9.0122											5 B Boron 10.811	6 C Carbon 12.011	7 N Nitrogen 14.007	8 O Oxygen 15.999	9 F Fluorine 18.998	10 Ne Neon 20.180
3	11 Na Sodium 22.990	12 Mg Magnesium 24.305											13 Al Aluminium 26.982	14 Si Silicon 28.086	15 P Phosphorus 30.974	16 S Sulphur 32.065	17 Cl Chlorine 35.453	18 Ar Argon 39.948
4	19 K Potassium 39.098	20 Ca Calcium 40.078	21 Sc Scandium 44.956	22 Ti Titanium 47.867	23 V Vanadium 50.942	24 Cr Chromium 51.996	25 Mn Manganese 54.938	26 Fe Iron 55.845	27 Co Cobalt 58.933	28 Ni Nickel 58.693	29 Cu Copper 63.546	30 Zn Zinc 65.39	31 Ga Gallium 69.723	32 Ge Germanium 72.64	33 As Arsenic 74.922	34 Se Selenium 78.96	35 Br Bromine 79.904	36 Kr Kypton 83.798
5	37 Rb Rubidium 85.468	38 Sr Strontium 87.62	39 Y Yttrium 88.905	40 Zr Zirconium 91.224	41 Nb Niobium 92.906	42 Mo Molybderum 95.96	43 Tc Technetium (98)	44 Ru Ruthenium 101.07	45 Rh Rhodium 102.91	46 Pd Palladium 106.42	47 Ag Silver 107.87	48 Cd Cadmium 112.41	49 In Indium 114.82	50 Sn Tin 118.71	51 Sb Antimony 121.76	52 Te Tellurium 127.60	53 I Iodine 126.90	54 Xe Xenon 131.29
6	55 Cs Caesium 132.91	56 Ba Barium 137.33	57-71 La-Lu Lanthanide	72 Hf Hafnium 178.49	73 Ta Tantalum 180.95	74 W Tungsten 183.21	75 Re Rhenium 186.21	76 Os Osmium 190.23	77 Ir Iridium 192.22	78 Pt Platinum 195.08	79 Au Gold 196.97	80 Hg Mercury 200.59	81 Tl Thallium 204.38	82 Pb Lead 207.2	83 Bi Bismuth 209.98	84 Po Polonium 208.98	85 At Astatine (210)	86 Rn Radon (222)
7	87 Fr Francium (223)	88 Ra Radum (226)	89-103 Ac-Lr Actinide	104 Rf Rutherfordium (267)	105 Db Dubnium (268)	106 Sg Seaborgium (271)	107 Bh Bohrium (272)	108 Hs Hassium (277)	109 Mt Meitneium (276)	110 Ds Darmstadtium (281)	111 Rg Roentgenium (280)	112 Cn Copernicium (285)	113 Nh Nihonium (286)	114 Fi Flerovium (289)	115 Mc Moscovium (290)	116 Lv Livermorium (293)	117 Ts Tennessine (294)	118 Og Oganesson (294)

* Lanthanide Series	58 Ce Cerium 140.12	59 Pr Praseodymium 140.91	60 Nd Neodymium 144.24	61 Pm Promethium (145)	62 Sm Samar ium 150.36	63 Eu Europium 151.96	64 Gd Gadolinium 157.25	65 Tb Terbium 158.93	66 Dy Dysprosium 162.50	67 Ho Holmium 164.93	68 Er Erbium 167.26	69 Tm Thulium 168.93	70 Yb Ytterbium 173.05	71 Lu Lutetium 174.97
** Actinide Series	90 Th Thorium 232.04	91 Pa Protactinium 231.04	92 U Uranium 238.03	93 Np Neptunium (237)	94 Pu Plutonium (244)	95 Am Americium (243)	96 Cm Curium (247)	97 Bk Berkelium (247)	98 Cf Californium (251)	99 Es Einsteinium (252)	100 Fm Fermium (257)	101 Md Mendelevium (258)	102 No Nobelium (259)	103 Lr Lawrencium (262)

Example 1

Give reason for the following:

(A) Ionisation potential decreases down a group.

(B) Ionic compounds do not conduct electricity in solid state.

Ans. (A) Ionisation potential decreases as we go down the group because as we go down, atomic size increases, so the force of attraction between nucleus and valence electron reduces.

It becomes easy to remove the electron.

(B) In solid state the ions are not free to move. While in molten state, they can easily travel in solution, thus allowing electricity to flow.

Example 2

Give the number of the group and the period of the element having three shells with three electrons in valence shell.

Ans. No. of shells = 3

No. of electrons in valence shell = 3

The number of shells is equal to the number of periods. So, the element belongs to third period.

The number of valence electrons shows the group of the element. So, the element belongs to 10 + 3 = 13, i.e., thirteenth group.

Key Terms

➢ **Periodic Table:** It is an arrangement of various elements in such a way that the similar property bearing elements are grouped together and dissimilar property bearing elements are separated from each other.

➢ **Modern Periodic Law:** The physical and chemical properties of the elements are the periodic functions of their atomic numbers.

➢ **Periodic Properties:** These are the properties related to the electronic configuration of elements that change periodically down a group and across a period.

➢ **Periodicity:** The recurrence of elements with similar properties after certain regular intervals when they are arranged in the increasing order of their atomic number is called periodicity.

➢ **Valency:** It is the combining capacity of an atom of the element. For normal elements, valency is equal to the number of valence electrons in an atom of the element (example: groups 1, 2 and 3) or eight minus number of electrons in the valence shell of an atom (example: groups 15,16 and 17).

Key Facts

➢ In the modern periodic table,

- Elements are arranged in order of increasing atomic number.
- The vertical columns of elements with similar properties are called groups.
- The horizontal rows are called periods.

➢ Valency depends on the number of electrons in the outermost shell.

- If number of electrons in the outermost shell = 1, 2, 3 or 4 then their valency = 1, 2, 3 or 4, respectively.
- If the number of electrons present in the outermost shell = 5, 6 or 7 then their valency = 8 – 5 = 3, 8 – 6 = 2 and 8 – 7 = 1.

➢ Valency is the combining capacity, so it is always positive.

TOPIC-2: PERIODICITY AND RELATED PROPERTIES

Concepts covered: • Periodicity on the basis of: 1. Atomic number 2. Nuclear charge
• Study of different groups in periodic table

Revision Notes

➢ The **atomic number (Z)** of an element is equal to the number of protons in the nucleus. However, it's a unique property of an element because no two elements have the same atomic number.

This question is for practice and its solution is available at the end of the chapter.

Atomic Number (Z) = Number of protons (p)

Number of electrons (e)

(i) It gives the electronic configuration of an element, e.g., an element with atomic number 13 will have electronic configuration of 2, 8, 3.

(ii) It helps in finding the position of element in periodic table. For example, the element with atomic number 17 will have electronic configuration of 2, 8, 7. This element will be placed in the 3rd period of group 17, i.e., VIIA, because it has (i) three energy shells and (ii) seven electrons in its outermost shell.

➢ The **mass number (A)** of an element is the sum of number of protons and neutrons in the nucleus of an atom of that element.

Mass Number (A) = Number of protons (p) + Number of neutrons (n)

However, if we look at the electronic configurations of lighter elements, it reveals that those elements which have even number of protons, for example, atomic numbers like $^{4}_{2}He$, $^{12}_{6}C$, etc., have their mass numbers twice the atomic numbers, except for $_4Be^9$ and $_{18}Ar^{40}$.

Elements which have odd number of protons like $^{7}_{3}Li$, $_5B^{11}$, $_9F^{19}$, $_{11}Na^{23}$, etc., have their mass numbers twice the atomic numbers +1 (A = 2Z+1) except for $_7^{14}N$ and $^{1}_{1}H$.

(i) Elements with n/p (neutron/proton) ratio around 1 are stable, e.g., light metals like sodium, potassium, calcium, etc.

(ii) Elements with n/p ratio of 1.5 and above are radioactive, i.e., they emit radiation. They are unstable elements, e.g., heavy metals like uranium.

➢ **Ionisation Energy or Ionisation Potential:** **[Board, 2024]**

Ionisation enthalpy is the amount of energy required to remove one valency electron from an isolated neutral gaseous atom.

Trends in Ionisation Energy: **[Board, 2023]**

(i) Down a group: Ionisation energy decreases due to an increase in atomic number and atomic size as one moves down a group.

(ii) Across a period: Ionisation energy tends to increase across a period. This is because as one moves from left to right, the atomic size decreases due to increase in the nuclear charge. Thus, more energy is required to remove the electron(s).

➢ **Electron Affinity (EA) or Electron Gain Enthalpy:** **[Board, 2023]**

The amount of energy released when one electron is added to a neutral isolated gaseous atom to form a monovalent negative ion.

$$X(g) + e^- \rightarrow X^-(g) + EA$$

X is any element taken in its gaseous state.

Trends in Electron Affinity:

(i) Down a group: As we move from top to bottom in a group, the atomic size increases more than the nuclear charge, so there is a net decrease in electron affinity.

(ii) Across a period: As we move from left to right, the atomic size decreases, and nuclear charge increases and the electron affinity increases.

➢ **Electronegativity:**

Electronegativity

Scan Me!

Electronegativity is the relative tendency of an atom in a molecule to attract the shared pair of electrons towards itself. Noble gases have complete octet, so they do not have a tendency to attract electrons.

Trends in Electronegativity

(i) Down a group: Electronegativity decreases down a group.

(ii) Across a period: Electronegativity increases from left to right in a period. This is due to an increase in atomic number and nuclear charge.

Key Terms

➢ **Metals:** The elements, which possess a tendency to lose their valence electrons and form a positive ion, are considered as metals.

➢ **Non-metals:** The elements, which possess a tendency to gain electrons in order to attain octet in their outermost orbit, are considered non-metals.

➢ **Atomic Number:** The number of protons in the nucleus.

➢ **Mass Number:** It is the sum of the number of protons and neutrons in the nucleus of the atom of that element.

Key Facts

➢ 1 pm = 10^{-12} m

➢ In group 17, the size of fluorine is smallest.

F <	Cl <	Br <	I <	At
64 pm	99 pm	114 pm	133 pm	140 pm

➢ In the third period, sodium atom is the largest in size and chlorine atom is the smallest.

➢ Exceptionally, the atomic size of inert gases are bigger. It is due to the fact that the outer shell of inert gas is complete. They have the maximum number of electrons in the outermost orbit, so the electronic repulsions are maximum. (Noble gases have fully filled outer shells, due to which their radius is expressed as Van der Waals radius, which represents the overall size of the atom, which includes its valence shell in a non-bonded situation.) Thus, the size of the atom of an inert gas is bigger.

Na >	Mg >	Al >	Si >	P >	S >	Cl
186 pm	160 pm	143 pm	117 pm	110 pm	104 pm	99 pm

➢ Cation is always smaller than the parent atom because cation is formed by the loss of electrons, so protons are more than electrons in a cation. Therefore, electrons are strongly attracted by the nucleus, hence, the size decreases.

This question is for practice and its solution is available at the end of the chapter.

- Anion is larger than the parent atom because anion is formed by the gain of electrons, so the number of electrons is more than protons. The effective positive charge in the nucleus is less, so less inward pull takes place. Hence, the size of anion increases. **[Board, 2024]**
- The ions having the same number of electrons are called isoelectronic ions. The greater the nuclear charge, the smaller the size.

  For example, Mg^{2+} (10 electrons)
  Na^{+} (10 electrons)
  F^{-} (10 electrons)
  O^{2-} (10 electrons)
- Elements which lose electrons to complete their octet are called reducing agents. Metals are good reducing agents. Greater the tendency to lose electrons (s), stronger the reducing agent.
- The oxides of elements in a particular period show decreasing basic nature and finally become acidic.

For example,

Na_2O	MgO	Al_2O_3	SiO_2	P_2O_5	SO_3	Cl_2O_7
Strongly basic	Basic	Amph-oteric	Feebly acidic	Acidic	More acidic	Most acidic

- The basic nature of oxides of metals increases down the group.
- Helium will have the highest ionisation energy (I.E. 2372 kJ mol^{-1}), whereas caesium will have the lowest ionisation energy (375 kJ mol^{-1}).
- Metals usually have low I.E., while non-metals have high I.E.
- The elements of the second period exhibit resemblance in properties with the elements of the next group of the third period due to very small electronegativity difference. This leads to a diagonal relationship, e.g., Li and Mg, Be and Al, B and Si. These elements are known as **bridge elements**.

Group →	1	2	13	14
Period 2	Li	Be	B	C
	⤡	⤡	⤡	
Period 3	Na	Mg	Al	Si

- Atomic number (Z) = Number of protons (p)
  = Number of electrons (e)
- Mass number (A) = Number of protons (p) + Number of neutrons (n)

Example 3

(i) What is meant by a group in the periodic table?
(ii) Within a group, where would you expect to find the element with
(A) the greatest metallic character?
(B) the largest atomic size?
(iii) State whether the ionisation potential increases or decreases on going down a group.
(iv) How many elements are there in period 2?
[Board, 2002]

Ans. (i) Groups: The vertical columns in a periodic table are called groups. The modern periodic table has eighteen vertical columns / groups.

(ii) (A) Within a group, metallic nature increases as one moves down a group, e.g., in group 1 Caesium is the most metallic element.

(B) Within a group, the size of an atom increases as one proceeds from top to bottom. This is due to the successive addition of shells. For example, in group 1, caesium (Cs) is largest in size. Similarly, in group 17, astatine (At) is largest in size.

(iii) Ionisation energy decreases down a group.

(iv) In period 2 there are 8 elements.

Course of Action
Maximum Time = 3 Hrs
Maximum Questions = 65

2 CHEMICAL BONDING

Learning Objectives

Students will be able to learn:
- *Electrovalent, covalent and co-ordinate bonding.*
- *Electron dot structures of different elemental and compound molecules.*

TOPIC-1: ELECTROVALENT AND COVALENT BONDING

Concepts covered: • Electron dot structures. • Electrovalent bonding and its properties. • Covalent bonding and its properties.

Revision Notes

➢ Stability of an atom means possessing the electron arrangement of an inert gas, i.e., octet or duplet in its outermost shell.

➢ To attain stability, atoms tend to combine chemically by the redistribution of electrons in the outermost shell or valence electrons, so that they attain a stable electronic configuration (duplet or octet).

➢ Chemical combination occurs due to tendency of elements to acquire the nearest noble gas configuration in their outermost orbit.

➢ An atom is electrically neutral. The number of positively charged particles (i.e., protons) is equal to the number of negatively charged particles (i.e., electrons). Every atom tries to attain the stable configuration of nearest inert gas, i.e., eight electrons in the valence shell or two electrons in the K shell. Atoms attain the stable electronic configuration by either losing, gaining or sharing electrons.

➢ The force of attraction that holds two atoms together to maintain the stability of a molecule is called **chemical bond.** Atoms form chemical bonds in order to attain stable electronic configuration of the nearest noble gas with the release of energy.

A stable electronic configuration is achieved by:

(i) Transfer of valence electrons from one atom to another (ionic bond or electrovalent bond).

(ii) Sharing of electrons between two atoms (covalent bond).

(iii) Sharing of a pair of electrons contributed entirely by one of the combining atoms only. (coordinate bond).

➢ **Electrovalent Bond**

Electrovalent Bond or Ionic Bond

A chemical bond formed by the transfer of valence electrons from one atom (metal) to another atom (non-metal) is called an **electrovalent** or an **ionic bond.**

➢ **Electrovalency**

The number of electrons lost or gained by an atom to form an electrovalent bond is called its **electrovalency.**

Conditions for the formation of an electrovalent bond are:

(i) Low ionisation potential of metal.

(ii) High electron affinity of non metal.

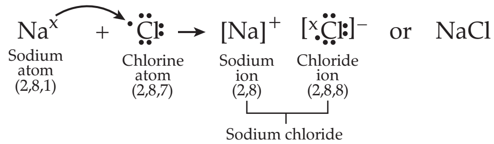
(iii) Large electronegativity difference between the two elements.

➢ **Electron Dot Structure**

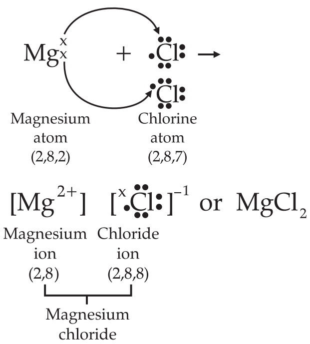
The symbolic representation of an element indicating their valence shell electrons is called **electron dot structure** or **Lewis structure.** For example, electron dot structure for hydrogen is H and for oxygen is $:\ddot{O}:$

➢ Electron Dot Structure of Some Electrovalent Compounds:

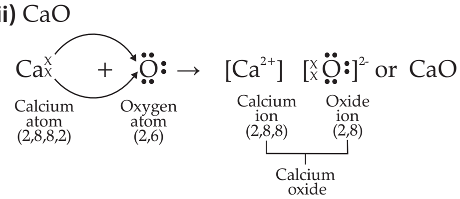
(i) NaCl



(ii) $MgCl_2$



(iii) CaO



➢ Properties of Electrovalent Compounds:

(i) Electrovalent or ionic compounds are hard crystalline solids. As their constituent particles are ions that are held by strong electrostatic forces of attraction, they cannot be separated easily.

(ii) Ionic compounds have high melting point and boiling point as they require a large amount of energy to weaken their strong electrostatic forces of attraction. They are non-volatile solids.

(iii) Ionic compounds do not conduct electricity in their solid state as the ions are not free. They are held by strong electrostatic forces of attraction. However, they can conduct electricity in their fused, (molten) and aqueous solutions. In their molten or aqueous state, the strong forces of attraction gets weakened, and thus, the ions become free to conduct electricity. so they also act as strong electrolytes.

(iv) Ionic compounds are soluble in polar solvents like water, but they are insoluble in non-polar organic solvents. As water has very high dielectric constant, the force of attraction decreases between the ions and thus it forms free ions and hence they dissolve.

(v) On passing electric current through molten and fused are same state. Use either word or put one in bracket and aqueous solutions of ionic compounds, the ions dissociate and migrate towards electrodes.

(vi) Ionic compounds undergo fast reactions in their aqueous solutions.

➢ Covalent Bond **[Board, 2024]**

Covalent Bond
Scan Me!

The bond formed as a result of or equal sharing of electrons between two combining atoms is called **covalent bond**. The covalent bond formation takes place between two **non-metallic** elements.

➢ Conditions for the Formation of a Covalent Bond

The combining atoms should have:

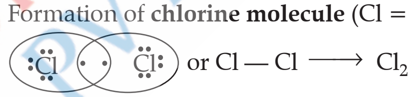
(i) Four or more electrons in outermost shell
(ii) High electronegativity
(iii) High electron affinity
(iv) High ionisation energy
(v) Electronegativity between the two atoms should be **zero or negligible**.

➢ Types of covalent bonds

(i) A **single covalent bond** is formed by the sharing of one pair of electrons between the atoms, each atom is contributing only one electron. A single covalent bond is denoted by putting a single short line (—) between the two atoms.

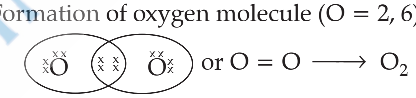
(A) Formation of **hydrogen molecule** (H = 1)

(B) Formation of **chlorine molecule** (Cl = 2, 8, 7)



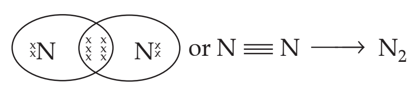
(ii) A **double covalent bond** is formed by the sharing of two pairs of electrons between the two atoms to acquire stable electronic configuration. A double covalent bond is denoted by putting a double line (=) between the two atoms.

Formation of oxygen molecule (O = 2, 6)



(iii) A **triple covalent bond** is formed by the sharing of three pairs of electrons between the two atoms. A triple covalent bond is denoted by putting three short lines (≡) between the two atoms.

Formation of nitrogen molecule (N = 2, 5) **[Board, 2024]**

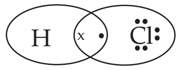


- Oxygen has double covalent bond
- Nitrogen has triple covalent bond
- Hydrogen has single covalent bond

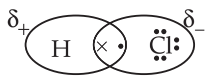
(iv) Polar covalent bonds: The covalent compounds in which the combining elements have large difference in their electronegativity and sharing of the electrons in the covalent bond is unequal are called polar covalent compounds.

Example: Hydrogen chloride gas (HCl), water (H_2O) and ammonia (NH_3).

(v) In case of hydrogen chloride gas, chlorine is more electronegative than hydrogen; therefore, chlorine attracts the shared pair of electrons towards its side. Hence, chlorine acquires a partial negative charge and hydrogen acquires a partial positive charge. Thus, the bonds become polar.



Mutual sharing



Chlorine is more electronegative than hydrogen

(vi) The compounds formed by the mutual sharing of electrons are called covalent compounds.

➢ **Properties of covalent compounds:**

(i) Covalent compounds are liquids or gases. Their constituent particles are held together by weak van der Waal forces.

(ii) Covalent compounds are volatile with low melting points and boiling points. As in these compounds, the molecules are held by weak van der Waal forces, so a lesser amount of energy is required to overcome these forces of attraction.

(iii) Covalent compounds do not conduct electricity, as they do not contain ions. They only contain molecules.

(iv) On passing electric current through the covalent compounds, they do not ionise as they contain only molecules and not ions.

However, polar covalent compounds on dissolving in water produce ions and thus act as electrolytes.

(v) Non polar covalent compounds and are insoluble in water but soluble in organic solvents. Polar covalent compounds are soluble in water and insoluble in organic solvents.

(vi) Polar covalent compounds only contain molecules, and they do not ionise in water.

(vii) Covalent compounds undergo slow speed reactions.

MNEMONICS

1. Concept: Bonding

Mnemonics:

Share **C**oconuts **T**ogether **I**ndia

Interpretation:

S- Shared pair of electrons

C- Covalent bond

T – Transfer of electrons

I – Ionic bonding

Key Terms

➢ **Ion:** Charged atoms which exist freely in the solution.

➢ **Cation:** When an atom loses electron, it gets converted to positively charged particle **called cation.**

➢ **Anion:** When an atom gains electron, it gets converted to negatively charged particle **called anion.**

➢ **Octet rule:** It is the tendency of the atoms of the element to have eight electrons in its valence shell.

➢ **Duplet rule:** It is the tendency of the atoms of an element to have two electrons in its valence K shell.

➢ **Noble Gases:** Gases which are chemically unreactive and have eight electrons in their valence shells. (except helium, which has only two electrons in the first orbit).

➢ **Electropositive elements:** The elements which easily lose electrons and acquire positive charge are called electropositive elements. Metals are electropositive elements.

Example: Na^+, K^+, Ca^{2+}, Mg^{2+}, Al^{3+}, etc.

➢ **Electronegative elements:** The elements which easily gain electrons and acquire negative charge are called electronegative elements. Non-metals are electronegative elements.

Example: Cl^-, Br^-, I^-, F^-, O^{2-}, S^{2-}, N^{3-}, etc.

➢ **Electrovalency:** It is the number of electrons lost or gained by an atom of the element during the formation of ionic bond.

➢ **Crystal Lattice:** It is the definite, three-dimensional geometrical arrangement of constituent atoms or ions in space.

➢ **Covalency:** It is the number of electrons contributed by an atom of the element for sharing during the formation of a covalent bond.

Key Facts

➢ Sharing of one, two or three pairs of electrons between two atoms form a covalent or molecular bond.

➢ The cation and anion being oppositely charged attract each other and form a chemical bond. Since this chemical bond formation is due to the electrostatic force of attraction between a cation and an anion, it is known as electrovalent or ionic bond.

➢ Caesium fluoride (CsF) is the most ionic compound.

➢ Bonds formed between metals and non-metals are ionic or electrovalent.

➢ The electropositive atom undergoes oxidation while the electronegative atom undergoes reduction. This is known as redox process.

➢ The process by which polar covalent compounds forms ions is known as **ionisation.**

➢ The molecule that possesses both partial positive charge (δ^+) and partial negative charge (δ^-) is called a **dipole molecule** and the bond present is polar covalent bond.

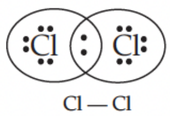
➢ Hydrogen can combine with all non-metals of Group 14 A to 17 A with the help of covalent bonds.

Example 1

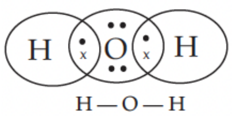
Give electron dot representation of:

(i) Cl_2 (ii) H_2O (iii) NH_3

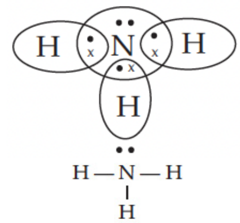
Ans. (i)



(ii)



(iii)



TOPIC-2: COORDINATE BONDING

Concepts covered: • Introduction of coordinate bond • Characteristic properties of coordinate bond • Formation of ammonium ion, hydronium ion and hydroxyl ion

Revision Notes

➢ **Coordinate Bond:**

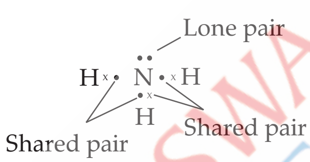
- The bond formed between two atoms by sharing a pair of electrons, both of which are contributed by only one of the bonding atoms but shared by both the atoms, is called **coordinate bond.** **[Board, 2024, 2025]**
- The atom which provides the electron pair for the formation of a coordinate bond is known as **donor.** The atom or ion sharing the donated electron pair is known as **acceptor.**
- Coordinate bond has properties of both covalent and ionic bonds, thus, it is also known as **co-ionic or dative bond.**
- A pair of electrons is not shared with any other atom for the formation of coordinate bond.
- The coordinate bond is shown by an arrow with its head pointing away from the donor to the acceptor atom ($\longrightarrow$).

➢ **Conditions for the Formation of Coordinate Bond**

One of the two atoms must have at least one lone pair of electrons. While another atom should be **deficient** of at least one lone pair of electrons.

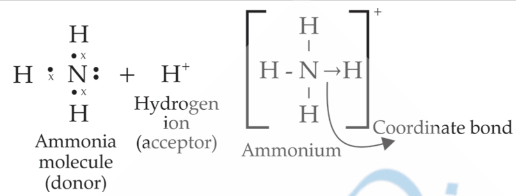
➢ **Formation of Ammonium Ion** $(NH_4)^+$

Ammonium ion (NH_4^+) is formed from ammonia molecule and hydrogen ion.



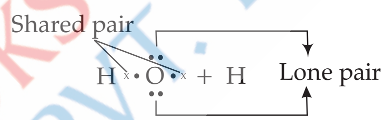
Formation of Ammonia

In ammonia (NH_3), N-atom has one lone pair of electrons, which is used for donation to H^+ ion for the formation of NH_4^+ ion.

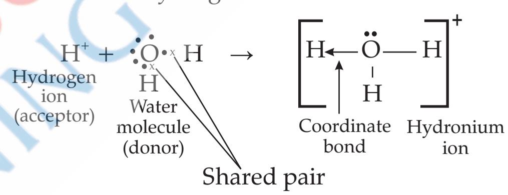


➢ **Formation of H_3O^+ ion**

Hydronium ion is formed from water molecules and hydrogen ions. The oxygen atom of a water molecule possesses **two lone pairs of electrons**.



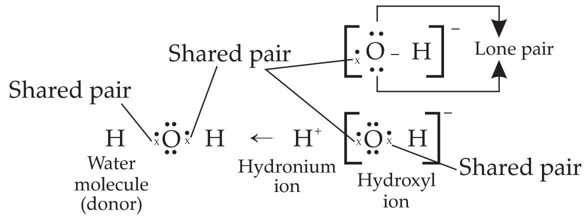
Out of these two lone pairs of electrons, one lone pair is shared with hydrogen ion.



➢ **Formation of Hydroxyl ion (OH^-)**

The hydroxyl ion or hydroxide ion is formed when one hydrogen ion (H^+) is removed from water molecule.

After removing one hydrogen ion from water, the shared pair of electrons remains with oxygen, as oxygen is more electronegative, and thus hydroxyl ion has negative charge.



➤ **Self-ionisation of water**

$$\underset{}{H_2O} + H_2O \rightleftharpoons \underset{\text{Hydronium ion}}{H_3O^+} + \underset{\text{Hydroxide ion}}{OH^-}$$

MNEMONICS

Concept: Examples of coordinate bond

Mnemonics: **A**ishwarya **M**ust **H**ave **N**ew **H**eels **X**tra **L**arge

Interpretation:

A M – Ammonium

H N – Hydronium

hydroxyl / hydroxide ion is not an example of coordinate bond. It have lone pair of electrons, no coordinate bond.

Key Terms

➤ **Bond pair:** Bond pair of electrons is the pair of electrons shared between two atoms.

➤ **Lone pair:** Lone pair of electrons is present in the valence shell on one atom which does not take part in sharing.

➤ **Coordinate bond:** It is a special case of covalent bond formed between two atoms in which both the shared pair of electrons are contributed by one of the atoms, whereas the other atom simply takes part in sharing.

➤ **Oxidation:** It is a chemical process in which an atom or an ion loses one or more electrons.

➤ **Reduction:** It is a chemical process in which an atom or ion gains one or more electrons.

Key Facts

➤ Addition of released H^+ ion to a lone pair of electrons of the oxygen atom of the polar water molecule leads to the formation of a hydronium ion.

➤ Hydronium ion is a hydrated proton.

$$\underset{\text{Proton}}{H^+} + \underset{\text{Water}}{H_2O} \longrightarrow \underset{\text{Hydronium ion}}{H_3O^+}$$

➤ When ammonium chloride (NH_4Cl) is formed, the cation NH_4^+ (possessing three covalent bonds and one coordinate bond) and anion Cl^- are attracted towards each other due to electrical charge between them, and then ionic bond or electrovalent bond is formed. Hence, NH_4Cl is a good example of compounds having all three types of bonds, i.e., covalent, coordinate and ionic bond. structure of Ammonium Chloride should be shown here for better understanding.

Example 2

How many covalent bonds and coordinate bonds are present in: **[OEB]**

(i) Hydronium ion?

(ii) Ammonium ion?

Ans. (i) Hydronium ion contains two covalent bonds and one coordinate bond.

(ii) Ammonium ion has three covalent bonds and one coordinate bond.

Course of Action
Maximum Time = 3:45 Hrs
Maximum Questions = 76

3 ACIDS, BASES AND SALTS

Learning Objectives

Students will able to learn:
- *Concept of molecules and their characteristic properties.*
- *Characteristics of acids, alkalis and salts.*
- *Relevant equations for the preparation of normal salts.*
- *Test for acidity and alkalinity.*

TOPIC-1: ACIDS AND BASES

Concepts covered: • Acids - Physical and chemical properties. • Bases - Physical and chemical properties. • pH and its application.

Revision Notes

Acids
Scan Me!

- **Acids:** The 'acid' comes from the Latin word '*acidus*' meaning 'sour'.
- Acids are defined as compounds which contain one or more hydrogen atoms and, when dissolved in water, produce hydronium ions (H_3O^+), the only positively charged ions, for example, hydrochloric acid (HCl), sulphuric acid (H_2SO_4) and nitric acid (HNO_3).
- **Ions present in acids:** Mineral acids (inorganic acids) like HCl, H_2SO_4 and HNO_3 ionise completely in the solution. So, they contain high concentration of hydronium ions (H_3O^+). Hydronium ion is hydrated hydrogen ion. Organic acids like acetic acid (CH_3COOH) and oxalic acid $(COOH)_2$ do not ionise completely in solution. So, they contain ions as well as molecules.

➢ **Classification of Acids:**

1. Depending on their sources - (i) Organic acids **(ii)** Inorganic acids

(i) Organic acids: Acids which are obtained usually from plants are called **organic acids.** They contain carbon atoms also along with hydrogen atoms.

Example: CH_3COOH (Acetic acid).

(ii) Inorganic acids: Acids which are obtained usually from minerals are known as **inorganic acids.** They do not contain carbon (except carbonic acid H_2CO_3).

Example: H_2SO_4 (Sulphuric acid), HCl (Hydrochloric acid)

2. Depending on their strength – **(i)** Strong acid **(ii)** Weak acid

(i) Strong acid: A strong acid vigorously ionises in aqueous solution, thereby producing a high concentration of hydronium ion $[H_3O]^+$. **[Board 2022]**

Example: HCl, H_2SO_4

(ii) Weak acid: Weak acids ionise partially in aqueous solution, and thus they produce ions as well as molecules.

Example: CH_3COOH (Acetic acid).

3. Depending on their concentration :

(i) Concentrated acid (Contains less water)

(ii) Dilute acid (Contains more water)

4. Depending on molecular composition: (i) Oxy-acids **(ii)** Hydro-acids

(i) Oxy-acids : Oxy-acids are those acids which contain oxygen along with hydrogen and some other element.

Example: Nitric acid, sulphuric acid

(ii) Hydro-acids: Hydro-acids are those acids which contain hydrogen and a non-metallic element and no oxygen. **[Board 2022]**

Example: Hydrochloric acid.

5. Depending on their basicity - The **basicity** of an acid is defined as the number of hydronium ions $(H_3O)^+$ that can be produced by ionisation of one molecule of that acid in aqueous solution.

Acid	Basicity
$HCl \longrightarrow H^+ + Cl^-$	Monobasic acid **[Board 2023]**
$H_2SO_4 \longrightarrow 2H^+ + SO_4^{2-}$	Dibasic acid
$H_3PO_4 \longrightarrow 3H^+ + PO_4^{3-}$	Tribasic acid **[Board 2023]**

➢ **Physical properties of acids :**

(A) Acids have a sour taste in their aqueous solutions.

(B) Acids turn blue litmus paper or solution to red, orange or yellow colour of methyl orange solution to red and Phenolphthalein is a colorless indicator and remains unchanged in acidic conditions.

(C) The aqueous solution of all mineral acids contains hydrogen ions.

(D) Mineral acids such as HCl, HNO_3, H_2SO_4, etc., are highly corrosive and cause painful burns on the skin. Conc. H_2SO_4 corrosive the skin, while conc. HCl stains the skin amber-coloured.

➢ **Chemical properties of acids :**

(A) Metals which are more electropositive than hydrogen react with dil. HCl and dil. H_2SO_4 to liberate hydrogen gas. **[Board 2023]**

$$Zn(s) + 2HCl(aq) \longrightarrow ZnCl_2(aq) + H_2(g)$$

Nitric acid (HNO_3) does not liberate hydrogen gas on reaction with metals. This is because of its oxidising property where liberated hydrogen is oxidised to water. However, Mg and Mn react with very dilute nitric acid to produce hydrogen gas.

(B) Acids react with metal oxides to form the corresponding salt and water.

Type of reaction the acid undergoes is **Neutralisation**.

$$CaO(s) + 2HCl(aq) \longrightarrow CaCl_2(aq) + H_2O(l)$$

When MnO_2 is heated with conc. HCl, a greenish-yellow chlorine gas is produced.

$$MnO_2(s) + 4HCl(aq) \longrightarrow MnCl_2(aq) + 2H_2O(l) + Cl_2(g)$$

(C) Bicarbonates and carbonates are decomposed by dilute acids with the liberation of carbon dioxide (CO_2) gas with brisk effervescence.

$$NaHCO_3(s) + HCl(aq) \longrightarrow NaCl(aq) + H_2O(l) + CO_2(g)$$

$$CaCO_3(s) + 2HNO_3(aq) \longrightarrow Ca(NO_3)_2(aq) + H_2O(l) + CO_2(g)$$

(D) Bisulphites and sulphites are decomposed by dilute acids to liberate sulphur dioxide (SO_2) gas having a pungent and choking odour.

$$NaHSO_3(s) + HCl(aq) \longrightarrow NaCl(aq) + H_2O(l) + SO_2(g)$$

$$Na_2SO_3(s) + H_2SO_4(aq) \longrightarrow Na_2SO_4(aq) + H_2O(l) + SO_2(g)$$

(E) Dilute acids react with metal sulphides and liberate hydrogen sulphide gas having rotten egg smell.

$$FeS(s) + H_2SO_4(aq) \longrightarrow FeSO_4(aq) + H_2S(g)$$

(F) Hydrogen chloride gas is produced when concentrated sulphuric acid is added to common salt.

$$2\,NaCl(s) + H_2SO_4(aq) \longrightarrow Na_2SO_4(aq) + 2HCl(g)$$

(G) When sodium or potassium nitrate is heated with concentrated sulphuric acid, vapours of nitric acid are evolved.

$$KNO_3(s) + H_2SO_4(aq) \xrightarrow{\text{below } 200^\circ C} KHSO_4(aq) + HNO_3(g)$$

General Uses of Some Acids:

(A) Boric acid - Antiseptic for eye wash

(B) Oxalic acid - Ink stain remover

(C) Tartaric acid - Baking powder

(D) Citric acid - Food preservative

(E) Carbonic acid - Flavoured drink

(F) Phosphoric acid - Fertilisers.

➢ **Bases :** A base is either a **metallic oxide** or a metallic hydroxide which reacts with hydronium ions of an acid to form salt and water only.

➢ **Alkali :** An alkali is a basic hydroxide which, when dissolved in water, produces hydroxyl (OH^-) ions as the only negatively charged ions.

$$NaOH(aq) \longrightarrow Na^+ + OH^-$$

➢ **Classification of Bases :**

1. On the basis of strength

(i) A strong base : It undergoes almost complete ionisation in aqueous solution to produce a high concentration of OH^-.

Example: Sodium hydroxide (NaOH) and potassium hydroxide (KOH).

(ii) A weak base : A weak base is a substance which ionises only to a small extent when dissolved in water.

Example: NH_4OH, $Ca(OH)_2$, etc.

2. On the basis of acidity : The number of hydroxyl ions $[OH]^-$ which can be produced per molecule of the base in aqueous solution or the number of hydrogen ions (of an acid) with which a molecule of that base will react to produce salt and water is known as **acidity of the base**.

Preparation of Acids, Bases and Salts

Scan Me!

Base	Acidity
$NaOH \longrightarrow Na^+ + OH^-$	Monoacidic base
$Ca(OH)_2 \longrightarrow Ca^{2+} + 2OH^-$	Diacidic base
$Al(OH)_3 \longrightarrow Al^{3+} + 3OH^-$	Triacidic base

3. Concentration of a base depends on the amount of water present in it. A concentrated base has more quantity of base and a little or no water, while a dilute base has more water and less base.

Preparation of Bases : The common methods of preparing bases are as follows:

(i) From metals : Metals when reacts with oxygen give bases.

$$4Na + O_2 \longrightarrow 2Na_2O$$

$$2Mg + O_2 \longrightarrow 2MgO$$

(ii) By the action of water on metals; like sodium, potassium and calcium :

$$2Na + 2H_2O \longrightarrow 2\,NaOH + H_2\uparrow$$

$$2K + 2H_2O \longrightarrow 2\,KOH + H_2\uparrow$$

$$Ca + 2H_2O \longrightarrow Ca(OH)_2 + H_2\uparrow$$

(iii) By the action of water on soluble metallic oxides:

$$Na_2O + H_2O \longrightarrow 2\,NaOH$$

$$K_2O + H_2O \longrightarrow 2\,KOH$$

(iv) By double decomposition: Aqueous solution of salts with a base (alkali) precipitates the respective metallic hydroxide.

$$FeCl_3 + 3NaOH \longrightarrow Fe(OH)_3\downarrow + 3\,NaCl$$

$$FeSO_4 + 2NaOH \longrightarrow Na_2SO_4 + Fe(OH)_2\downarrow$$ **[ICSE 2023]**

$$CuSO_4 + 2\,NaOH \longrightarrow Cu(OH)_2\downarrow Na_2SO_4$$

(v) By the action of oxygen on metal sulphides :

$$2\,ZnS + 3O_2 \longrightarrow 2\,ZnO + 2SO_2$$

$$2\,PbS + 3O_2 \longrightarrow 2\,PbO + 2SO_2$$

(vi) By decomposition of salts: **[Board 2023]**

$$CaCO_3 \longrightarrow CaO + CO_2\uparrow$$

$$CuCO_3 \longrightarrow CuO + CO_2\uparrow$$

$$2Ca(NO_3)_2 \longrightarrow 2CaO + 4NO_2 + O_2\uparrow$$

$$2Zn(NO_3)_2 \longrightarrow 2ZnO + 4NO_2 + O_2$$

➢ **Physical Properties of Bases**

(A) Alkalis have a bitter taste and are soapy to touch.

(B) Alkalis turn red litmus paper or solution blue, colourless phenolphthalein pink, orange colour of methyl orange to yellow colour and turmeric powder to brown colour.

➢ **Chemical Properties of Bases**

(A) Alkalis react with acids to form salt and water.

Type of reaction it undergoes is **Neutralisation**.

$$2\,NaOH\,(aq) + H_2SO_4\,(aq) \longrightarrow Na_2SO_4\,(aq) + 2\,H_2O$$

$$CuO\,(s) + H_2SO_4\,(aq) \longrightarrow CuSO_4\,(aq) + H_2O\,(l)$$

(B) Alkalis react with ammonium salts to liberate ammonia gas with a pungent smell.

$$2\,NaOH\,(aq) + (NH_4)_2SO_4\,(aq) \longrightarrow Na_2SO_4\,(aq) + 2\,H_2O\,(l) + 2NH_3\,(g)$$

(C) Alkalis react with solutions of salts of heavy metals (copper, iron, zinc, lead, etc.) to form insoluble precipitates of metal hydroxides. This can be used to detect the presence of hydroxide ions in solution.

$$CuCl_2\,(aq) + 2\,NaOH\,(aq) \longrightarrow Cu(OH)_2\downarrow + 2NaCl\,(aq)$$

$$FeCl_3\,(aq) + 3\,NaOH\,(aq) \longrightarrow Fe(OH)_3\downarrow + 3NaCl(aq)$$

$$ZnCl_2(aq) + 2\,NaOH\,(aq) \longrightarrow Zn(OH)_2\downarrow + 2NaCl(aq)$$

Chemical Properties of Bases

Scan Me!

(D) Hydroxides of zinc, lead and aluminium behave like acids in the presence of strong alkalis, while they behave like **bases** in the presence of strong acids, so as to form salt and water.

$$Zn(OH)_2\,(aq) + 2\,NaOH\,(aq) \longrightarrow Na_2ZnO_2\,(aq) + 2H_2O\,(l)$$

$$Zn(OH)_2\,(aq) + H_2SO_4\,(aq) \longrightarrow ZnSO_4\,(aq) + 2H_2O\,(l)$$

$$Al(OH)_3\,(aq) + NaOH\,(aq) \longrightarrow NaAlO_2\,(aq) + 2H_2O\,(l)$$

$$Al(OH)_3\,(aq) + 3\,HCl\,(aq) \longrightarrow AlCl_3\,(aq) + 3H_2O\,(l)$$

(E) All bases except NaOH and KOH decompose on heating to form their respective oxides.

$$Ca(OH)_2\,(aq) \xrightarrow{Heat} CaO\,(aq) + H_2O\,(l)$$

$$2Al(OH)_3\,(aq) \xrightarrow{Heat} Al_2O_3\,(aq) + 3H_2O\,(l)$$

(F) Caustic alkalis (NaOH and KOH) are corrosive to the skin as they combine with the oils and fats in the skin.

➢ **Uses of Bases**

(1) Aluminium hydroxide is used as a foaming agent in fire extinguishers.

(2) Sodium hydroxide (caustic soda) is used in the manufacture of soaps.

(3) Ammonium hydroxide is used for removing grease and stains from woollen clothes.

(4) Potassium hydroxide is used in the manufacturing of soft soaps and alkaline batteries.

(5) Calcium hydroxide (Slaked lime) is used in the manufacturing of bleaching powder, softening of hard water, neutralising acidity of soil, as a general germicide and in making mortar.

Neutralisation Reaction

Scan Me!

➢ **Neutralisation Reaction:** The reaction between an acid and a base to form salt and water is known as a neutralisation reactions. **[Board 2022]**

$$HCl\,(aq) + NaOH\,(aq) \longrightarrow NaCl\,(aq) + H_2O\,(l)$$

$$H^+ + Cl^- + Na^+ + OH^- \xrightarrow{Water} Na^+ + Cl^- + H_2O\,(l)$$

$$H^+ + OH^- \longrightarrow H_2O\,(l)$$

Or,

When ammonia is dissolved in water, it acts as a base and water as an acid. The following are the applications of neutralisation reaction:

(A) Farmers reduce acidity of the soil by adding slaked lime (calcium hydroxide) to it.

(B) Antacid tablets containing magnesium hydroxide are given to persons suffering from acidity to neutralise excess acid produced in the stomach.

(C) Cold milk, which is alkaline, also helps a person in neutralising the hydrochloric acid present in the stomach.

(D) The sting of ants and bees contains formic acid. This can be neutralised by rubbing soap which contains free sodium hydroxide, on the affected area of the skin or by applying paste of baking soda+ water.

(E) The sting of yellow wasps contains an **alkali**. This is neutralised by rubbing acetic acid (vinegar) on the affected area of the skin.

- The acidity or alkalinity of a liquid is expressed in terms of pH. It is a measure of hydrogen ion concentration of the solution.

pH of a solution is the number by which negative power of 10 has to be raised in order to express the hydrogen ion concentration of the solution.

$$\left[H^+\right] = 10^{-x}; \quad \therefore \text{pH} = x$$

It is also defined as the negative logarithm to the base 10 of the hydrogen ion concentration, i.e.,

$$\text{pH} = -\log[H^+]$$

A pH of 7 indicates a neutral solution, i.e., pure water. Numbers less than 7, i.e., pH 6, 5, 4, ..., 1, indicate acidic solutions. The acidity increases as the pH number decreases. Numbers greater than 7, i.e., pH 8, 9, 10, ..., 14, indicate alkaline solutions. The alkalinity increases as the pH number increases. **[Board 2023]**

A reasonably accurate value of the pH of the solution can be determined by putting two or three drops of the solution on a wide range pH paper. A colour will appear and by matching this colour with the chart provided by the supplier, the pH can be determined.

MNEMONICS

Concept: Properties of acid
Mnemonics:
A Student **T**old **T**eacher **B**us **R**ider **C**alled "**H**I"
Interpretation:
A—Acid
S—Sour
T—Taste
T—Turns
B—Blue
R—Red
C—Contains
HI—Hydrogen ion

Key Terms

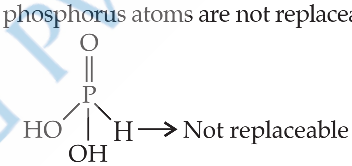
- The bases having an acidity of 1, 2 and 3 are called **monoacidic, diacidic and triacidic bases,** respectively.
- **Concentration of Acid :** It means the amount of acid present in a definite amount of its aqueous solution.
- **Monobasic Acid :** It is an acid which on ionisation in water produces one hydronium ion per molecule of the acid.
- **Dibasic Acid:** It is an acid which on ionisation in water produces two hydronium ions (H_3O^+) per molecule of the acid.
- **Tribasic Acid:** It is an acid which on ionisation in water produces three hydronium ions per molecule of the acid.
- **Indicator:** It is a complex substance that indicates separate colours in acidic and basic mediums.
- **Monoacidic Base:** It is a base that dissociates in molten (fused) state or in aqueous solution to produce one OH^- ion per molecule of that base.
- **Diacidic Base:** It is a base that dissociates in molten (fused) state or in aqueous solution to produce two OH^- ions per molecule of that base.
- **Triacidic Base:** It is a base that dissociates in molten (fused) state or in aqueous solution to produce three OH^- ions per molecule of that base.
- **pH:** The pH of solution is the negative logarithm to the base 10 of the hydrogen ion concentration expressed in moles per litre.

Key Facts

- The hydrated hydrogen ion that exists in the solutions of acids is known as hydronium ion (H_3O^+).

$$H^+ + H_2O \longrightarrow H_3O^+$$

- Carbonic acid (H_2CO_3) is a weak mineral acid. It contains ions as well as molecules. It turns blue litmus red. It is non-corrosive and so used in soft drinks.
- Basicity of an acid depends not on the number of hydrogen atoms in one molecule of that acid but on the number of ionisable hydrogen atoms that it has per molecule.
- H_3PO_3 is a dibasic acid because in oxyacid of phosphorus, hydrogen atoms which are attached to oxygen atoms are replaceable. Hydrogen atoms directly bonded to phosphorus atoms are not replaceable.



- In order to dilute an acid, pour acid into water in small amounts and stir constantly.

It is noted that water is not added to acid, as it is an exothermic processwater might splash and cause accident. However acid being denser settles down slowly in water.

- Strength of an acid is the measure of concentration of hydronium ions it produces in its aqueous solution.
- Dilute HCl is stronger acid than highly concentrated acetic acid.
- The strength of an acid depends on the degree of ionisation (α) and concentration of H_3O^+ ions produced by that acid in aqueous solution.
- Degree of ionisation (α) =

$$\frac{\text{Number of acid molecules ionised}}{\text{Total number of acid molecules present in aqueous solution}} \times 100$$

- If the value of α for an acid or base is greater than 30%, it is strong and if it is less than 30%, it is weak.
- All mineral acids have corrosive action on the skin and cause painful burns.

For example, Conc. H_2SO_4 — stains the skin black
Conc. HNO_3 — stains the skin yellow
Conc. HCl — stains the skin amber

- Those substances whose smell changes in acidic or basic solutions are known as olfactory indicators. For example, onion, vanilla and clove oil.

Indicator	Colour change in acidic medium	Colour change in basic medium
Litmus	Blue to red	Red to blue
Methyl orange	Orange to red	Orange to yellow
Phenolphthalein	Remains colourless	Colourless to pink

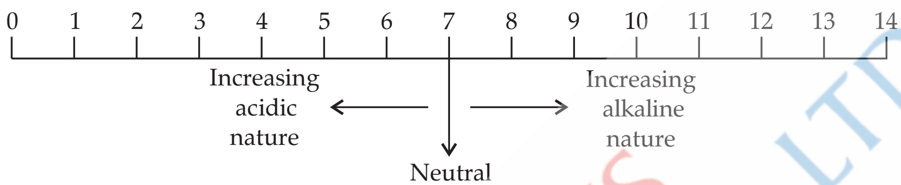
- Nitric acid is a very strong oxidising agent, so it is not used in the preparation of hydrogen. It oxidises hydrogen and forms water. Only magnesium and manganese can produce hydrogen with very dil. HNO_3 (1%).

$$\underset{}{Mn} + \underset{(1\%)}{2HNO_3} \longrightarrow Mn(NO_3)_2 + H_2\uparrow$$

- An alkali is a base soluble in water.
- All alkalis form OH^- ions in aqueous solution as the only negative ions. They turn red litmus blue.
- All alkalis are bases but all bases are not alkalis.

For example, $Fe(OH)_3$ and $Cu(OH)_2$ are bases, but not alkalis because they are insoluble in water.

- **pH Scale :**



- The universal indicator is a **mixture of indicator dyes** that gives a spectrum of colours depending on how acidic or alkaline a solution is.
- Universal indicators give different colours at different concentrations of hydrogen ions in a solution.

For example, a universal indicator produces green colour in a neutral solution, i.e., when pH = 7. It changes in a basic solution progressively from 7 to 14.

- If the pH is less than 5.6 of rainwater, it is said to be acid rain.
- To get rid of acidity in stomach, antacids like milk of magnesia $[Mg(OH)_2]$ are generally used to adjust the pH.
- When the pH of mouth falls to 5.5, tooth decay starts.
- Bee sting leaves acid in the body. If baking soda ($NaHCO_3$), a base, is applied to the stung area, it gives relief.

Revision Notes

➢ **Salts:** The chemical compounds which, on dissolving in water, produce **positively** charged particles other than hydrogen ions and negatively charged particles other than hydroxyl ions are called salts.

$$NaCl \rightleftharpoons Na^+ + Cl^-$$

$$KCl \rightleftharpoons K^+ + Cl^-$$

➢ **Classification of Salt :** There are six kinds of salts.

(A) Normal salt : The salt formed by complete replacement of replaceable hydrogen ions of an acid by a basic radical or metallic ion. Some examples of normal salts are sodium chloride (NaCl), potassium nitrate (KNO_3), copper sulphate ($CuSO_4$) and sodium acetate (CH_3COONa). **[Board 2022]**

$$\text{Acid + Base} \longrightarrow \text{Salt + Water}$$

$$HCl + NaOH \longrightarrow NaCl + H_2O$$

$$HNO_3 + KOH \longrightarrow KNO_3 + H_2O$$

$$H_2SO_4 + Cu(OH)_2 \longrightarrow CuSO_4 + 2H_2O$$

$$CH_3COOH + NaOH \longrightarrow CH_3COONa + H_2O$$

These salts, on dissolving in water, dissociate into their constituent ions. Normal salts have no ionisable hydrogen atoms.

(B) Acid Salt: A salt formed by incomplete or partial replacement of replaceable hydrogen ions of an acid by a basic radical or metallic ion. Some examples of acid salts are sodium hydrogen sulphate ($NaHSO_4$), potassium bisulphate ($KHSO_4$), sodium hydrogen sulphite ($NaHSO_3$).

$$\text{Acid + Alkali} \longrightarrow \text{Salt + Water}$$

$$H_2SO_4 + NaOH \longrightarrow NaHSO_4 + H_2O$$

$$H_3PO_4 + NaOH \longrightarrow NaH_2PO_4 + H_2O$$

$$H_2SO_3 + NaOH \longrightarrow NaHSO_3 + 2H_2O$$

Acid salts are formed when the basicity of the acid taken is more than the acidity of a base. In aqueous solution, acid salts furnish hydrogen ion or hydronium ion (H_3O^+).

(C) Basic Salt: The salt formed by the partial or incomplete replacement of replaceable hydroxyl ions of diacidic or triacidic bases by an acid radical. Some examples of basic salts are basic lead chloride [Pb(OH)Cl] and basic copper chloride [Cu(OH)Cl], basic magnesium chloride [Mg(OH)Cl]. **[Board 2023]**

Salts and its types

Scan Me!

$$\text{Base + Acid} \longrightarrow \text{Salt + Water}$$

$$Pb(OH)_2 + HCl \longrightarrow Pb(OH)Cl + H_2O$$

$$Cu(OH)_2 + HCl \longrightarrow Cu(OH)Cl + H_2O$$

$$Mg(OH)_2 + HCl \longrightarrow Mg(OH)Cl + H_2O$$

The basic salts are only formed when the acidity of the base taken is more than the basicity of an acid. The basic salts contain a metallic cation, a hydroxyl ion from base and an anion obtained from the acid.

(D) Double salt: The salts that contain more than one cation or anion are known as double salts. They are obtained by the combination of two different salts but differ in their crystalline structure. For example, potash alum ($K_2SO_4 \cdot Al_2(SO_4)_3 \cdot 24H_2O$), Mohr's salt ($FeSO_4 \cdot (NH_4)_2 \cdot 6H_2O$).

(E) Mixed salts: The salt that consists of a fixed proportion of two salts, often sharing either a common cation or common anion is known as mixed salt.

For example, bleaching powder ($CaOCl_2$), sodium potassium carbonate ($NaKCO_3$).

(F) Complex salts : The salts which contain different types of metal atoms which, on hydrolysis, produce complex ions along with simple ions are called complex salts. For example, silver amino chloride $[Ag(NH_3)_2]Cl$, tetraammine copper (II) sulphate $[Cu(NH_3)_4]SO_4$, sodium argentocyanide $\{Na[Ag(CN)_2]\}$.

➢ **Preparation of Soluble Salts :**

(A) By synthesis: Several soluble salts are prepared by heating the constituent elements together.

$$2Fe + 3Cl_2 \longrightarrow 2FeCl_3$$

$$2Na + Cl_2 \longrightarrow 2NaCl$$

(B) Simple displacement: Soluble salts of active metals like Mg, Zn, Fe, etc., can be prepared by the simple displacement reactions involving an active metal and dilute acid.

$$Mg + \text{dil. } 2HCl \rightarrow \underset{\text{(Magnesium chloride)}}{MgCl_2} + H_2\uparrow$$

$$Zn + \text{dil. } H_2SO_4 \rightarrow \underset{\text{Zinc sulphate}}{ZnSO_4} + H_2\uparrow$$

(C) Double decomposition: Insoluble or soluble bases react with acids to form salt and water. By this reaction only soluble salts are prepared. (Neutralisation reaction)

$$NH_4OH + \text{dil. } HCl \rightarrow \underset{\text{(Ammonium chloride)}}{NH_4Cl} + H_2O$$

$$2KOH + \text{dil. } H_2SO_4 \rightarrow \underset{\text{Potassium sulphate}}{K_2SO_4} + 2H_2O$$

(D) By the reaction of metallic carbonates, bicarbonates, sulphites, bisulphites, sulphides and bisulphides with dilute acids:

$$CaCO_3 + \text{dil. } 2HCl \rightarrow \underset{\text{Calcium chloride}}{CaCl_2} + H_2O + CO_2\uparrow$$

$$ZnSO_3 + \text{dil. } H_2SO_4 \rightarrow \underset{\text{Zinc sulphate}}{ZnSO_4} + H_2O + SO_2\uparrow$$

(E) By the reaction of metals with alkali:

$$Zn + 2NaOH \xrightarrow{\text{Boiling}} \underset{\text{Sodium Zincate}}{Na_2ZnO_2} + H_2\uparrow$$

$$Pb + 2KOH \xrightarrow{\text{Boiling}} \underset{\text{Potassium metaplumbate}}{K_2PbO_2} + H_2\uparrow$$

$$2Al + 2NaOH + 2H_2O \xrightarrow{\text{Boiling}} \underset{\text{Sodium aluminate}}{2NaAlO_2} + 3H_2\uparrow$$

➢ **Preparation of Insoluble Salts :**

(i) By direct combination

$$Pb + S \longrightarrow PbS$$

(ii) Combination of an acidic oxide with a basic oxide

$$CO_2 + CaO \longrightarrow CaCO_3$$

(iii) By the precipitation reaction (double decomposition)

$$BaCl_2 + H_2SO_4 \longrightarrow BaSO_4\downarrow + 2HCl$$

➢ **Laboratory Preparation of Some Salts :**

(Normal and Acid Salts)

● **Preparation of an Acid Salt (Sodium Bicarbonate)**

Method of preparation: Sodium bicarbonate is obtained by passing carbon dioxide gas into a cold solution of sodium carbonate (Na_2CO_3).

$$Na_2CO_3 + CO_2 + H_2O \longrightarrow 2NaHCO_3$$

Procedure: Dissolve anhydrous sodium carbonate (5g) in distilled water (nearly 25 cm^3) in a flask. Cool the solution by keeping the flask in a freezing mixture, then pass CO_2 gas in the solution. After some time, crystals of sodium bicarbonate will precipitate out, filter the crystals and dry them.

● **Preparation of Sodium Sulphate Crystals ($Na_2SO_4 . 10H_2O$)**

Method of preparation: By neutralisation of caustic soda with dilute sulphuric acid.

$$2NaOH + H_2SO_4 \longrightarrow Na_2SO_4\downarrow + 2H_2O$$

Titration is conducted to determine the completion of the neutralisation reaction because the reactants as well as the products are soluble.

➢ **General Properties of Salts:**

(i) Salts are electrovalent compounds, which conduct electricity in molten state as well as in their aqueous solution.

(ii) Non-volatile solids that form crystals.

(iii) Most of the salts are soluble in water. Degree of solubility in water varies with temperature.

(iv) Hydrolysis of salts

Salt formed by a strong acid and weak base or by a weak acid and a strong base, which reacts with water to give acidic or basic solution, is known as hydrolysis of salts.

(A) Salts of strong acids and weak bases on hydrolysis give acidic solutions (pH<7), e.g., iron (III) chloride ($FeCl_3$), copper sulphate ($CuSO_4$).

(B) Salts of weak acids (H_2CO_3, CH_3COOH) and strong bases (KOH, NaOH) on hydrolysis give alkaline solutions (pH > 7).

(C) Salts of strong acids and strong bases on hydrolysis give neutral solutions.

(D) Salts of weak acids and weak bases on hydrolysis may give acidic, alkaline or neutral solutions depending upon the dissociated ions and undissociated molecules. For example, ammonium acetate [CH_3COONH_4] and ammonium carbonate [$(NH_4)_2CO_3$].

➢ **Water of Crystallisation:**

It is that definite amount of water with which the substance is associated when crystallising out from an aqueous solution, e.g., copper (II) sulphate crystallises out as a pentahydrate, $CuSO_4 . 5H_2O$. Thus, one mole of copper (II) sulphate is associated with five moles of water.

➢ **Hydrated Salts - Water of Crystallisation:**

● The compounds which crystallise out of their saturated solutions with fixed number of molecules of water of crystallisation are called hydrated salts.

● These molecules of water of crystallisation are in loose chemical combination with the salt.

● The hydrated salts owe their crystalline nature to the molecules of water of crystallisation.

Salts containing water of crystallisation

Common Name	Chemical Name	Formula
White vitriol	Zinc sulphate heptahydrate	$ZnSO_4.7H_2O$
Glauber's salt	Sodium sulphate decahydrate	$Na_2SO_4.10H_2O$
Washing soda	Sodium carbonate decahydrate	$Na_2CO_3.10H_2O$
Potash alum	Potassium aluminium sulphate	$KAl(SO_4)_2.12H_2O$
Epsom salt	Magnesium sulphate heptahydrate	$MgSO_4.7H_2O$
Green vitriol	Iron(II) sulphate heptahydrate	$FeSO_4.7H_2O$
Blue vitriol	Copper(II) sulphate pentahydrate	$CuSO_4.5H_2O$
Lime saltpetre	Calcium nitrate	$Ca(NO_3)_2.4H_2O$

➢ **Efflorescence:**

It is the property of some hydrated salts to lose wholly or partly their water of crystallisation when their crystals are exposed to atmosphere. These substances crumble into powder and are called efflorescent salts.

For example :

$$\underset{\text{Washing Soda crystals}}{Na_2CO_3 . 10H_2O} \xrightarrow{\text{Dry air}} \underset{\text{Washing Powder}}{Na_2CO_3 . H_2O} + 9H_2O$$

Higher the temperature of the air, higher the efflorescence because that air absorbs more water with rising temperature and decreasing moisture.

Deliquescence: Certain water soluble substances, when exposed to the atmosphere at ordinary temperature, absorb moisture from the atmospheric air, become moist, lose their crystalline form and ultimately dissolve in the absorbed water, forming a saturated solution. Such substances are called deliquescent substances and this property is called deliquescence.

Examples:

- Anhydrous calcium chloride, $CaCl_2$
- Caustic potash KOH
- Caustic soda NaOH
- Ferric chloride $FeCl_3$

Hygroscopic Substances: Certain substances, when exposed to the atmosphere at ordinary temperature, absorb moisture from the atmosphere without dissolving in it. These are called hygroscopic substances, and this property is called hygroscopy. These substances are generally used for drying gases in laboratory, e.g., conc. sulphuric acid (H_2SO_4), quick lime (CaO), silica gel and phosphorus pentoxide (P_2O_5).

Drying or desiccating agents: Certain substances remove moisture from other substances and are therefore called desiccating agents or drying agents or desiccants. Almost all hygroscopic substances are **desiccating** agents.

For example, conc. sulphuric acid, phosphorus pentoxide, and quick lime.

Dehydrating agents: The substances that can **remove** water molecules even from compounds are called dehydrating agents. For example, concentrated sulphuric acid can remove water molecules from blue vitriol ($CuSO_4.5H_2O$), so it is a dehydrating agent.

$$\underset{\text{Hydrated Salt}}{CuSO_4.5H_2O} + \underset{\text{Conc.}}{H_2SO_4} \rightarrow \underset{\text{Anhydrous Salt}}{CuSO_4} + 5H_2O$$

Key Terms

➢ **Displacement:** It is a chemical change in which a more active element displaces a less active element from its salt solution.

➢ **Decomposition:** It is a chemical change in which a compound breaks up into its elements or simpler compounds.

➢ **Hydrolysis:** The phenomenon due to which salt formed by a weak acid and a strong base, or by a strong acid and a weak base, reacts with water to give an acidic or alkaline solution is called hydrolysis.

Key Facts

➢ Acid salts ionise in water solution to give hydronium ions, and hence, they exhibit all the properties of an acid.

➢ Soluble salts are obtained by the evaporation of water, followed by crystallisation.

➢ Some soluble salts, like zincates and aluminates, are found only in solution. These salts can not be obtained in their pure forms by crystallisation.

➢ An insoluble salt can be obtained from another insoluble salt by double decomposition.

➢ If at least 1 g of a substance can be dissolved in 100 mL of water at 298 K, it is called a soluble salt.

➢ If 0.1 g to 1 g of a substance can be dissolved in 100 mL of water at 298 K, it is called a partially **soluble salt.**

➢ If less than 0.1 g of a substance can be dissolved in 100 mL of water at 298 K, it is called an **insoluble** salt.

➢ All metallic oxides and hydroxides are insoluble except for sodium, potassium and ammonium. Calcium hydroxide is slightly soluble. $NaHCO_3$ is sparingly soluble, but $KHCO_3$ is fairly soluble in water.

➢ Efflorescent substances lose their weight, while hygroscopic and deliquescent substances gain weight when exposed to atmosphere.

➢ Table salt (NaCl) turns moist and ultimately forms a solution on exposure to air, especially during the rainy season. Though pure sodium chloride is not deliquescent, the commercial version of the salt contains impurities like $MgCl_2$ and $CaCl_2$, which are deliquescent substances.

Example 1

Define the terms: **[OEB]**

(i) Acid
(ii) pH scale
(iii) Neutralisation
(iv) Di-acidic base

Ans.(i) Acid: Acid is the chemical compound which, on dissolving in water, produces hydronium ions as the only positively charged particles.

(ii) pH Scale: The scale used for determining the acidic or alkaline character of the solution.

(iii) Neutralisation: When H^+ ions derived from an acid combine with hydroxyl ions of a base to form unionised water molecule is called as neutralisation.

(iv) Di-acidic base is a base that dissociates in molten (fused) state or in aqueous solution to produce two OH^- ions per molecule of that base.

Example 2

Barium chloride solution can be used to distinguish between a sodium sulphate solution and a sodium nitrate solution. Justify? [AI]

Ans. Barium chloride solution, when it reacts with sodium sulphate solution, forms a white precipitate which is insoluble in all mineral acids, whereas when barium chloride solution reacts with sodium nitrate, no visible reaction takes place.

$$\underset{\text{(Soln)}}{BaCl_2} + Na_2SO_4 \longrightarrow BaSO_4 + 2NaCl$$

$$\underset{\text{(Soln)}}{BaCl_2} + 2NaNO_3 \longrightarrow Ba(NO_3)_2 + 2NaCl$$

Course of Action
Maximum Time = 3:00 Hrs
Maximum Questions = 54

4 ANALYTICAL CHEMISTRY

Learning Objectives

The students will be able to:

- *Identify the action of NH_4OH and NaOH on solutions of Ca, Mg, Fe, Cu, Zn and Pb salts drop by drop in excess.*
- *Observe the colour of salt and its solution.*
- *Understand the special action of ammonium hydroxide on solutions of copper salts and sodium hydroxide on ammonium salts.*
- *Observe action of NaOH and NH_4OH on solutions of salts of metals.*
- *Understand action of alkalis (NaOH, KOH) on certain metals (including Al, Zn, Pb), their oxides and hydroxides and show their amphoteric nature.*

Revision Notes

➢ Qualitative analysis is carried out with the help of **reagents**. A reagent is a substance that reacts with another substance to initiate or test a chemical reaction or to test the presence or absence of a specific substance in the chemical reaction system. Alkalis such as NaOH, KOH and NH_4OH are important laboratory reagents. These alkalis give characteristic tests with various metal ions, from which these metal ions can be identified.

➢ **Colour of the Salts and Their Solutions:**

The salts of normal elements, i.e., the elements of groups 1, 2 and 13 to 17, are generally colourless. Salts of transition elements, i.e., the elements of groups 3 to 12, are generally coloured.

Colour of Ions

Cation	**Symbol**	**Colour**
Ammonium ion	NH_4^+	Colourless
Sodium ion	Na^+	
Potassium ion	K^+	
Calcium ion	Ca^{2+}	
Magnesium ion	Mg^{2+}	
Aluminium ion	Al^{3+}	
Lead ion	Pb^{2+}	
Zinc ion	Zn^{2+}	
Cupric ion	Cu^{2+}	Blue
Ferrous ion	Fe^{2+}	Light green
Ferric ion	Fe^{3+}	Yellow/brown
Nickel ion	Ni^{2+}	Green
Chromium ion	Cr^{3+}	Green
Manganese ion	Mn^{2+}	Pink

Colourless ions	**Symbol**	**Coloured ions**	**Symbol**	**Colour**
Chloride ion	Cl^-	Permanganate ion	MnO_4^-	Pink or Purple
Sulphate ion	SO_4^{2-}			
Carbonate ion	CO_3^{2-}	Dichromate ion	$Cr_2O_7^{2-}$	Orange

Nitrate ion	NO_3^-			
Hydride ion	H^-	Chromate ion	CrO_4^{2-}	Yellow
Bicarbonate ion	HCO_3^-			
Sulphide ion	S^{2-}			
Bromide ion	Br^-			
Acetate ion	CH_3COO^-			

➢ **Action of Sodium hydroxide (NaOH) on solution of salts of metals:**

Ion	Salt (Colour)	Reaction	Precipitate formed	Colour of the precipitate	Solubility of the precipitate in an excess of NaOH
Fe^{2+}	$FeSO_4$ (Ferrous sulphate) (Green)	$FeSO_4 + 2NaOH \rightarrow Fe(OH)_2\downarrow + Na_2SO_4$	$Fe(OH)_2$ (Ferrous hydroxide or Iron (II) hydroxide)	Dirty green	Insoluble
Fe^{3+}	$FeCl_3$ (Ferric chloride) (Brown)	$FeCl_3 + 3NaOH \rightarrow Fe(OH)_3\downarrow + 3NaCl$	$Fe(OH)_3$ (Ferric hydroxide or Iron (III) hydroxide)	Reddish brown	Insoluble [Board, 2022]
Cu^{2+}	$CuSO_4$ (Copper sulphate) (Blue)	$CuSO_4 + 2NaOH \rightarrow Cu(OH)_2\downarrow + Na_2SO_4$	$Cu(OH)_2$ (Cupric hydroxide)	Pale blue [Board, 2023/2022]	Insoluble [Board, 2023/2022]
Zn^{2+}	$ZnSO_4$ (Zinc sulphate) (Colourless)	$ZnSO_4 + 2NaOH \rightarrow Zn(OH)_2\downarrow + Na_2SO_4$ $Zn(OH)_2 + 2NaOH \rightarrow$ (Excess) $Na_2ZnO_2 + 2H_2O$ Sodium zincate (Soluble)	$Zn(OH)_2$ (Zinc (II) hydroxide)	White gelatinous	Soluble
Pb^{2+}	$Pb(NO_3)_2$ (Lead nitrate) (White)	$Pb(NO_3)_2 + 2NaOH \rightarrow Pb(OH)_2\downarrow + 2NaNO_3$ $Pb(OH)_2 + 2NaOH \rightarrow$ (Excess) $Na_2PbO_2\downarrow + 2H_2O$ Sodium plumbite (Soluble)	$Pb(OH)_2$ (Lead (II) hydroxide)	Chalky White [Board 2023]	Soluble [Board 2023]
Ca^{2+}	$Ca(NO_3)_2$ (Calcium nitrate)	$Ca(NO_3)_2 + 2NaOH \rightarrow Ca(OH)_2\downarrow + 2NaNO_3$	$Ca(OH)_2$ (Calcium hydroxide)	White	Insoluble
Mg^{2+}	$MgCl_2$ (Magnesium Chloride) (White)	$MgCl_2 + 2NaOH \rightarrow Mg(OH)_2$ (ppt.) $+ 2NaCl$	$Mg(OH)_2$ (Magnesium Hydroxide)	Dull White	Insoluble

➢ **Action of Ammonium hydroxide (NH_4OH)** on solution of salts of metals:

Ion	Salt (Colour)	Reaction	Precipitate formed	Colour of the precipitate	Solubility of the precipitate in an excess of NH_4OH
Ca^{2+}	$Ca(NO_3)_2$ (Calcium nitrate) (White)	$Ca(NO_3)_2 + 2NH_4OH \rightarrow Ca(OH)_2\downarrow + 2NH_4NO_3$	[conc. of OH^- ions is low and cannot ppt. $Ca(OH)_2$, so $Ca(OH)_2$ remains soluble]	No precipitate	No precipitate
Fe^{2+}	$FeSO_4$ (Iron (II) sulphate) (Green)	$FeSO_4 + 2NH_4OH \rightarrow Fe(OH)_2\downarrow + (NH_4)_2SO_4$	$Fe(OH)_2$ (Iron (II) hydroxide)	Dirty green	Insoluble

Fe^{3+}	$FeCl_3$ (Iron (III) chloride) (Brown)	$FeCl_3 + 3NH_4OH \rightarrow Fe(OH)_3\downarrow + 3NH_4Cl$	$Fe(OH)_3$ Iron (III) hydroxide)	Reddish brown	Insoluble
Cu^{2+}	$CuSO_4$ (Copper sulphate) (Blue)	$CuSO_4 + 2NH_4OH \rightarrow Cu(OH)_2\downarrow + (NH_4)_2SO_4$ $Cu(OH)_2 + (NH_4)_2SO_4 + \underset{\text{(Excess)}}{2NH_4OH} \rightarrow [Cu(NH_3)_4]SO_4 + 4H_2O$	$Cu(OH)_2$ (Copper (II) hydroxide) Tetraammine copper (II) sulphate	Pale blue Deep inky blue solution **[Board 2023]**	Soluble **[Board 2023]**
Zn^{2+}	$ZnSO_4$ (Zinc sulphate) (Colourless)	$ZnSO_4 + 2NH_4OH \rightarrow Zn(OH)_2\downarrow + (NH_4)_2SO_4$ $Zn(OH)_2 + (NH_4)_2SO_4 + \underset{\text{(Excess)}}{2NH_4OH} \rightarrow [Zn(NH_3)_4]SO_4 + 4H_2O$	$Zn(OH)_2$ (Zinc(II)hydroxide) Tetraammine zinc(II) sulphate	White gelatinous Colourless solution	Soluble Soluble
Pb^{2+}	$Pb(NO_3)_2$ (Lead (II)nitrate) (White)	$Pb(NO_3)_2 + 2NH_4OH \rightarrow Pb(OH)_2\downarrow + 2NH_4NO_3$	$Pb(OH)_2$ (Lead(II) hydroxide)	Chalky white	Insoluble

➢ **Action of sodium hydroxide (NaOH) on zinc, aluminium and lead (action of alkalis on metals):**

$$Zn + \underset{\text{Conc.}}{2NaOH} \longrightarrow \underset{\substack{\text{Sodium zincate}\\\text{(colourless)}}}{Na_2ZnO_2} + H_2$$

$$2Al + \underset{\text{Conc.}}{2NaOH} + 2H_2O \longrightarrow \underset{\substack{\text{Sodium}\\\text{meta-aluminate}\\\text{(colourless)}}}{2NaAlO_2} + 3H_2$$

$$Pb + \underset{\text{Conc.}}{2NaOH} \longrightarrow \underset{\substack{\text{Sodium}\\\text{plumbite}\\\text{(colourless)}}}{Na_2PbO_2} + H_2$$

Similarly, we can write reactions with potassium hydroxide (KOH).

➢ **Action of alkalis on amphoteric metal oxides / metal hydroxides (action with sodium hydroxide):**

$$\underset{\text{(White)}}{ZnO} + \underset{\text{Conc.}}{2NaOH} \longrightarrow \underset{\substack{\text{Sodium}\\\text{zincate}\\\text{(colourless, soluble)}}}{Na_2ZnO_2} + H_2O$$

$$\underset{\text{(White)}}{Zn(OH)_2} + \underset{\text{Conc.}}{2NaOH} \longrightarrow \underset{\substack{\text{Sodium}\\\text{zincate}\\\text{(colourless, soluble)}}}{Na_2ZnO_2} + 2H_2O$$

$$\underset{\text{(Yellow)}}{PbO} + \underset{\text{Conc.}}{2NaOH} \longrightarrow \underset{\substack{\text{Sodium}\\\text{plumbite}\\\text{(colourless, soluble)}}}{Na_2PbO_2} + H_2O$$

$$\underset{\text{(Yellow)}}{Pb(OH)_2} + \underset{\text{Conc.}}{2NaOH} \longrightarrow \underset{\substack{\text{Sodium}\\\text{plumbite}\\\text{(colourless, soluble)}}}{Na_2PbO_2} + 2H_2O$$

$$\underset{\text{(White)}}{Al_2O_3} + \underset{\text{Conc.}}{2NaOH} \longrightarrow \underset{\substack{\text{Sodium}\\\text{meta-aluminate}\\\text{(colourless, soluble)}}}{2NaAlO_2} + H_2O$$

$$\underset{\text{(White)}}{Al(OH)_3} + \underset{\text{Conc.}}{NaOH} \longrightarrow \underset{\substack{\text{Sodium}\\\text{meta-aluminate}\\\text{(colourless, soluble)}}}{NaAlO_2} + 2H_2O$$

Similarly, we can write reactions with KOH.

Key Terms

- **Analysis:** Involves the determination of chemical components present in a given sample in case of chemistry.
- **Analytical chemistry:** A branch of chemistry which deals with the experimental study of sample by qualitative as well as quantitative means.
- **Qualitative analysis:** Deals with the **identification of unknown substances** in a given sample by chemical tests.
- **Quantitative analysis:** Deals with the **determination of composition** of a mixture.
- **Precipitation:** It is the process of formation of an insoluble solid substance (precipitate) which is formed in a chemical reaction and settles down in the reaction solution.
- **Reagent:** It is a substance which reacts with another substance.

Key Facts

- When the sodium hydroxide solution is added drop by drop to the solution of some heavy metallic salts, the metal hydroxide formed gets **precipitated**.
- **Colour of the precipitate** identifies the specific metal ion.
- When ammonium hydroxide solution is added drop-wise to the solutions of some heavy metallic salts, precipitates of their hydroxides are formed, which are identified by their distinct colours.
- Some precipitated metallic hydroxides are soluble in excess of NH_4OH due to the formation of soluble complex compounds on further reaction with excess of NH_4OH.
- Certain metals like Zn, Al and Pb react with hot concentrated caustic alkalis (NaOH or KOH) to form the corresponding soluble salt and **liberate hydrogen**.
- **Amphoteric oxides** and **hydroxides** are those metal compounds which react with both acids and alkalis to form salt and water.

Example 1

1. (i) Name a yellow monoxide that dissolves in hot and conc. alkali. Write the chemical equation involved.

(ii) Name a white metal oxide that dissolves when fused with caustic soda or caustic potash. **[OEB]**

Ans. (i) Lead monoxide (PbO) dissolves in hot and concentrated alkali. (l)

$$PbO(s) + 2NaOH(conc.) \rightarrow \underset{\text{Sodium plumbite}}{Na_2PbO_2(aq)} + H_2O(l)$$

(ii) Aluminium oxide, Al_2O_3 dissolves when fused with caustic soda or caustic potash.

$$Al_2O_3(s) + 2NaOH(aq) \rightarrow \underset{\text{Sodium aluminate}}{2NaAlO_2(aq)} + H_2O(l)$$

2. Distinguish between the following pairs of compounds using the test given within the brackets.

(i) Calcium sulphite and calcium carbonate (using dil. HCl).

(ii) Calcium nitrate and potassium nitrate (using a flame test).

(iii) Lead nitrate solution and zinc nitrate solution (using an alkali).

Ans. (i)

Substance	Test	Gas evolved	Odour
Calcium carbonate	dil. HCl	CO_2	Odourless
Calcium sulphite	dil. HCl	SO_2	Choking odour

(ii)

Substance	Test	Flame Colour
Calcium nitrate	Flame test	Flickering brick red
Potassium nitrate	Flame Test	Pink / Purple lilac

(iii)

Substance	NH_4OH in excess
Lead nitrate solution	Chalky white precipitate formed is insoluble.
Zinc nitrate solution	Gelatinous white precipitate formed is soluble.

Course of Action
Maximum Time = 6 Hrs
Maximum Questions = 116

5 MOLE CONCEPT AND STOICHIOMETRY

Learning Objectives

The students will be able to learn:

- *Gay Lussac's Law of combining volumes; Avogadro's Law.*
- *Atomicity of hydrogen, oxygen, nitrogen and chlorine.*
- *Vapour density and its relation to relative molecular mass.*
- *Mole and its relation to mass.*
- *Calculations based on chemical equations.*

TOPIC-1: GAY LUSSAC'S LAW

Concepts covered: • Gay-Lussac's law of combining volumes, Avogadro's law, volume statement

Revision Notes

➢ **Gay-Lussac's law of combining volumes :**

- It states that whenever gases react, they always do so in volumes which bear a simple whole number ratio to one another and to the volumes of gaseous products. All volumes are being measured under similar conditions of temperature and pressure. In simple words, it states that "for a given mass and constant volume of an ideal gas, the pressure exerted on the sides of its container is directly proportional to its absolute temperature". **[Board 2023]**

Gay Lussac's Law
Scan Me!

- Gay-Lussac's Law is valid only for gases, and the volumes of solids and liquids are considered to be zero. It expresses the relationship between pressure and temperature.

➢ **Example of Gay-Lussac's Law :**

Hydrogen chloride : One volume of hydrogen mixed with one volume of chlorine gives two volumes of hydrogen chloride gas.

$$\underset{1\ \text{Vol.}}{H_2} + \underset{1\ \text{Vol.}}{Cl_2} \rightarrow \underset{2\ \text{Vol.}}{2HCl}$$

(At same temperature and pressure)

Simple ratio is 1: 1: 2

➢ **Avogadro's law explains Gay-Lussac's law**

Avogadro's law states that, under the same conditions of temperature and pressure, equal volumes of different gases have the same number of molecules. As substances react in simple ratio by the number of molecules, volumes of gaseous reactants and products will also bear a simple ratio. This is the statement of Gay-Lussac's law.

$$\underset{1\ \text{vol}}{H_2} + \underset{1\ \text{vol}}{Cl_2} \rightarrow \underset{2\ \text{vol}}{2HCl}$$

[by Gay-Lussac's law] [by Avogadro's law]

MNEMONICS

1. Concept : Gas Laws

Mnemonics :

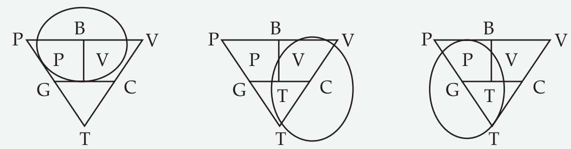
Can These **G**uys **P**ossibly **B**e **V**ictorious?

C—Charles's
T—Temperature

G—Gay-Lussac's Law
P—Pressure

B—Boyles's
V—Volume

Interpretation :



Boyle's Law $= P_1V_1 = P_2V_2$

Charles's Law $= \frac{V_1}{T_1} = \frac{V_2}{T_2}$

Gay-Lussacs's Law $= \frac{P_1}{T_1} = \frac{P_2}{T_2}$

Key Terms

- **Absolute Scale or Kelvin Scale :** A temperature scale with absolute zero (zero Kelvin) as the starting point is called the absolute scale or Kelvin scale.
- **Limiting reagent or reactant :** The reactant which is completely used up in a reaction is called limiting reagent or limiting reactant.
- **Stoichiometry :** It measures quantitative relationships which are used to determine the amount of products or reactants that are produced or required in a given reaction.
- **Atom :** It is the smallest particle of an element that may or may not take part in a chemical reaction. It may or may not exist independently.
- **Molecule :** It is the smallest particle of a compound that can exist by itself. It never breaks up except for taking part in a chemical reaction.

Key Facts

- Since the volume of a gas changes remarkably with change of temperature and pressure, it becomes necessary to choose standard values of temperature and pressure to which gas volumes can be referred.
- **Standard values :**
  - **0°C or 273 K for temperature**
  - 1 atmospheric pressure or **760 mm or 76 cm of Hg** for pressure.
- Absolute zero = 0K = – 273°C
- Gay-Lussac's law is valid only for gases. The volume of solids and liquids is considered to be zero.
- The relative atomic mass of any element is the weighted average of the relative atomic masses of its natural isotopes. For example, chlorine consists of a mixture of two isotopes of masses 35 and 37 in the ratio of 3 : 1, and hence atomic mass of chlorine is 35.5.
- Atomicity $= \dfrac{\text{Molecular Mass}}{\text{Atomic Mass}}$

Example 1

i. State Gay-Lussac's law.

Ans. According to this law for a given mass and constant volume of an ideal gas, the pressure exerted on the sides of its container is directly proportional to its absolute temperature.

ii. What is atomicity?

Ans. The number of atoms present in a molecule of an element is called atomicity.

iii Calculate atomicity of oxygen.

Ans. Atomicity = Molecular mass/atomic mass
For oxygen: Molecular mass = 32
Atomic mass = 16
So atomicity will be = 32/16 = 2

TOPIC-2: RELATIVE MOLECULAR MASS AND MOLE CONCEPT

Concepts covered: • Relative molecular mass • Vapour density • Mole concept

Revision Notes

- The **relative molecular mass** of an element or a compound is the mass of one molecule of an element or compound compared with the mass of an atom of carbon - 12, which is arbitrarily assigned a value of 12.00.
- Relative molecular mass is a number dealing with a molecule.
- Gram molecular mass is the mass of one mole of a substance.

Relative molecular mass

Scan Me!

For example,

(i) Gram molecular mass of oxygen gas = 32 g

(ii) Gram molecular mass of nitric acid = 63 g.

- The relative molecular mass of a gas or vapour is twice its vapour density (VD).
- Relative molecular mass of a gas = 2 × VD.
- The molar mass of any gas is its relative molecular mass expressed in g (grams).
- The molar volume of any gas at STP is equal to 22.4 dm^3.
- The volume of a gas, which contains 6.023×10^{23} molecules, is always 22.4 dm^3 at STP.
- The number of molecules contained in one mole of any substance is equal to 6.023×10^{23}.
- Mole is the mass of a substance containing 6.023×10^{23} particles. The particles may be atoms, molecules, ions, electrons, etc. **[Board 2024]**

Thus, the mole of substance is related to

(A) the mass of a substance,

(B) particles contained in a substance.

Number of moles of a substance =

$$\frac{\text{Mass of the substance}}{\text{Gram molecular mass of the substance}}$$

$$= \frac{\text{Number of particles in the substance}}{6.023 \times 10^{23}}$$

- **Mole Concept :** Mole is simply like a dozen or a gross. A dozen is a collection of 12 objects, a gross is a collection of 144 objects. Similarly, a mole is a collection of 6.023×10^{23} (Avogadro's number) elementary particles (atoms or ions or molecules).
- **Mole :** A mole is defined as the amount of a substance containing elementary particles like atoms, molecules or ions in 12 g of carbon (C-12).

Mole Concept

Scan Me!

- **Mole of Atoms (Relating Mole and Atomic Mass) :** One mole of atoms contains 6.023×10^{23} atoms having mass equal to its gram atomic mass. One mole of oxygen atoms contains 6.023×10^{23} atoms of oxygen and weighs 16 g.
- **Mole of Molecules (Relating Mole and Molecular Mass):** One mole contains 6.023×10^{23} molecules and is equivalent to the given molecular mass of any given substance. For example, 1 mole of O_2 contains 6.023×10^{23} molecules and weighs 32 g.

- **Mole and Molar volume:** One mole of any gaseous molecule occupies 22.4 (dm^3) litres or 22400 cm^3 (mL) at STP. This volume is known as **molar volume**. The molar volume of a gas can be defined as the volume occupied by one mole of a gas at STP.

[Board 2024]

Gram molecular volume

$$\textbf{(Molar volume)} = \frac{\text{Gram molecular weight}}{\text{Weight per litre of gas at STP}}$$

For example :

1 mole of $H_2SO_4 = (2 \times 1) + 32 + (4 \times 16) = 98$ g

$= 6.022 \times 10^{23}$ molecules of H_2SO_4

1 mole of $NH_3 = 14+(3\times1) = 17$ g $= 6.022 \times 10^{23}$ molecules of NH_3

1 mole of Na $= 23$ g $= 6.023 \times 10^{23}$ atoms of Na.

Volume occupied by 1 mole of NH_3 = 22.4 litres at STP

Volume occupied by 1 mole of O_2 = 22.4 litres at STP

- **Relative Vapour Density :** The relative vapour density of a gas is the ratio between the masses of equal volumes of gas (or vapour) and hydrogen under the same conditions of temperature and pressure.

$$\text{Relative VD} = \frac{\text{Mass of volume V of the gas under similar conditions}}{\text{Mass of volume V of hydrogen gas under similar conditions}}$$

- According to Avogadro's law, volume at the same temperature and pressure have equal number of molecules.
- 2 × Relative vapour density (VD) = Relative molecular mass of a gas. The relative molecular mass of gas or vapour is twice its vapour density.

MNEMONICS

Concept : Relation in molecular mass and vapour density

Mnemonics :

My Mom is equal to 2 Vestern Dancer

Interpretation :

M—molecular
M—mass
2—2
V—vapour
D—density
Molecular mass = 2 x Vapour density

Key Terms

- **Relative Atomic Mass :** The relative atomic mass or atomic weight of an element is the number of times one atom of the element is heavier than 1/12 time of the mass of an atom of C-12.
- **Atomic Mass Unit :** It is defined as 1/12 the mass of carbon atom, C-12.
- **Relative Molecular Mass :** The relative molecular mass or molecular weight of an element or a compound is the number that represents how many times one molecule of the substance is heavier than 1/12 of the mass of an atom of carbon-12.
- **Gram Atomic Mass :** It is defined as the atomic mass of an element **expressed in grams.**
- **One Gram Atom :** The quantity of the element which weighs equal to its gram atomic mass is known as one gram atom of that element.
- **Gram Molecular Mass :** It is defined as the molecular mass of a substance expressed in grams.
- **Avogadro's Number :** It is defined as the number of atoms present in 12 g (gram atomic mass) of C-12 isotope, i.e., 6.023×10^{23} atoms.

Key Facts

- One mole of oxygen atoms contains 6.023×10^{23} atoms of oxygen and weighs 16 g.
- 1 G.M.M. of the substance = 1 mole of that substance = 6.02×10^{23} molecules of that substance = 22.4 L at S.T.P.
- The relative vapour density of a gas (or a vapour) is the ratio between the masses of equal volumes of gas (or vapour) and hydrogen under the same conditions of temperature and pressure.
- The relative molecular mass of a gas or vapour is twice its vapour density.
- Mass of one atom $= \dfrac{\text{Atomic mass}}{6.023 \times 10^{23}}$
- Mass of one molecule $= \dfrac{\text{Molecular mass}}{6.023 \times 10^{23}}$
- Number of molecules = moles × 6.023×10^{23}
- Number of atoms = moles × 6.023×10^{23}

Example 3

4. Concentrated nitric acid oxidises phosphorus to phosphoric acid according to the following chemical equation:

$$P + 5HNO_3 \rightarrow H_3PO_4 + H_2O + 5NO_2$$

(A) What mass of phosphoric acid can be prepared from 6.2 g of phosphorus?

(B) What mass of nitric acid will be consumed at the same time?

(C) What would be the volume of steam produced at the same time if measured at 760 mm Hg pressure and 273°C?

(Atomic mass: H = 1, N = 14, O = 16 and P = 31)

Ans. $\underset{31\,g}{P} + \underset{\substack{5(1+14+3\times16) \\ 315\,g}}{5HNO_3} \rightarrow \underset{98\,g}{H_3PO_4} + H_2O + 5N_2O$

(A) 31 g of phosphorus produces 98 g of phosphoric acid. 6.2 g of phosphorus produces $\frac{98}{31} \times 6.2 = 19.6$ g of phosphoric acid.

(B) 31 g of phosphorus consumes (5 x 63) = 315 g of nitric acid.

6.2 g of phosphorus consumes nitric acid

$$= \frac{315}{31} \times 6.2 = 63 \text{ g}$$

(C) Moles of steam formed from 31 g of phosphorus = $\frac{18}{18} = 1$ mole

Moles of steam formed from 6.2 g of phosphorus

$$= \frac{1 \text{ mol}}{31 \text{ g}} \times 6.2 \text{ g} = 0.2 \text{ mol.}$$

Volume of steam produced at STP = (0.2 mol) × (22.4 L/mol) = 4.48 L

Pressure remains constant but temperature (273 + 273 = 546 K) is doubled, the volume of steam produced also gets doubled.

Hence, volume of steam produced at 760 mm Hg pressure and 273°C = 4.48×2 = 8.96 L.

5. Ammonia may be oxidised to nitrogen monoxide in the presence of a catalyst according to the following chemical equation :

If 27 litres of ammonia is consumed, what volume of nitrogen monoxide will be produced at same temperature and pressure?

Ans. $\underset{4\text{vol}}{4NH_3} + \underset{5\text{vol}}{5O_2} \rightarrow \underset{4\text{vol}}{4NO} + 6H_2O$

4 litres of ammonia is produced from 4 litres of nitrogen monoxide gas.

So, volume of nitrogen monoxide gas produced from 27 litres of ammonia will be = 27 litres of nitrogen oxide gas.

TOPIC-3: PERCENTAGE COMPOSITION, EMPIRICAL AND MOLECULAR FORMULA

Concepts covered: • Molecular formula • Empirical formula • To calculate percentage of element in a compound • Relationship between molecular formula and empirical formula

Revision Notes

➢ **Molecular Formula :** It is the **symbolic representation** of a molecule of a substance which indicates the actual number of atoms of various elements present in one molecule of it.

For example, **molecular formula** of benzene = C_6H_6.

➢ To Calculate Percentage of an Element in a Compound:

Percentage of an element in a compound =

$$\frac{\text{Total mass of the element}}{\text{Molecular mass of the compound}} \times 100$$

➢ **To Calculate Percentage Composition of a Compound:**

● In order to calculate the **percentage composition** of a compound, we have to calculate the percentage of each element in the compound.

➢ **Empirical Formula :** Empirical formula of a compound gives us the simplest whole number ratio between the atoms of various elements present in one molecule of it.

[Board, 2022/2024]

● Empirical formula of benzene = C_1H_1

● Empirical formula of glucose = $C_1H_2O_1$

➢ Relationship between Empirical and Molecular Formula :

● Molecular formula of a compound is a simple whole number multiple of its empirical formula.

Molecular formula = n × empirical formula

Where n is a simple whole number

Molecular weight = n × empirical weight

or $n = \dfrac{\text{Molecular formula}}{\text{Empirical formula}}$

or $n = \dfrac{\text{Molecular Mass}}{\text{Empirical Formula Mass}}$

$= \dfrac{2 \times \text{Vapour density of a gas or vapour}}{\text{Empirical formula mass}}$

MNEMONICS

Concept : Relation between molecular formula and empirical formula

Mnemonics:

MF Tyres **N**ot **E**asy To **F**ail

M—Molecular
F—Formula
N—n
E—Empirical
F—Formula

Interpretation :
Molecular Formula = n × E F

Key Terms

➢ Molecular formula of blue vitriol is $CuSO_4 \cdot 5H_2O$.

➢ The empirical formula of glucose ($C_6H_{12}O_6$) is CH_2O. It indicates that the ratio of C, H and O atoms in a molecule of glucose is 1 : 2 : 1.

Examples

6. Define following:

(i) Molecular formula

(ii) Empirical formula

Ans. (i) Molecular formula is the symbolic representation of a molecule of a substance which indicates the actual number of atoms of various elements present in one molecule of it.

(ii) Empirical formula is the simplest whole number ratio between the atoms of various elements present in one molecule.

7. Calculate percentage of carbon in carbon dioxide.

Ans. Mass of carbon in carbon dioxide = 12

Molecular mass of carbon dioxide = 44 g

% of carbon = Atomic mass of carbon/ Molecular mass of carbon dioxide x 100

= 12/44 x 100 = 27.27 %

Course of Action
Maximum Time = 3:00 Hrs
Maximum Questions = 97

6 ELECTROLYSIS

Learning Objectives

The students will be able to:

- *Understand electrolytes and non-electrolytes and their examples.*
- *Comprehend electrolysis and related terms such as electrolyte, electrode, anode, cathode, anion, cation, oxidation and reduction.*
- *Analyse the migration of ions during electrolysis and explain factors influencing selective discharge of ions using examples and application of selective discharge theory.*
- *Understand practical applications of electrolysis, such as electroplating and electro-refining of metals, and identify reasons and conditions for their use.*
- *Develop problem-solving skills by applying electrolysis principles to solve numerical problems and interpret experimental data.*
- *Improve critical thinking and communication skills by evaluating the environmental and economic impact of electrolytic processes and effectively presenting findings from electrolysis experiments.*

TOPIC-1: ELECTROLYTES AND NON-ELECTROLYTES

Concepts covered: • Electrochemical changes • Conductors of electric current • Electrolytes and non-electrolytes

Revision Notes

Electrolytes and Non-Electrolytes
Scan Me!

➢ **Electrochemical Changes**: The chemical changes which are accompanied by the flow of **electricity** are called **electrochemical changes**.

➢ **Electrochemical changes are of two types:**

(i) The electrochemical changes in which electricity from outside source is used to carry out a reaction called **electrolysis**.

(ii) Those electrochemical changes in which electricity is produced as a result of indirect redox reactions.

➢ **Conductors are of two types:**

(i) Metallic Conductors: These are the substances that conduct electric current in the solid state, and at the same time, these are not chemically decomposed. Examples: copper, silver, gold, graphite, etc.

(ii) Electrolytes: These are the substances which conduct electric current in the solution or molten state, and at the same time, these are chemically decomposed. For example, sodium chloride (NaCl), hydrogen chloride (HCl), sodium hydroxide (NaOH), etc.

➢ **Non-electrolytes:** The substances that do not conduct **electric current** in the molten or dissolved state are called non-electrolytes. For example, urea, sucrose, glucose, fructose, etc.

➢ **Differences between metallic conductors and electrolytes:**

Metallic conductors	Electrolytes
(i) Flow of electricity is due to the movement of electrons.	Flow of electricity is due to the movement of ions.
(ii) The electrical conductivity decreases with the increase in temperature.	The electrical conductivity increases with the increase in temperature.

(iii) Flow of electricity takes place in the solid state.	Flow of electricity takes place in the molten or dissolved state.
(iv) There is no flow of matter.	There is a flow of matter.
(v) There is no chemical decomposition. It involves a physical change.	There is chemical decomposition of the substance. It involves a chemical change.

Theory of Electrolytic Dissociation:

Svante Arrhenius, in 1887, gave the theory of electrolytic dissociation. According to him:

(i) An electrolyte on dissolving in water dissociates into free mobile ions, i.e., cations and anions, and allows the flow of an electric current through it.

(ii) All ions carry an electric charge and are responsible for the flow of current through the solution.

(iii) The conductivity of the electrolyte depends upon the concentration of ions in the solution.

(iv) The number of positive charges equals the number of negative charges in the solution, and thus the solution is in **electrolyte equilibrium**.

The equilibrium is also established between the unionised molecules and the ions produced.

(v) In non-electrolytes, such as sugar solution, there is no ionisation, and therefore, only molecules are present in solution.

(vi) The degree of dissociation is to the extent that an electrolyte dissociates or breaks up into ions.

➢ **Electrolysis** is the process by which ionic substances are decomposed into simpler substances when an electric current is passed through them.

➢ **Characteristics of Electrolysis:**

- The passage of electricity through an electrolyte causes the metallic ion (cation) to migrate towards the cathode and the non-metallic ion (anions) to migrate towards the anode. The number of electrons gained by the anode is equal to the number of electrons donated by the cathode.
- Only hydrogen gas and metals are liberated at the cathode and hence are called **electropositive elements**. Only non-metals are liberated at the anode and are called **electronegative elements**. The process of electrolysis is a **redox reaction.**
- The reaction at the cathode involves the **reduction** of cations as they gain electrons to become neutral atoms, while that at anode involves **oxidation** of anions as they lose electrons to become neutral.

For example: $NaCl \rightarrow Na^+ + Cl^-$

At Cathode: $Na^+ + e^- \rightarrow Na$ (Reduction)

At Anode: $Cl^- \rightarrow Cl + e^-$ (Oxidation)

$Cl + Cl \rightarrow Cl_2$

Overall reaction: $2NaCl \rightarrow 2Na + Cl_2$

- The alternating current (AC) does not cause any chemical change and therefore does not help electrolysis to occur. Electrolysis is caused by direct current (DC).

➢ **Electrochemical Series:**

The metals are arranged in a series known as the **electrochemical series**. The arrangement is that the elements that ionise most readily (discharged with great difficulty) are placed at the top of the series and the other elements in the descending order.

Metal	Cation		At Cathode	At Anode		Anion
K	K^+	Increasing ease of reduction	Discharged with the utmost difficulty.	Discharged with the utmost difficulty.	Increasing ease of oxidation	SO_4^{2-}
Ca	Ca^{2+}		Positive ions (cations) discharged at the cathode by gain of electrons.	Negative ions (anions) discharged at the anode by loss of electrons. The tendency of the anions to get oxidised at the anode increases on descending the electrochemical series.		
Na	Na^+					NO_3^-
Mg	Mg^{2+}		The tendency of the cations to get reduced at the cathode increases on descending the electrochemical series.			
Al	Al^{3+}					Cl^-
Zn	Zn^{2+}					
Fe	Fe^{2+}			Discharged most easily.		Br^-
Pb	Pb^{2-}		Discharged most easily.			
H	H^+					I^-
Cu	Cu^{2+}					
Hg	Hg^{2+}					OH^-
Ag	Ag^+					

➢ **Preferential or Selective Discharge of Ions at Cathode:**

The preferential discharge of ions present in an electrolyte at the respective electrodes is also known as selective discharge of ions. If a solution contains more than two cations and anions, the selective discharge of ions depends upon following three functions.

(i) The relative position of the cations or anions in electrochemical series lowers the position, making the discharge at the electrode easier.

(ii) Concentration of the ions in the electrolyte.

(iii) The nature of the electrode.

➢ **Concentration of Ions in the Electrolyte:**

Higher the concentration of the ion, greater the probability of it being discharged at the respective electrode. For example, electrolysis of concentrated NaCl solution.

$$NaCl \rightleftharpoons Na^+ + Cl^-$$

$$H_2O \rightleftharpoons H^+ + OH^-$$

Electrolysis

Scan Me!

Cathode: Na^+ ions discharged for dilute solution of NaCl.

Anode: OH^- ions discharged.

The concentration of Na^+ and Cl^- ions is much higher than that of H^+ and OH^- ions.

Electrolysis of dilute NaCl solution.

$$NaCl \rightleftharpoons Na^+ + Cl^-$$

$$H_2O \rightleftharpoons H^+ + OH^-$$

Exception: H_2 is deposited at cathode in brine solution, as its discharge potential is very low as compared to Na^+, even in low concentrations.

Cathode - Na^+ ions discharged for dilute solution of NaCl.

Anode - OH^- ions discharged.

H^+ and OH^- ions are lower in the electrochemical series than Na^+ and Cl^-.

➢ **Nature of the Electrode:**

Electrodes used in the process of electrolysis are either inert or active. If an active electrode (i.e., Cu, Ni, Ag) is used, it will take part in a chemical reaction. Then, the anions (e.g., SO_4^{2-} and OH^- ions) migrate to the anode but are not discharged. The active electrode (Cu, Ag) itself loses electrons and forms ions, e.g., Cu^{2+}, Ag^+ at the anode. If inert, electrodes like iron, graphite and platinum are used, they will not take part in electrolyte reactions.

I. Electrolysis of Molten Lead Bromide

Electrolyte: Molten Lead Bromide ($PbBr_2$)

Temperature: Above 380°C

Electrolytic cell: Crucible made of silica.

Electrodes: The cathode and anode are both made of graphite plates which are inert.

Ions present: Pb^{2+} and Br^-, i.e., lead ions and bromide ions.

Electrode Reactions:

At the cathode: $Pb^{2+} + 2e^- \rightarrow Pb$

At the anode: $Br^- - e^- \rightarrow Br$

$Br + Br \rightarrow Br_2$

Overall reactions: $PbBr_2 \rightleftharpoons Pb^{2+} + 2Br^-$ **[Board 2024]**

Observation: (a) Dark reddish-brown fumes of bromine evolved at the anode.

(b) Greyish white metal lead is formed on the cathode.

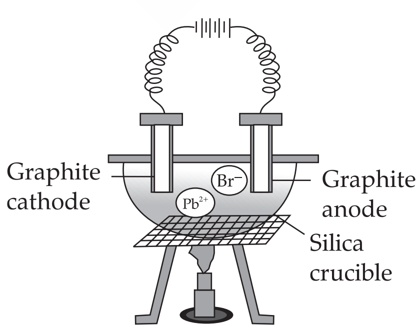


Fig. 6.1 Electrolysis of molten lead bromide

II. Electrolysis of Acidified Water Using Platinum Electrodes

Electrolysis of Acidified Water Using Platinum Electrodes

Scan Me!

Electrolyte: Acidified water

(Water + Dilute H_2SO_4)

Electrodes: Platinum foils, connected by corner wires to the two terminals of a battery.

Ions present: H^+, OH^-, SO_4^{2-}

Electrode reactions:

At the cathode: $H^+ + e^- \rightarrow H$

$H + H \rightarrow H_2$

At the anode:

$OH^- \rightarrow OH + e^-$

$OH + OH \rightarrow H_2O + [O]$

$[O] + [O] \rightarrow O_2$ **[Board 2024]**

Electrolysis of Copper Sulphate Solution Using Platinum Anode and Copper or Platinum Cathode

Scan Me!

Overall reaction:

$$2H_2O \xrightarrow[\text{Electric current}]{H_2SO_4} 2H_2(g) + O_2(g)$$

The ratio of hydrogen and oxygen liberated at the cathode and anode is in the ratio 2: 1 by volume.

III. Electrolysis of Copper Sulphate Solution Using Platinum Anode and Copper or Platinum Cathode

➢ **Electrolyte:** Saturated solution of copper (II) sulphate proposed in distilled water with a small amount of conc. H_2SO_4.

➢ **Dissociation of aqueous copper sulphate:**

$$CuSO_4 \rightleftharpoons Cu^{2+} + SO_4^{2-}$$

$$H_2O \rightleftharpoons H^- + OH^-$$

Particles present: Cu^{2+}, H^+, SO_4^{2-}, OH^- and H_2O molecules

Ions present: Cu^{2+}, H^+, SO_4^{2-}, and OH^-

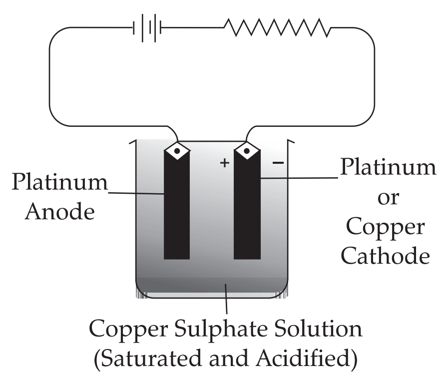
Reactions at cathode: $Cu^{2+} + 2e^- \rightarrow Cu$

Reactions at anode: $4OH^- - 4e^- \rightarrow 4OH$

$2OH - 2OH \rightarrow 4H_2O + O_2$

Product at anode: Oxygen

Product at cathode: Pink or reddish brown copper is deposited.



IV. Electrolysis of Aqueous Copper Sulphate Using Copper Electrodes

Electrolytic cell ⟶ Iron crucible/glass voltmeter

Electrolyte ⟶ Aqueous copper sulphate

Electrode ⟶ **Cathode:** Copper, **Anode:** Copper

- Dissociation $\underset{\text{Copper sulphate}}{CuSO_4} \rightleftharpoons Cu^{2+} + SO_4^{2-}$

[ions present Cu^{2+}, H^+, SO_4^{2-}, OH^- of copper]

Sulphate (aq.) $H_2O \rightleftharpoons H^+ + OH^-$

- **At cathode:** $\underset{\text{reduction reaction}}{Cu^{2+}} + 2e^- \rightarrow Cu$
- **At anode:** $\underset{\text{oxidation reaction}}{Cu} \rightarrow Cu^{2+} + 2e^-$
- **Product at cathode:** Brownish pink metal Cu is deposited at the cathode during electrolysis of $CuSO_4$ solution.
- **Product at anode:** Nil [copper ions formed]
- At cathode, Cu^{2+} ions and H^+ ions migrate to the cathode. Cu^{2+} ions are discharged in preference to H^+. (Cu^{2+} is lower in the series).
- If copper anode is used, the blue colour of aq.$CuSO_4$ remains unchanged during electrolysis. **[Board 2024]**

Example

1. (i) Give one example in each case of a substance which contains: R

(A) Ions only

(B) Molecules only

(C) Both ions and molecules

(ii) What is meant by the term electrolyte?

Ans. (i) (A) Sodium chloride solution

(B) Carbon tetrachloride

(C) Acetic acid

(ii) Electrolyte is an ionic compound which, in fused, molten or aqueous solution, allows the passage of electric current through it.

2. Differentiate between electrical conductivity of copper sulphate solution and copper metal. U&R

Ans.

Copper metal	Electrolyte- $CuSO_4$ Solution
(i) The flow of electricity takes place by flow of electrons, which have negligible mass.	**(i)** The flow of electricity takes place by flow of ions, which are dense particles as compared to electrons.
(ii) There is no decomposition of the parent metal, and thus the chemical properties of the metal are intact.	**(ii)** Decomposition of the electrolytic solution takes place, and thus the chemical properties of the electrolyte are altered.
(iii) Metals are good conductors of electricity in the solid state and in the molten state.	**(iii)** Electrolytes are good conductors of electricity in aqueous solution or molten state but do not conduct electricity in the solid state.
(iv) During metallic conduction, there is no transfer of matter. The flow of electricity only produces heat energy and no new products are formed.	**(iv)** Electrolytic conduction involves transfer of ions. The electrolyte is decomposed and new products are formed.

3. Three different electrolytic cells A, B and C, are converted in separate circuits. Electrolytic cell A contains sodium chloride solution. When the circuit is completed, a bulb in the circuit glows brightly. Electrolytic cell B contains acetic acid solution and in this case the bulb in the circuit glows dimly. The electrolytic cell C contains sugar solution and the bulb does not glow. Give a reason for each of these observations. A

Ans. In case of electrolytic cell A, which contains sodium chloride, the bulb glows brightly. It means that it is a strong electrolyte. In case of electrolytic cell B, which contains acetic acid, the bulb glows dimly, it means it is a weak electrolyte. In case of electrolytic cell C, which contains sugar solution, the bulb does not glow. It means it is a non-electrolyte.

MNEMONICS

1. Concept: Electrochemical Series

Mnemonics:

KINGS **C**AN **N**OT **M**OVE **A Z**EBRA. **I**N **L**ONG **H**UT

Interpretation:

K ings- POTASSIUM

C an- CALCIUM

N ot- (Na)SODIUM

M ove- MAGNESIUM

A - ALUMINIUM

Z ebra- ZINC

I n– Fe - IRON

L ong- LEAD

H ut is- HYDROGEN

2. Concept: Electrode reactions

Mnemonics:

AN **O**IL **R**IG **CA**T:

Interpretation:

At the **A**node, **O**xidation **I**nvolves **L**oss of electrons.

Reduction **I**nvolves **G**aining electrons at the **C**athode.

Key Terms

- **Electrochemistry:** It is the branch of chemistry which deals with the study of the relationship between electrical and chemical energy.
- **Conductors:** The substances that conduct electric current when present in one form or the other are called conductors. Examples: copper, silver, gold, graphite, etc.
- **Insulators:** The substances that do not conduct electric current are called non-conductors or insulators.

Examples:
Rubber, wood, plastic, glass, sulphur, etc.

- **Ionisation:** The process by which a polar covalent compound (molecule) is converted into ions in aqueous solutions is called ionisation.

For example: $HCl \xrightleftharpoons{\text{water}} H^+ + Cl^-$

$$H-Cl + H_2O \rightleftharpoons H_3O^+ + Cl^-$$

- **Dissociation:** It is the process of separation of ions of the ionic solid in the aqueous solutions.

For example:

$$NaCl(s) \xrightarrow{\text{water}} Na^+(aq) + Cl^-(aq)$$

- **Degree of ionisation or dissociation:** It is the fraction of the total number of moles of the electrolytes which undergo ionisation or dissociation in solution at equilibrium.
- **Electrolytic cell**: It is the vessel in which electrolysis is carried out. It is also called voltmeter. The vessel is made up of glass or any other insulating material.
- **Electrodes:** It is a pair of rods or plates of metals (or graphite) dipped in the molten or dissolved electrolyte through which the electric current enters or leaves the electrolytes. **[Board 2024]**
- **Anode:** The electrode connected to the positive terminal of the battery is called anode. During electrolysis, anode acquires a positive charge, and hence ions in a solution which are negatively charged (i.e., anions) migrate to the anode. The anions donate excess electrons to the anode and are oxidised to neutral atoms. It is an oxidising electrode. Oxidation takes place at the anode.
- **Cathode:** The electrode connected to the negative terminal of the battery is called cathode. During electrolysis, cathode acquires a negative charge, and hence ions in a solution which are positively charged (i.e., cations) migrate to the cathode. The cations gain excess electrons from the cathode and are reduced to neutral atoms. It is a reducing electrode. Reduction takes place at the cathode.
- **Ions:** Atoms or groups of atoms which carry a positive or a negative charge are known as ions. The charge on an ion, positive or negative, is equal to the valency of the atom or the ion. The ion that carries a positive charge is called cation, and the one which carries a negative charge is called anion.
- **Anions:** Anions migrate to the anode during electrolysis. Anions donate or lose electrons to the anode (oxidation process) and get oxidised to neutral atoms.

For example: **At anode** $\underset{\text{Anion}}{Cl^-} \rightarrow \underset{\text{Atom}}{Cl} + e^-$

- **Cations:** Cations migrate to the cathode during electrolysis. Cations accept or gain electrons from the cathode (reduction process) and get reduced to neutral atoms.

For example: **At Cathode** $\underset{\text{cation}}{Na^+} + e^- \rightarrow \underset{\text{Atom}}{Na}$

Key Facts

- The term electrolysis is made up of two words:
'electro' — means flow of electrons of electricity
'lysis' — means separating
- Electrolysis establishes a relationship between electrical energy and chemical change.
- Metals and alloys are conductors and non-metals are non-conductors.
- Non-electrolytes do not have ions even in solution. They contain only molecules.
- All salts are strong electrolytes. They form acidic, basic or neutral solutions after complete dissociation.
- Electrolytic cell (voltmeter) converts electrical energy into chemical energy.
- Electrochemical cell converts chemical energy into electrical energy.
- Oxidation is defined as a process which involves:
  - loss of electrons (s)
  - addition of oxygen
  - removal of hydrogen
- Reduction is defined as a process which involves:
  - gain of electron (s)
  - removal of oxygen
  - addition of hydrogen
  - Electrolysis is a redox reaction
- For making **acidified water**, dil. H_2SO_4 is preferred because it is non-volatile while dil. HNO_3 or HCl are volatile acids.

TOPIC-2: APPLICATIONS OF ELECTROLYSIS

Concepts covered: • Applications of electrolysis • Electroplating with silver and nickel • Electro-refining of copper

Revision Notes

➢ **Electrolysis has several technical and commercial applications. These are as follows:**

1. Electroplating with metals
2. Electro-refining of metals
3. Extraction of metals (Electrometallurgy)

➢ **I. Electroplating of an Article with Silver:**

Electrolyte: Sodium argentocyanide, $Na[Ag(CN)_2]$

Cathode: Article to be plated.

Anode: Pure plate of silver metals

Dissociation of $Na[Ag(CN)_2]$

$$Na[Ag(CN)_2] \rightleftharpoons Na^+ + Ag^+ + 2CN^-$$

Dissociation of H_2O

$$H_2O \rightleftharpoons H^+ + OH^-$$

Reaction at the cathode: $Ag^+(aq) + e^- \rightarrow Ag(s)$

Reaction at the anode: $Ag(s) \rightarrow Ag^+(aq) + e^-$

➢ **II. Electrolyte: Nickel Sulphate, $NiSO_4$**

Cathode: Pure plate of nickel metal

Anode: Pure plate of nickel metal

Dissociation of $NiSO_4$

$$NiSO_4 \rightarrow Ni^{2+} + SO_4^{2-}$$

Dissociation of H_2O:

$$H_2O \rightleftharpoons H^+ + OH^-$$

Reaction at the cathode: $Ni^{2+}(aq) + 2e^- \rightarrow Ni(s)$

Reaction at the anode: $Ni(s) \rightarrow Ni^{2+}(aq) + 2e^-$

[Board 2024]

➢ **Reasons for Electroplating**

Reasons for Electroplating

Scan Me!

It prevents corrosion or rusting, e.g., iron plated with nickel or chromium for decoration purposes and makes the article attractive and gives it an expensive appearance, e.g., brass metal plated with silver or gold.

Conditions to be maintained during electroplating:

(i) The article to be deposited is always made the **cathode** in electrolysis since the metal is deposited at the cathode.

(ii) The metal to be plated on the article is always made of since metal continuously dissolves into the solutions; it has to be replaced periodically. **[Board 2024]**

(iii) The electrolyte solution must contain metal ions like Ag, Au and Ni since these ions travel towards cathode and get deposited on the metal placed at cathode.

(iv) Less amount of current is supplied for longer time.

(v) Only direct current is to be used because alternating current causes discharge or ionisation on alternate electrodes. No effective coating takes place on cathode.

[Board 2024]

➢ **Electro-refining**

● In electro-refining, the impure metal makes the anode in an electrolytic tank containing some suitable electrolyte of the metal. A thin sheet of pure metal makes the cathode. On passing electric current, metallic cations from the electrolyte move to the cathode, lose their charge and get deposited in the form of pure metal. At the same time, the anions travel to the anode and dissolve an equivalent amount of metal from it. The impurities present in the crude metal either pass into solutions or settle down as **anode mud**.

● **Reaction at cathode:** $Cu^{2+} + 2e^- \rightarrow Cu$ [deposited]

● Cu^{2+} ions are discharged at the cathode. Thus, pure copper is deposited on the thin sheet of pure copper placed at the cathode. [Cu-active electrode]. **[Board 2023]**

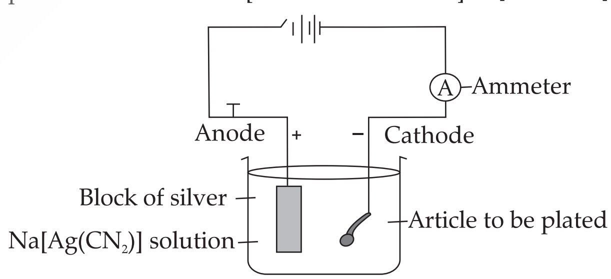


Fig. 6.3 Electroplating an article with silver

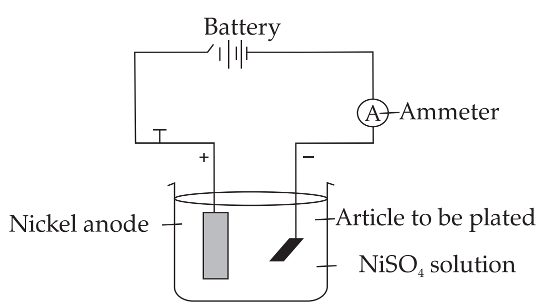


Fig. 6.4 Electroplating an article with nickel

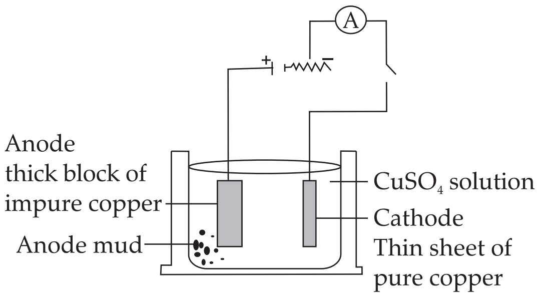


Fig. 6.5 Electrorefining of copper

- **Reaction at anode:** $Cu - 2e^- \rightarrow Cu^{2+}$ [cation]
- SO_4^{2-} and OH^- migrate to the anode, but neither are discharged. Instead, the copper anode itself loses electrons to give Cu^{2+} ions in solution. Hence, anode diminishes in mass.
- The process of extraction of metals from their fused or molten ores by the use of electrolysis or electric current is called **electro-metallurgy.**

MNEMONICS

Interpretation:

Reduction at cathode and anode for oxidation

Concept: Reaction at cathode

Mnemonics:

Red cat attack **an ox**:

Example 1

1. The following questions relate to the electroplating of an article with silver: R&A

(i) Name the electrode formed by the article to be plated.

(ii) What ions must be present in the electrolyte?

(iii) Name the electrolyte used.

Ans. (i) Cathode.

(ii) The ions of the metal, i.e., Ag to be electroplated on the spoon.

(iii) Aqueous sodium argentocyanide.

2. Electrolysis is used in the purification of metals. Name the following with reference to the electro-refining of copper. R

(i) Anode and cathode used.

(ii) Electrolyte used.

(iii) The product formed at anode and cathode.

Ans. (i) Anode – Impure block of copper

Cathode – Pure thin sheet of copper

(ii) Acidified copper sulphate solution.

(iii) Anode - Nil Cathode: Pure copper.

Key Facts

- **Electroplating:** A process in which a thin film of a metal like gold, silver, nickel chromium, etc., gets deposited on another metallic article with the help of electricity.
- **Electro-refining:** A process by which metals such as Al, Pb, Cu, Ni and Zn extracted from their ores by chemical method are freed from impurities electrolytically to obtain metals in their pure state.
- **Electro-metallurgy:** It is the process of extraction of metals which are higher in the electrochemical series, such as Na, K, Al, Mg, etc., from their fused **ores by** electrolysis.
- Brass objects are frequently plated with silver to give them shining appearance and beauty of a silver article.
- Iron tools are often electroplated with Ni, Cr or Zn to protect against rusting.
- The article to be electroplated is always placed at the cathode.
- During electrolytic reaction, the metal is always deposited at the cathode by gain of electrons.
- The thickness of the coating will depend on the duration of the current passed.
- Metals like Zn, Pb, Hg, Ag and Cu are refined by electrolysis.

OBJECTIVE TYPE QUESTIONS

[A] Multiple Choice Questions

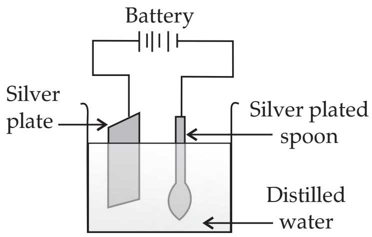
1. An aqueous solution of copper sulphate turns colourless on electrolysis. Which of the following could be the electrodes? U **[Board, Specimen Paper 2024]**

P. anode: copper; cathode: copper
Q. anode: platinum; cathode: copper
R. anode: copper; cathode: platinum

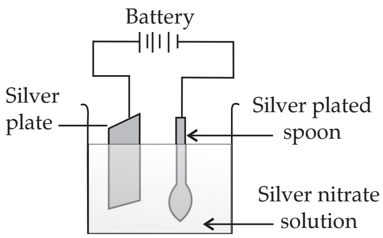
(A) Only P ✓ **(B)** Only Q
(C) Only R **(D)** Both Q and R

Exp In option Q, the platinum anode is inert and does not replace Cu^{2+} ions, while the copper cathode deposits Cu, causing the solution to lose its blue color.In P and R, the copper anode dissolves and replenishes Cu^{2+}, so the solution remains blue.

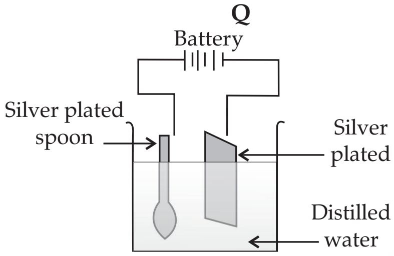
2. In which of the following electrolytic cells [P, Q, R or S] will silver plating be done on the spoon? **[Board, Specimen Paper 2024]**



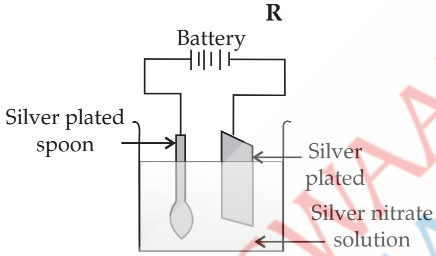
P



Q



R



S

(A) P
(B) Q ✓
(C) R
(D) S

Exp Due to transfer of electrons from anode to cathode, silver nitrate is used as an electrolyte. Spoon acts as cathode (-) and silver deposits at the spoon.

3. Cathode is a reducing electrode because **[Board, Specimen Paper 2021]**

(A) it has fewer electrons.
(B) it has deficiency of electrons.
(C) cations gain electrons from cathode. ✓
(D) anions lose electrons to cathode.

Exp Cathode involves reduction of cations as they gain electrons to become neutral atoms, while that at anode involves oxidation of anions as they lose electrons to become neutral.

4. The condition that is most appropriate for electroplating with nickel: **[Board, Specimen Paper 2021]**

(A) Electrolyte is molten copper sulphate.
(B) Anode should be made of impure nickel plate. ✓
(C) Alternating current is used.
(D) Periodic replacement of cathode is needed.

Exp Molecule of water combines with proton (H^+) to form hydronium ion.

$$H_2O + H^+ \rightarrow H_3O^+$$

5. Which of the following reactions takes place at the anode during the electroplating of an article with silver?

(A) $Ag^{-} 1e^{-} \rightarrow Ag^{1+}$ ✓
(B) $Ag^{+} 1e^{-} \rightarrow Ag^{1-}$
(C) $Ag - 1e^{-} \rightarrow Ag$
(D) None of the above **[Board, 2023]**

Exp Oxidation takes place at anode.

[B] Assertion Reason Questions

Assertion (A) and Reason (R). Answer these questions by selecting the appropriate option given below:

(A) Both A and R are true, and R is the correct explanation of A.
(B) Both A and R are true, and R is not the correct explanation of A.
(C) A is true, but R is false.
(D) A is false, but R is true.

1. Assertion (A): Storing $AgNO_3$ solution in a copper vessel is not recommended. **[OEB]**
Reason (R): Copper reacts with $AgNO_3$ to form insoluble copper (II) nitrate and silver precipitate.

Ans. Option (A) is correct.

Exp Storing silver nitrate ($AgNO_3$) solution in a copper vessel is discouraged due to the risk of contamination. Though copper does react with $AgNO_3$ to form insoluble copper (II) nitrate and silver precipitate, the primary concern is solution purity, as copper can corrode, compromising the solution's integrity.

2. Assertion (A): The metal to be plated is placed at cathode, which needs to be replaced from time to time.

Reason (R): The metal is continuously dissolved as ions in solution and hence needs to be replaced.

Ans. Option (D) is correct.

Exp The metal to be plated is placed at anode, and anode is replaced periodically.

[C] Fill in the blanks

i. In electroplating with silver, CN-1 and OH-1 ions migrate to the _________(anode/ cathode) but are not discharged.

ii. The electrolyte used in electroplating with nickel is ____________ (aq. nickel nitrate/ aq. nickel sulphate).

iii. The article to be electroplated is always kept at ____________(anode/cathode).

iv. The metal whose oxide, which is amphoteric, is reduced to metal by carbon reduction _________. (Fe / Mg / Pb / Al) **[Board 2017]**

Ans. i. anode
ii. aq. nickel sulphate.
iii. cathode
iv. Pb

[D] Give reason

i. In electrorefining of copper, the anode mud contains impurities like gold and silver and not iron and zinc.

This question is for practice and its solution is available at the end of the chapter.

ii. Electroplating should be done with low current.

Ans. i. Iron and zinc ionise and dissolve in the electrolyte copper sulphate solution, as they are high in electrochemical series. While gold and silver are insoluble impurities, so they settle down.

ii. A low current for a longer period of time gives smooth, firm, uniform and long lasting deposition.

[E] Give one word or phrase for the following

i. Electrode used as cathode in electro-refining of impure copper. **[Board, 2020]**

ii. Electrolyte used in electroplating with silver.

Ans i. Pure copper metal

ii. Aqueous solution of sodium silver cyanide OR sodium argentocyanide.

[F] State one appropriate observation for the following

i. At the anode when aqueous copper sulphate solution is electrolysed using copper electrodes. **[Board 2015]**

ii. At the cathode, during electrolysis of copper sulphate using active electrodes.

Ans. i. The copper of the anode dissolves, and therefore it becomes thin or is consumed gradually.

Examiner's Comment

- Instead of focusing on the anode, candidates stated the effect on the electrolyte.

Answering Tips

- Remember all times that naming the product is of no use when observations are asked.
- In the course of practical work, write observations and then discuss them at the end of the practical session.

ii. A brownish pink metal, copper is deposited at cathode.

Course of Action
Maximum Time = 2 Hrs
Maximum Questions = 73

7 METALLURGY

Learning Objectives

The students will be able to:

- *Understand the occurrence of metals in nature, distinguishing between minerals and ores, and identifying common ores of iron, aluminium, and zinc.*
- *Learn stages involved in the extraction of metals.*
- *Explain the extraction of aluminium through both chemical and electrolytic methods.*
- *Understand the Bayer's process for purifying bauxite using NaOH.*
- *Describe the Hall-Héroult process for electrolytic extraction, including the structure of the electrolytic cell, components of the electrolyte, electrodes, and electrode reactions.*
- *Identify common alloys such as stainless steel, duralumin, brass, bronze, and fuse metal/solder.*
- *Understand the purposes of alloying, such as improving strength, corrosion resistance, or melting properties.*

Revision Notes

The **elements** are the basic building blocks of matter. There are about 120 elements known till now. To study the characteristics and the properties of all these elements in a simpler way, they are divided into two categories i.e., metals and non-metals.

➢ **Minerals:** Minerals are naturally occurring compounds of metals which are generally mixed with other substances such as silica, limestone, etc. These earthy impurities are called gangue or matrix. **[Board, 2022]**

➢ **Ores:** Ores are those minerals from which metals are extracted commercially at a comparatively lower cost and with minimum effort. **[Board, 2024]**

➢ **Occurrence of metals:** Most of the metals are reactive, so they occur in combined state in the form of their oxides, carbonates, halides, sulphides, sulphates, etc., mixed with mud, clay, sand and stone.

- **Stages involved in the extraction of Metals:**

Extraction of a metal from its ore consists of the following processes:

Extraction of Metals
Scan Me!

1. Crushing and grinding: Ores are crushed into fine powder in big jaw crushers and ball mills. This process is called **pulverisation**.

2. Dressing of the ore: Hydrolytic method, magnetic separation, froth flotation method and chemical method. These processes removes the rocky impurities like SiO_2 called gangue.

3. Conversion of concentrated ore to its oxide: Roasting and calcination. **[Board, 2024]**

Roasting: The process of heating the concentrated ore in the presence of excess air to a high temperature before it is reduced to a metal.

Calcination: The process of heating the concentrated ore in the absence of air or in a limited supply of air.

Eg. Roasting: $2ZnS + 3O_2 \rightarrow 2ZnO + 2SO_2$

Calcination: $ZnCO_3 \rightarrow ZnO + CO_2$

Reduction of metal oxides
Scan Me!

4. Reduction of metallic oxides: Some can be reduced by hydrogen, carbon and carbon monoxide (e.g., copper oxide, lead (II) oxide, iron (III) oxide and zinc oxide) and some cannot (e.g., Al_2O_3, MgO).

5. Electro refining: On electrolysis of the impure metal, pure metal is collected at cathode and impurities settle down in the form of anode mud.

➢ **Chemical Separation Bayer's Process**

This method makes use of difference between the chemical properties of the ore and the gangue. Bauxite ore is impure aluminium oxide ($Al_2O_3.2H_2O$) containing iron (III) oxide (Fe_2O_3) and silica (SiO_2) as the main impurities. Bayer's process is used to obtain pure aluminium oxide from bauxite ore. In this method of chemical purification, the finely powdered ore is treated with hot sodium hydroxide solution.

$$\underset{\text{[Impure bauxite]}}{Al_2O_3.2H_2O} + \underset{\text{[Conc.solution]}}{2NaOH} \xrightarrow{150-200^\circ C} \underset{\text{[Sodium aluminate]}}{2NaAlO_2} + 3H_2O$$

$$\underset{\text{[sodium aluminate]}}{NaAlO_2} + 2H_2O \xrightarrow{50-60^\circ C} \underset{\text{[Sodium hydroxide]}}{NaOH} + \underset{\text{[Aluminium hydroxide]}}{Al(OH)_3 \downarrow}$$

$$\underset{\text{[Aluminium hydroxide]}}{2Al(OH)_3} \xrightarrow{1100^\circ C} \underset{\text{[Pure alumina]}}{Al_2O_3} + 3H_2O$$

Reduction by aluminium: The oxides of metals of chromium or manganese are very stable and cannot be reduced by conventional reducing agents. These metal oxides can be reduced by aluminium powder. This method of reduction is also known as **Thermite process.**

$$3MnO_2 + 4Al \xrightarrow{Heat} 2Al_2O_3 + 3Mn + \text{Heat energy}$$

$$Cr_2O_3 + 2Al \xrightarrow{Heat} Al_2O_3 + 2Cr + \text{Heat energy}$$

➤ **Chemical Separation Bayer's Process**

Ores of Al	Chemical Name	Formula
(A) Ores of Al		
Bauxite	Hydrated aluminium oxide	$Al_2O_3 . 2H_2O$
Cryolite	Sodium aluminium fluoride	Na_3AlF_6
Corundum	Anhydrous aluminium oxide	Al_2O_3
(B) Ores of Fe		
Red haematite	Anhydrous ferric oxide	Fe_2O_3
Brown haematite	Hydrated ferric oxide	$2Fe_2O_3 . 3H_2O$
Magnetite	Triferric tetraoxide	Fe_3O_4
Iron pyrite	Iron disulphide	FeS_2
Siderite	Ferrous carbonate	$FeCO_3$
(C) Ores of Zn		
Zinc blende	Zinc sulphide	ZnS
Zincite	Zinc oxide	ZnO
Calamine	Zinc carbonate	$ZnCO_3$

➤ **Metallurgy of Aluminium**

(i) Occurrence: It is a highly electropositive metal and it does not occur in free state.

(ii) Purification of Ore: Most of aluminium is extracted from bauxite which contains ferric oxide (Fe_2O_3) and silica (SiO_2) as impurities. The crushed ore is subjected to electromagnetic separation to remove ferric oxide and then concentrated by Bayer's process or Hall's process.

Extraction of Aluminium

Scan Me!

(iii) Bayer's Process: prepares the alumina.

Hall-Heroult's Process: Alumina to produce the aluminum metal.

(iv) Electrolytic Vessel: Iron tank is lined with heat resistant material like carbon with a sloping floor for the removal of molten aluminium.

(v) Electrolyte: It is a molten mixture of alumina, cryolite, fluorspar.

(vi) Electrodes — Anode: Thick carbon rods

Cathode: Tank with carbon lining

(vii) Temperature: 950°C

A layer of powdered coke is sprinkled over the surface of the electrolytic mixture. This reduces the heat loss by radiation and prevents the carbon anode from burning in air.

(viii) Reactions:

Alumina

$$Al_2O_3 \rightleftharpoons 2Al^{3+} + 3O^{2-}$$

At cathode: Carbon lining (gas carbon) of the cell.

$$2Al^{3+} + 6e^- \longrightarrow 2Al$$

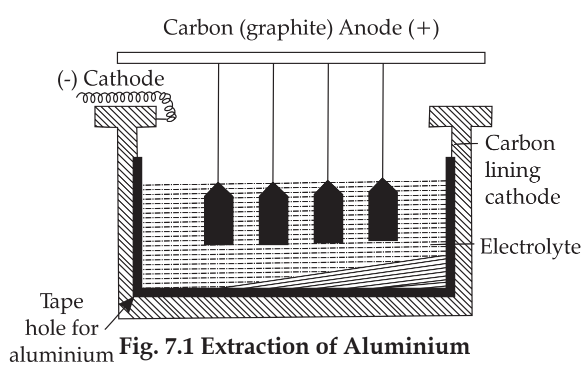


Fig. 7.1 Extraction of Aluminium

At anode: Thick rods of graphite are suspended into the fused electrolyte.

$$3O^{2-} - 6e^- \longrightarrow 3[O]$$ **[Board, 2023]**

$$3O + 3O \longrightarrow 3O_2$$

Anode is oxidised to carbon monoxide and then to carbon dioxide.

$2C + O_2 \longrightarrow 2CO$

$2CO + O_2 \longrightarrow 2CO_2$

The oxygen evolved at the anode escapes as a gas or reacts with carbon anode. The carbon anode is thus oxidised. The carbon anode is, hence consumed and renewed periodically after a certain period of usage. Further purification can be done by Hoope's process.

➢ **Refining of Aluminium (Hoope's electrolytic process):**

The process uses an electrolytic cell which contains three layers of molten substance of differing specific gravity. Molten impure aluminium forms the bottom layer. The bottom layer has carbon lining and serves as anode. Pure molten aluminium with carbon electrodes serves as the cathode in the top layer. The middle layer consists of a mixture of fluorides of sodium barium and aluminium. The lower layer consists of impure Al at bottom along with carbon lining.

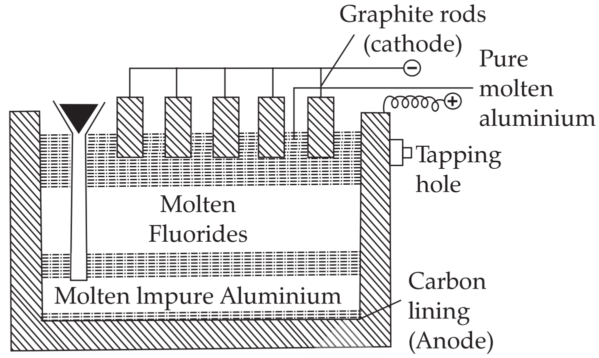


Fig. 7.2 Refining of Aluminium **[Board, 2023]**

Anode: $\underset{\text{(Impure)}}{Al} - 3e^- \rightarrow Al^{3+}$

Cathode: $Al^{3+} + 3e^- \rightarrow \underset{\text{(Pure)}}{Al}$

➢ Alloy: Alloy is a homogeneous mixture of two or more metals or of one or more metals with certain non-metallic elements. **[Board, 2024]**

Example 1

Answer the following questions with respect to the electrolytic process in the extraction of aluminium:

(i) Identify the components of the electrolyte other than pure alumina and the role played by each.

(ii) Explain why powdered coke is sprinkled over the electrolytic mixture.

Ans. (i) Cryolite – lowers the fusion temperature of the electrolyte.

Fluorspar – increases the conductivity of the electrolyte or acts as a solvent.

(ii) To prevent the heat loss from the electrolyte.

➢ **Common Alloys, their Compositions, Properties and Uses:**

Principal metal	Alloy's name	Composition	Properties	Used for making
Aluminium **[Board, 2023,24]**	1. Duralumin **[Board, 2023,24]**	95% Al 4% Cu 0.5% Mg 0.5% Mn	1. Light but as strong as steel. 2. Hard and resistant to corrosion. 3. Highly ductile.	1. Bodies of aircraft, buses and tube trains. 2. Light tools 3. Pressure cooker.
	2. Magnalium	90-95% Al 10-5% Mg	1. Resists corrosion 2. Light 3. Strong	1. Aircraft 2. Scientific instruments 3. Metal mirrors 4. Light tools 5. Beams of balance 6. Household appliances
	3. Alnico	Al,Ni,Co,Fe	1. Light 2. Shiny 3. Resists corrosion	Magnets
Iron	1. Stainless steel **[Board, 2023,24]**	73% Fe, 18% Cr, 8% Ni, 1% C	1. Resists corrosion 2. Lustrous, hard 3. Resistant to acids and alkalis	1. Utensils 2. Cutlery 3. Ornamental pieces 4. Surgical instruments
	2. Manganese steel	85% Fe, 1% C, 14% Mn	1. Durable, tough and hard	1. Safes 2. Rock drills 3. Armour plates
	3. Tungsten	84% Fe, 5% W, 1%C,	1. Very hard	1. Cutting tools for high speed lathes
	4. Nickel steel	95-98%Fe 5-3% Ni	1. Hard and elastic 2. Resistant to corrosion	1. Electric wire cables 2. Automobile parts

	5. Invar	Fe 63%, Ni 36% C 1%	1. Negligible expansion	1. Metre scales 2. Scientific instruments
Zinc and copper	1. Brass	60-70% Cu 40-30% Zn	1. Malleable and ductile. 2. Can be easily cast. 3. Resists corrosion. 4. Yellow/silvery in colour.	1. Decorative hardware, utensils. 2. Screws and handles. 3. Cartridge containers. 4. Parts of watches . 5. Musical instruments. 6. Electrical goods.
	2. Bronze **[Board, 2024]**	80% Cu 18% Sn 2% Zn	1. Hard and easily cast. 2. Can take up polish. 3. Resists corrosion.	1. Medals 2. Statues 3. Utensils 4. Bearings and 5. Coins
	3. German silver	50% Cu 30% Sn 20% Zn	1. White and light like silver. 2. Malleable and ductile. 3. High electrical resistance.	1. Decorative articles. 2. Electric heaters, rheostats. 3. Resistors
	4. Bell metal	78% Cu 22% Sn	1. Sonorous (produces sound). 2. Hard and brittle.	1. Bell, gongs. 2. Statues.
	5. Gun metal	Cu 88%, Sn 8%, Zn 1%, Pb 1%	1. Hard and brittle 2. Easily cut	1. Barrels of cannons. 2. Bearings, gears, etc.
Lead	1. Solder or Fuse metal	Pb, Sn	1. Low melting point, high tensile strength	1. Welding 2. Fuse
	2. Type metal	75%Pb, 15% Sb, 10% Sn	1. Low melting point, easily cast.	1. For printing blocks.

Example 2

A light and strong alloy is required for making the body of an aircraft. What should be its constituents?

Ans. The alloy used for making the body of an aircraft is duralumin and its constituents are aluminium and copper, and also contains traces of manganese and magnesium.

MNEMONICS

Concept: Ores of Iron

Mnemonics:

RB **He**ma **Ma**si is my **si**ster

Interpretation:

R: Red Haematite

B: Brown

He: Haematite

Ma: Magnetite

Si: Siderite

Concept: Ores of zinc

Mnemonics:

z z z......

Interpretation:

Zinc blende

Zinc calamine

Zincite

Key Terms

➢ **Metallurgy:** The process used for the **extraction of metals** in their pure form from their ores is referred to as metallurgy. It also deals with the production and purification of metals and manufacture of alloys.

➢ **Minerals:** The naturally occurring compounds of metals which are generally mixed with other matter such as soil, sand, limestone and rocks are known as minerals.

➢ **Ores:** These are the minerals from which the metals can be extracted conveniently, quantitatively and economically.

➢ **Flux:** It is a substance added to the charge in a furnace during smelting (or reduction) to remove gangue materials as slag. The flux normally added in any metallurgical process is lime (CaO).

➢ **Slag:** It is a fusible product formed when flux combines with gangue during the extraction of metals.

For example:

Gangue + Flux $\longrightarrow$ Slag

$SiO_2 + CaO \longrightarrow CaSiO_3$

➢ **Fusible alloy:** An alloy melting in the range of about 57°C to 260°C, usually contains bismuth, lead, tin, etc. These alloys are called fusible alloys.

➢ **Amalgam:** A homogeneous mixture or an alloy of mercury with a number of metals alloys such as sodium, zinc, gold and silver as well as with some non-metals is known as amalgam.

Key Facts

- **Flow Chart for extraction of metal from its ore:**

Ore
↓
Crushing and Grinding
↓
Concentration → • Gravity Separation • Froth floatation • Magnetic Separation • Chemical method or leaching
↓
Roasting and Calcination
↓
Reduction → • Chemical reduction • Electrolytic reduction
↓
Refining
↓
Pure metal.

- **Sulphide ores like Zinc blende (ZnS) and Galena (PbS)** are lighter than the impurities present. They are concentrated by froth flotation process.
- The selection of **reducing agents** depends upon the position of the metal in the reactivity series.
- Oxides of highly active metals like potassium, sodium, calcium, magnesium and aluminium **have great affinity towards oxygen** and so cannot be reduced by common reducing agents like coke (carbon), carbon monoxide or hydrogen.
- Reactive metals can not be extracted from their aqueous salt solutions by electrolysis as they can react with water.
- The highly reactive metals can be used as reducing agents because they can displace metals of lower reactivity from their compounds . For example, in order to obtain manganese, its oxide is heated with aluminium powder.

$3\,MnO_2\,(s) + 4Al\,(s) \longrightarrow 3Mn\,(l) + 2Al_2O_3\,(s) + \text{Heat}.$

- Aluminium is used as a powerful reducing agent. When the mixture of aluminium powder and iron oxide is ignited, the latter is reduced to metal. This process is called aluminothermy.

$Fe_2O_3 + 2Al \longrightarrow 2Fe + Al_2O_3 + \text{Heat}.$

- Refining is the process by which crust metal is purified.
- Aluminium is the most abundant metal in the earth crust.
- Aluminium is extracted from its main ore bauxite $Al_2O_3.2H_2O$. Bauxite contains 60% Al_2O_3 the rest being sand, ferric oxide and titanium oxide.
- **Cryolite lowers the fusion temperature from 2050°C to 950°C** and enhances conductivity.
- **Fluorspar** and cryolite **act as a solvent for the electrolytic mixture** and increases its conductivity since pure alumina is almost a non-conductor of electricity.
- Powdered coke is sprinkled over the surface of the electrolytic mixture. It reduces heat loss by radiation and prevent the burning of anode.
- An alloy melting in the range of about 51°C to 260°C, usually contains bismuth, lead, tin, etc. These alloys are called fusible alloys.
- The purpose of an alloy is to improve specific usefulness of the primary component and not to adulterate or degrade it.
- Al_2O_3 is an amphoteric oxide and reacts with NaOH, while the other impurities have no action.
- **Flow Chart for extraction of aluminium:**

Ore - Bauxite $Al_2O_3.2H_2O$
↓
Concentration by NaOH to obtain Al_2O_3

$$Al_2O_3.2H_2O + 2NaOH \xrightarrow{140°C-150°C} 2NaAlO_2 + 3H_2O$$

$$NaAlO_2 + 2H_2O \xrightarrow{50°C} NaOH + Al(OH)_3 \downarrow$$

$$2Al(OH)_3 \xrightarrow{1000°C} Al_2O_3 + 3H_2O$$

↓
Electrolytic reduction of alumina
Electrolyte: Alumina, Cryolite, Fluorspar (1: 3: 1)

$$\text{Cathode: } 4Al^{3+} + 12e^- \longrightarrow 4Al$$

$$\text{Anode: } 6O^{2-} - 12e^- \longrightarrow 3O_2$$

$$C + O_2 \longrightarrow CO_2$$

↓
Purification by electrolysis

$$\text{Cathode: } Al^{3+} + 3e^- \longrightarrow Al$$

$$\text{Anode: } Al - 3e^- \longrightarrow Al^{3+}$$

Course of Action
Maximum Time = 6 Hrs
Maximum Questions = 164

8 STUDY OF COMPOUNDS

Learning Objectives

The students will be able to:
- *Discuss the preparation and properties of hydrogen chloride.*
- *Explain the preparation and properties of ammonia.*
- *Discuss the laboratory preparation of nitric acid.*
- *Explain the oxidising property of nitric acid.*
- *Discuss the preparation and properties of sulphuric acid.*
- *Study all reactions in terms of reactants, products, conditions, equations and observations.*

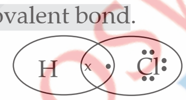
TOPIC-1: HYDROGEN CHLORIDE GAS

Concepts covered: • Preparation of hydrogen chloride gas in laboratory • Physical properties of hydrogen chloride gas

Revision Notes

➢ Hydrogen chloride gas was prepared by **Glauber in 1648** by heating common salt (NaCl) with conc. H_2SO_4. **Lavoisier** named it muriatic acid. Davy (1810) named it as hydrochloric acid.

➢ It has a polar covalent bond.



➢ It is synthesised by the reaction of hydrogen and chlorine in the presence of diffused sunlight.

$$H_2 + Cl_2 \xrightarrow[\text{sunlight}]{\text{diffused}} 2HCl$$

➢ **Preparation of Hydrogen Chloride Gas in Laboratory**

Hydrogen chloride is prepared in laboratory by the reaction of metallic chloride with concentrated sulphuric acid. (Fig. 8.1) **[Board, 2024]**

Preparation of HCl gas
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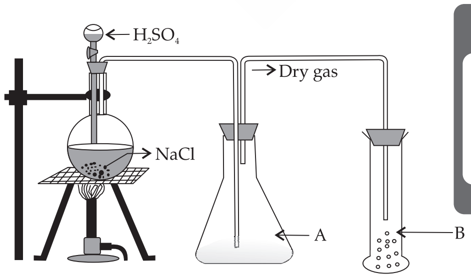


Fig. 8.1: Preparation of hydrogen chloride gas

Reactants used for the preparation of hydrogen chloride gas are sodium chloride and concentrated sulphuric acid. Sodium chloride is preferred over other metallic chlorides, as it is less expensive and easily available.

$$\underset{\text{Rock salt}}{NaCl} + H_2SO_4(\text{conc.}) \xrightarrow[200°C]{\text{below}} \underset{\substack{\text{Sodium}\\\text{bisulphate}}}{NaHSO_4} + \underset{\substack{\text{Hydrogen}\\\text{chloride gas}}}{HCl}$$

$$\underset{\text{Rock salt}}{2NaCl} + H_2SO_4(\text{conc.}) \xrightarrow[200°C]{\text{above}} \underset{\substack{\text{Sodium}\\\text{sulphate}}}{Na_2SO_4} + \underset{\substack{\text{Hydrogen}\\\text{chloride gas}}}{2HCl}$$

Conc. H_2SO_4 acts as a non-volatile acid and is used to prepare a volatile acid from its salt.

➢ **Precautions to be taken during the laboratory preparation of hydrogen chloride gas.**

(i) The reaction mixture should not be heated **beyond or above 200°C**.

- At above 200°C, a sticky mass of sodium sulphate is formed which sticks to glass apparatus and is difficult to remove.
- Heat energy is wasted if heated above 200°C.

➢ Hydrogen chloride gas is dried by passing through concentrated sulphuric acid. It is not dried by passing through phosphorous pentoxide or calcium oxide (which is a basic oxide), as both of these drying agents undergo chemical reactions with hydrogen chloride gas.

$$2P_2O_5 + 3HCl \rightarrow POCl_3 + 3HPO_3$$

$$CaO + 2HCl \rightarrow CaCl_2 + H_2O$$

- Hydrogen chloride gas is collected by upward displacement of air, as the gas is heavier than air. It is not collected over water, as the gas is extremely soluble in water and it forms hydrochloric acid. **[Board, 2024]**
- When the gas jar is completely filled with the gas then dense white fumes appear at the mouth of the jar.
- In order to know whether the gas jar is full of hydrogen chloride gas, a glass rod dipped in ammonium hydroxide is brought near the mouth of the jar. If dense white fumes appear immediately, then it shows that the jar is full of hydrogen chloride gas.
- Hydrogen chloride gas is dissolved in water with the help of special arrangement called **funnel arrangement** to prevent the back suction of water and gives greater surface area for the absorption of gas.

Physical properties of hydrogen chloride gas

(i) It is a colourless gas with pungent, suffocating smell which gives off dense fumes in moist air.

(ii) It is extremely soluble in water.

The extreme solubility of hydrogen chloride gas is demonstrated by the fountain experiment. (Fig. 8.2)

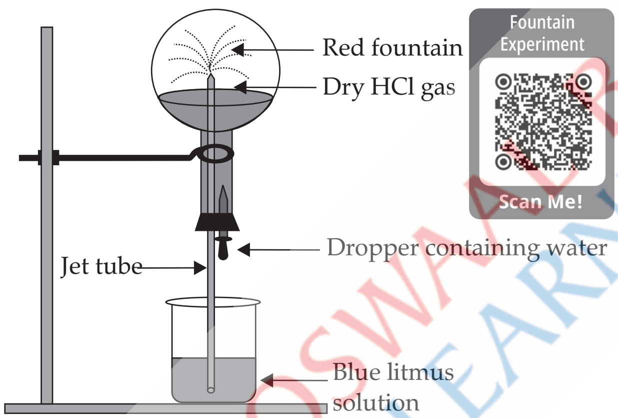
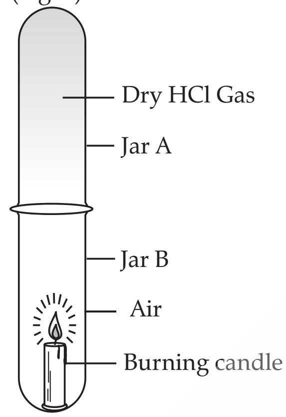


Fig. 8.2: Fountain experiment to demonstrate extreme solubility and acidic nature of HCl gas

(iii) Hydrogen chloride gas is heavier than air. Set up the apparatus as shown in fig. 8.3.

It is seen that the dense white fumes appear closer to the hydrogen chloride gas, suggesting that the gas is heavier than air, as it has travelled less distance as compared to air (Fig. 3).



Fountain Experiment

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Fig. 8.3: An Experiment to Show hydrogen Chloride Gas is Heavier than Air.

(iv) Hydrogen chloride gas is neither combustible nor a supporter of combustion. It extinguishes the burning splinter when brought near to it.

Chemical properties of Hydrogen chloride gas

➢ Hydrogen chloride gas is thermally decomposed to give chlorine.

$$2HCl \xrightarrow{500^\circ C} H_2 + Cl_2$$

➢ Metals which are lying above hydrogen in the metal activity series displace hydrogen when heated with hydrogen chloride gas.

$$Mg + 2HCl \xrightarrow{\Delta} MgCl_2 + H_2$$

$$Zn + 2HCl \xrightarrow{\Delta} ZnCl_2 + H_2$$

$$Fe + 2HCl \xrightarrow{\Delta} FeCl_2 + H_2$$

$$Ca + 2HCl \xrightarrow{\Delta} CaCl_2 + H_2$$

➢ Hydrogen chloride gas reacts with ammonia to form dense white fumes of ammonium chloride.

$$NH_3 + HCl \rightarrow \underset{\text{Dense white fumes}}{NH_4Cl}$$

➢ Hydrogen chloride gas is a polar covalent compound which, on dissolving in water, produces hydronium and chloride ions.

➢ Hydrogen chloride gas, on dissolving in water, produces strong, monobasic acid (hydrochloric acid).

$$HCl + H_2O \rightarrow H_3O^+ + Cl^-$$

➢ Dry hydrogen chloride gas has no effect on litmus, showing the neutral (non-acidic) character .

Key Term

➢ **Polar Covalent Bond:** A type of chemical bond where a pair of electrons is unequally shared between two different non-metals having different electronegativities.

Key Facts

- Sodium chloride is cheap, and therefore it is preferred for preparation of HCl over other metal chlorides.
- Conc. and dilute nitric acid is not used during the preparation of HCl because it is volatile and may volatilise out along with hydrogen chloride.
- When hydrogen chloride gas is exposed to air, it gives off white fumes due to the formation of hydrochloric acid on reacting with atmospheric water vapour.
- The fumes formed are a gas-liquid mixture of fine droplets of hydrochloric acid suspended in air.
- The blue litmus solution turns red due to the acidic nature of hydrogen chloride gas in presence of moisture or water vapours only.

MNEMONICS

Concept: Physical properties of HCl.

Mnemonics:

Call **G**auri **P**hone **S**e **ESHA**

Interpretation:

C – Colourless
G – Gas
P – Pungent
S – Smell
E – Extremely
S – Soluble (In water)
H – Heavier
A –Air

Example 1

Explain the following:

(i) Dry hydrogen chloride gas does not affect a dry strip of blue litmus paper, but it turns red in the presence of a drop of water.

(ii) Hydrogen chloride gas is not collected over water. A I **[OEB]**

Ans. (i) This is because dry HCl gas is a covalent compound and there are no hydronium $(H_3O)^+$ ions present. But in the presence of a drop of water, H_3O^+ ions are produced.

$$HCl + H_2O \rightarrow H_3O^+ + Cl^-$$

(ii) This is because it is highly soluble in water.

TOPIC-2: HYDROCHLORIC ACID

Concepts covered: • Preparation of hydrochloric acid • Properties of hydrochloric acid

Revision Notes

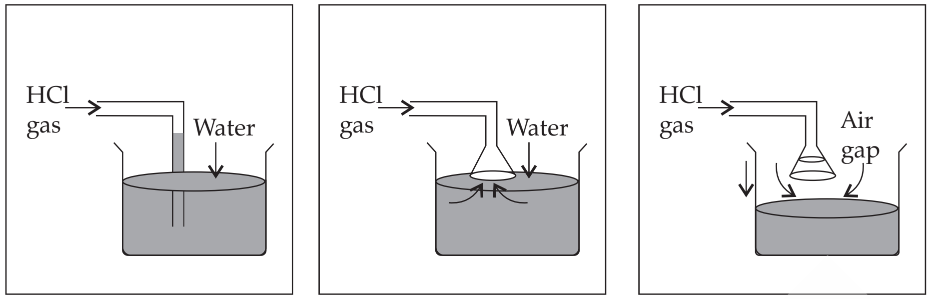
- Hydrogen chloride gas, when dissolved in water, produces **hydrochloric acid**.

$HCl + H_2O \rightleftharpoons H_3O^+ + Cl^-$

Hence, hydrochloric acid is a solution of hydrogen chloride in water.

➢ **Preparation of Hydrochloric Acid:**

Procedure: Hydrochloric acid is prepared by dissolving hydrogen chloride gas in water using a **special funnel** arrangement as shown in figure B.



A B(I) B(II)

Fig.8.4 Funnel arrangement

Since hydrogen chloride gas is extremely soluble in water, if the delivery tube through which HCl gas is passed is directly immersed in water, then a partial vacuum is created in the tube because rate of absorption of HCl gas is high. The pressure outside being higher causes the water to be pushed up into the delivery tube. This is called back-suction.

➢ **Mechanism by which back-suction is avoided or minimised:**

The funnel arrangement prevents or minimises back-suction of water. And it also provides a large surface area for the absorption of HCl gas.

The rim of the funnel is placed just touching the surface of the water. If back-suction occurs, the water rises up the funnel and the level outside the funnel falls, creating an air gap between the rim of the funnel and the surface of water.

The pressure outside and inside equalises, and the water which had risen in the funnel falls down again.

➢ **Precaution:** An empty flask (anti-suction device) is put between the generative flask and the water trough. In case the back-suction occurs, the water will collect in it and will not reach the generating flask.

Properties of Hydrochloric Acid:

Physical Properties

(i) Colour, odour and taste: Colourless, pungent choking smell and acidic.

(ii) Physiological action: Concentrated acid, which is corrosive.

(iii) Boiling point: 110°C

(iv) Acidic nature: The presence of H^+ in HCl imparts acidic properties. It is **monobasic** in nature and forms only one normal salt with a base.

$$HCl \xrightleftharpoons{H_2O} H^+(aq) + Cl^-(aq)$$

Chemical properties

(i) Action on metals: Hydrochloric acid reacts with metals which are above hydrogen in the activity series and forms metal chlorides and hydrogen.

$$Mg + 2HCl \rightarrow MgCl_2 + H_2\uparrow$$

(ii) Action on oxides and hydroxides: Hydrochloric acid reacts with oxides and hydroxides to form salt and water only.

$$ZnO + 2HCl \rightarrow ZnCl_2 + H_2O$$

$$Zn(OH)_2 + 2HCl \rightarrow ZnCl_2 + 2H_2O$$

(iii) Reaction with salts of weaker acids: HCl acid decomposes salts of **weaker acids**, e.g., carbonates, bicarbonates, sulphates and sulphides. **[Board, 2024]**

$$Na_2CO_3 + 2HCl \rightarrow 2NaCl + H_2O + CO_2\uparrow$$

$$NaHCO_3 + HCl \rightarrow NaCl + H_2O + CO_2\uparrow$$

$$Na_2SO_3 + 2HCl \rightarrow 2NaCl + H_2O + SO_2\uparrow$$

$$FeS + 2HCl \rightarrow FeCl_2 + H_2S\uparrow$$

(iv) Action on thiosulphates: It reacts with thiosulphate to produce SO_2 gas and yellow precipitates of sulphur.

$$Na_2S_2O_3 + 2HCl \rightarrow 2NaCl + H_2O + SO_2 + S\downarrow \text{ Yellow}$$

(v) Reaction with nitrates: Lead nitrate and mercury (I) nitrate react with hydrochloric acid to give white precipitate of lead chloride and mercury (I) chloride, respectively.

HCl acts as a precipitating age.

$$Pb(NO_3)_2 + 2HCl \rightarrow \underset{\text{white}}{PbCl_2}\downarrow + 2HNO_3$$

$$Hg_2(NO_3)_2 + 2HCl \rightarrow \underset{\text{white}}{Hg_2Cl_2}\downarrow + 2HNO_3$$

Silver nitrate gives thick, curdy white precipitate of silver chloride.

$$AgNO_3 + HCl \rightarrow AgCl\downarrow + HNO_3$$

$$AgCl + 2NH_4OH \rightarrow \underset{\substack{\text{Diammine}\\ \text{silver (I) chloride}}}{[Ag(NH_3)_2]^+Cl^-} + 2H_2O$$

$$[Ag(NH_3)_2]^+Cl^- + HNO_3 \rightarrow AgCl\downarrow$$

$$2AgCl \xrightarrow{\text{Sunlight}} 2Ag + Cl_2$$

(vi) Oxidation of hydrochloric acid: Powerful oxidising agents such as manganese dioxide (MnO_2), lead oxide (PbO_2), red lead (Pb_3O_4), potassium permanganate ($KMnO_4$) and potassium dichromate ($K_2Cr_2O_7$) oxidise conc. HCl, liberating chlorine gas.

$$MnO_2 + 4HCl \xrightarrow{\Delta} MnCl_2 + 2H_2O + Cl_2$$

$$PbO_2 + 4HCl \xrightarrow{\Delta} PbCl_2 + 2H_2O + Cl_2$$

$$Pb_3O_4 + 8HCl \xrightarrow{\Delta} 3PbCl_2 + 4H_2O + Cl_2$$

$$2KMnO_4 + 16HCl \xrightarrow{\Delta} 2KCl + 2MnCl_2 + 8H_2O + 5Cl_2$$

$$K_2Cr_2O_7 + 14HCl \xrightarrow{\Delta} 2KCl + 2CrCl_3 + 7H_2O + 3Cl_2$$

(vii) Formation of Aqua-regia

Mixture of 3 parts of conc. HCl and 1 part of conc. HNO_3 is called **aqua-regia.**

$$\underset{\text{Conc.}}{3HCl} + \underset{\text{Conc.}}{HNO_3} \rightarrow \underset{\text{Nitrosyl chloride}}{NOCl} + 2H_2O + \underset{\text{Nascent chlorine}}{2[Cl]}$$

Nascent chlorine reacts with noble metals like gold and platinum to give their soluble chlorides.

$$Au + 3[Cl] \rightarrow \underset{\text{Gold(III) chloride}}{AuCl_3}$$

$$Pt + 4[Cl] \rightarrow \underset{\text{Platinum (IV) chloride}}{PtCl_4}$$

➢ **Test for Hydrogen Chloride and Hydrochloric Acid**

(i) HCl gas gives thick white fumes of ammonium chloride (NH_4Cl) when a glass rod dipped in NH_3 is held near the mouth of the tube.

$$NH_3 + HCl \rightarrow NH_4Cl$$

(ii) With silver nitrate solution, both give a white precipitate of silver chloride.

$$AgNO_3 + HCl \rightarrow AgCl\downarrow + HNO_3$$

Precipitate is insoluble in nitric acid but soluble in ammonium hydroxide.

(iii) When hydrochloric acid is heated with manganese dioxide, then greenish yellow gas, i.e., chlorine is evolved which turns moist starch iodide paper blue-black.

$$MnO_2 + 4HCl \rightarrow MnCl_2 + 2H_2O + Cl_2$$

➢ **Hydrochloric acid is used as:**

- a laboratory reagent.
- in the manufacture of chlorine, chlorides, dyes, drugs, etc.
- in industry to pickle steel
- to remove rust from iron sheets
- in tanning and calico printing industry
- to decrease activity of gastric juice in patients.

Key Terms

➢ **Back Suction:** The effect in which the pressure in the delivery tube and flask is reduced and the atmospheric pressure from outside forces the water back up to the delivery tube is called back suction.

➢ **Pickling of metals:** It is a process of removal of oxide before they are painted, electroplated or galvanised.

Key Facts

➢ When hydrogen chloride gas is dissolved in water, hydrochloric acid is formed.

➢ HCl ionises in water due to its polar nature.

$$HCl + H_2O \rightarrow H_3O^+ + Cl^-$$

➢ The aqueous solution of HCl gas is called hydrochloric acid.

➢ A constant boiling mixture or azeotrope is a solution which boils without any change in its composition. HCl acid forms a constant boiling mixture at 110°C.

Example 2

Complete the given reactions and balance them.

(A) $Na_2CO_3 + HCl \rightarrow$

(B) $Na_2SO_3 + HCl \rightarrow$

(C) $ZnO + HCl \rightarrow$

(D) $Pb(NO_3)_2 + HCl \rightarrow$

(E) $PbO_2 + HCl \rightarrow$ **[OEB]**

Ans. (A) $Na_2CO_3 + 2HCl \rightarrow 2NaCl + H_2O + CO_2$

(B) $Na_2SO_3 + 2HCl \rightarrow 2NaCl + H_2O + SO_2$

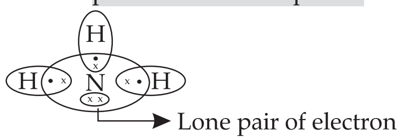
(C) $ZnO + 2HCl \rightarrow ZnCl_2 + H_2O$

(D) $Pb(NO_3)_2 + 2HCl \rightarrow PbCl_2 + 2HNO_3$

(E) $2PbO_2 + 4HCl \rightarrow 2PbCl_2 + 2H_2O + Cl_2$

Revision Notes

- The molecular formula of ammonia is NH_3. The relative molecular mass is 17.
- Ammonia is a polar covalent compound.



Electron dot diagram of Ammonia

- Ammonia is found in free state in traces.
- In combined state, ammonia is found in the form of ammonium salts like ammonium chloride, ammonium sulphate, etc.
- The smell in toilets is due to the bacterial decomposition of urea present in urine. As a result, ammonia gas is released.

$$\underset{\text{Urea}}{NH_2CONH_2} + 2H_2O \rightarrow \underset{\text{Ammonia}}{2NH_3} + H_2O + CO_2$$

- **Preparation of Ammonia:**

- In laboratory, ammonia is prepared by heating ammonium salts (except ammonium nitrate) with caustic alkalis like sodium hydroxide, potassium hydroxide or calcium hydroxide. (Fig. 8.5). **[Board 2024]**

$$\underset{\substack{\text{Calcium}\\\text{hydroxide}}}{Ca(OH)_2} + \underset{\substack{\text{Ammonium}\\\text{chloride}}}{2NH_4Cl} \xrightarrow{\Delta} \underset{\substack{\text{Calcium}\\\text{chloride}}}{CaCl_2} + \underset{\text{Ammonia}}{2NH_3}\uparrow + \underset{\text{Water}}{2H_2O}$$

$$\underset{\substack{\text{Sodium}\\\text{hydroxide}}}{NaOH} + \underset{\substack{\text{Ammonium}\\\text{chloride}}}{NH_4Cl} \xrightarrow{\Delta} \underset{\substack{\text{Sodium}\\\text{chloride}}}{NaCl} + \underset{\text{Ammonia}}{NH_3}\uparrow + \underset{\text{Water}}{H_2O}$$

(i) The reactants are finely ground and then taken in a round bottom flask. (Note: Finely ground reactants increase the surface area, hence increasing the speed of reaction.) The round bottom flask is fitted in a slanting position, bending in the downward direction.

(ii) Ammonia is dried by passing over quicklime, i.e., calcium oxide. It is not dried by passing through conventional drying agents like conc. sulphuric acid, anhydrous calcium chloride or phosphorus pentoxide because these drying agents undergo chemical reaction with ammonia.

[Board 2018]

$$2NH_3 + H_2SO_4 \rightarrow (NH_4)_2SO_4 \text{ salt is formed.}$$

$$CaCl_2 + 8NH_3 \rightarrow CaCl_2.8NH_3 \text{ addition compound formed}$$

$$6NH_3 + 3H_2O + P_2O_5 \rightarrow 2(NH_4)_3PO_4 \text{ salt is formed}$$

(iii) Ammonia is collected by downward displacement of air, as the gas is lighter than air and highly soluble in water.

[Board 2024/2018]

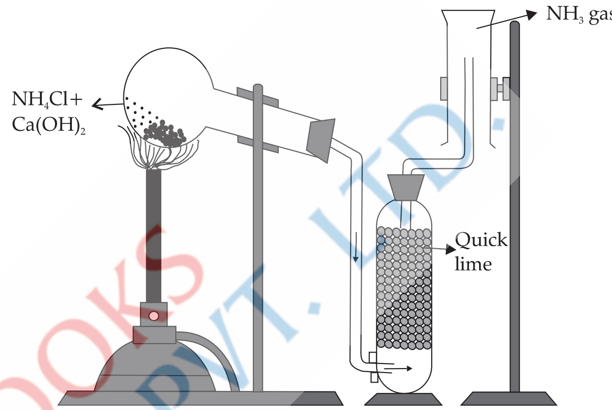


Fig. 8.5. Laboratory preparation of ammonia by ammonium chloride and calcium hydroxide.

(iv) Ammonium nitrate is not used for the preparation of ammonia, as it is explosive and gives nitrous oxide and water.

$$NH_4NO_3 \rightarrow N_2O + 2H_2O$$

- **Action of warm water on metallic nitrides (Fig. 8.6)**

Magnesium nitride, calcium nitride and aluminium nitride, on warming with water, produce their respective metallic hydroxides with the liberation of ammonia.

[Board 2017]

$$Mg_3N_2 + 6H_2O \xrightarrow{\text{Warm}} 3Mg(OH)_2 + \uparrow 2NH_3$$

$$Ca_3N_2 + 6H_2O \xrightarrow{\text{Warm}} 3Ca(OH)_2 + \uparrow 2NH_3$$

$$AlN + 3H_2O \xrightarrow{\text{Warm}} Al(OH)_3 + \uparrow NH_3$$

[Board 2024]

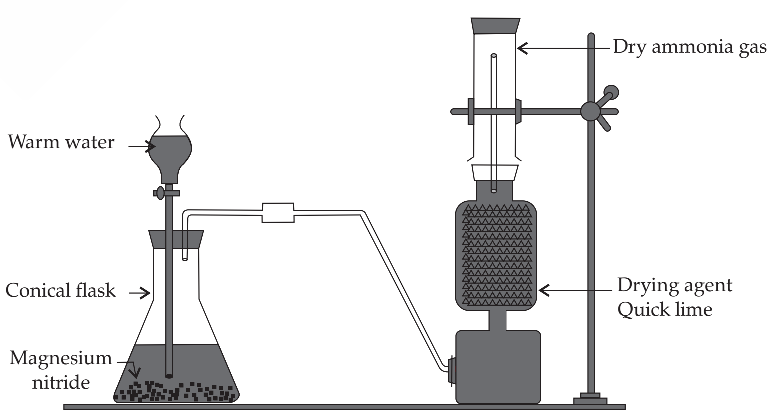


Fig. 8.6. Laboratory preparation of ammonia gas from magnesium nitride

- **Manufacture of Ammonia Gas by Haber's Process (Fig. 8.7)**

(i) Ammonia is industrially manufactured by **Haber's process**. The ratio by volume of nitrogen and hydrogen is 1:3.

(ii) Nitrogen is obtained by the fractional distillation of liquefied air. Hydrogen is obtained by Bosch process.

(iii) Ammonia is separated from unreacted nitrogen and hydrogen by:

(1) Liquefaction: Ammonia can be easily liquefied in comparison to nitrogen and hydrogen.

(2) By absorbing ammonia in water: As ammonia is highly soluble in water, whereas nitrogen and hydrogen are insoluble in water.

(iv) The speed of the reaction can be enhanced by taking finely divided iron as the **catalyst**. The efficiency of a catalyst is increased by using a promoter which is either molybdenum or aluminium oxide.

The reaction is reversible and exothermic.

$$N_2 + 3H_2 \rightleftharpoons 2NH_3 + \text{Heat}$$

The favourable conditions for the reaction are

Catalyst	Finely divided iron
Promoter	Molybdenum
Temperature	450-500°C
Atmospheric pressure	200-1000 atm.

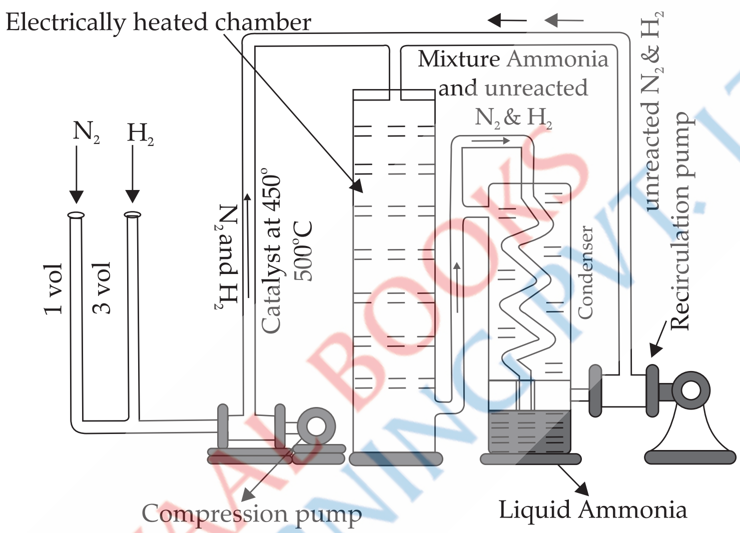


Fig. 8.7: Haber's process of manufacture of ammonia.

Haber's process
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Physical properties of ammonia are:

(i) It is a colourless gas having pungent irritating odour.

(ii) It is lighter than air.

(iii) It is highly or extremely soluble in water. The extreme solubility of ammonia is demonstrated by fountain experiment. (Fig. 8.8).

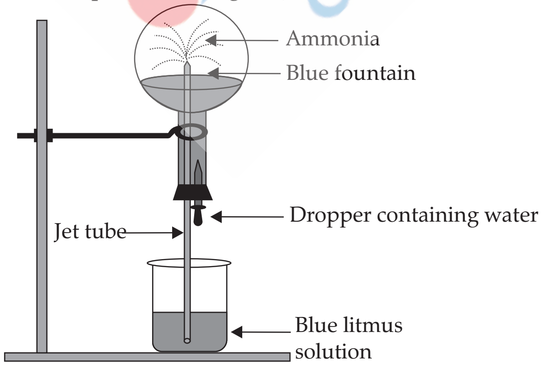


Fig. 8.8: Fountain experiment to demonstrate extreme solubility and basic nature of NH_3 gas.

Ammonia is dissolved in water with the help of funnel arrangement so as to prevent **back suction** of water, as the gas is highly soluble in water. Ammonia, when dissolved in water, produces ammonium hydroxide, which is alkaline in nature.

$$NH_3 + H_2O \rightarrow \underset{\text{Ammonium hydroxide}}{NH_4OH}$$

$$NH_4OH \rightarrow \underset{\text{Ammonium ion}}{NH_4^+} + \underset{\text{Hydroxide ion}}{OH^-}$$

Chemical properties of ammonia are as follows:

➢ Basic nature:

(i) Ammonia, on dissolving in water, produces hydroxyl ions as the only negatively charged particles.

$$NH_3 + H_2O \rightarrow NH_4OH \rightleftharpoons NH_4^+ + OH^-$$

(ii) Ammonia reacts with acids to form salts.

$$NH_3 + \underset{\text{Hydrochloric acid}}{HCl} \rightarrow \underset{\text{Ammonium chloride (Dense white fumes)}}{NH_4Cl}$$

$$NH_3 + \underset{\text{Nitric acid}}{HNO_3} \rightarrow \underset{\text{Ammonium nitrate}}{NH_4NO_3}$$

$$2NH_3 + \underset{\text{Sulphuric acid}}{H_2SO_4} \rightarrow \underset{\text{Ammonium sulphate}}{(NH_4)_2SO_4}$$

$$2NH_4OH + H_2SO_4 \rightarrow (NH_4)_2SO_4 + 2H_2O$$

$$NH_4OH + \underset{\text{Nitric acid}}{HNO_3} \rightarrow \underset{\text{Ammonium nitrate}}{NH_4NO_3} + H_2O$$

(iii) Ammonia, on dissolving in water, produces hydroxyl ions, the only negatively charged particle. Thus, the aqueous solution of ammonia called ammonium hydroxide, which is alkaline in nature, shows characteristic colour changes with indicators.

Red litmus solution — blue

Phenolphthalein — turns pink

Methyl orange — yellow

➢ **Burning of Ammonia in Oxygen:**

Ammonia is neither combustible nor a supporter of combustion. However, if jet of ammonia is ignited in atmosphere of oxygen, it burns with yellowish green flame.

$$4NH_3 + 3O_2 \rightarrow 2N_2 + 6H_2O$$

➢ **Catalytic oxidation of ammonia:**

Ammonia reacts with oxygen at 800°C in the presence of platinum catalyst to give nitric oxide and water vapour.

Catalytic oxidation of ammonia

Scan Me!

$$4NH_3 + 5O_2 \xrightarrow[800°C]{Pt} 6H_2O + 4NO\uparrow + \text{Heat}$$

$$2NO + O_2 \longrightarrow \underset{\text{Brown gas}}{2NO_2}$$

Procedure: Pass dry ammonia gas and oxygen gas through inlets over heated platinum, placed in combustion tube, which emits a reddish glow. Reddish brown vapours of NO_2 are seen in the flask due to the oxidation of NO.

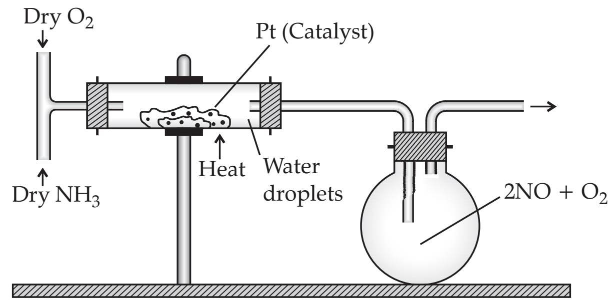


Fig. 8.9: Catalytic oxidation of ammonia

The reaction is exothermic in nature.

➢ **Reducing Nature of Ammonia:**

(i) Ammonia reduces heated metallic oxides such as copper oxide and lead(II) oxide to give metal, water vapour and nitrogen. **[Board 2024]**

$$2NH_3 + \underset{\text{black solid}}{3CuO} \longrightarrow \underset{\text{reddish brown solid}}{3Cu} + 3H_2O + N_2\uparrow$$

Procedure: Pass ammonia gas overheated CuO in hard glass tube. The black copper oxide is reduced to reddish brown copper. Nitrogen gas is collected over water in a gas jar. The water formed is collected in the **U-tube** and is tested with anhydrous $CuSO_4$, which turns blue.

This reaction proves that ammonia is made up of nitrogen and hydrogen, and so, it is a nitrogen hydride.

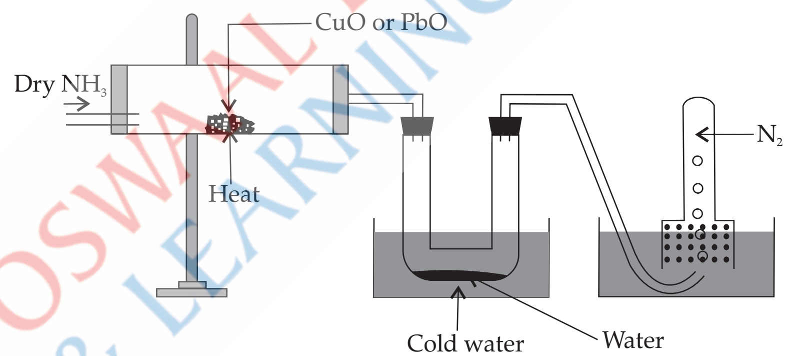


Fig. 8.10: Experiment to show reducing nature of NH_3

(ii) Reduction of Calcium Hypochlorite:

$$3CaOCl_2 + 2NH_3 \rightarrow 3CaCl_2 + N_2 + 3H_2O$$

Calcium hypochlorite

(iii) Reduction of Chlorine:

When ammonia is in excess: Greenish yellow chlorine disappears to give dense white fumes of ammonium chloride. **[Board 2024]**

$$\underset{\text{Excess}}{8NH_3} + 3Cl_2 \rightarrow 6NH_4Cl + N_2$$

When chlorine is in excess:

$$NH_3 + \underset{\text{Excess}}{3Cl_2} \rightarrow NCl_3 + 3HCl$$

These reactions show that.

(i) Ammonia is a reducing agent.

(ii) Chlorine has strong affinity for hydrogen.

[Board 2019/2020/2016]

➢ Ammonia reacts with carbon dioxide to form urea, which is a very important nitrogenous fertiliser containing 46.66% nitrogen.

$$2NH_3 + CO_2 \xrightarrow[150\,atm]{150°C} NH_2CONH_2 + H_2O$$

➢ Aqueous solution of ammonia precipitates metallic hydroxides from their soluble salts. NH_4OH acts as a precipitating agent.

$$FeCl_3 + 3NH_4OH \rightarrow \underset{\substack{\text{Ferric}\\\text{hydroxide}\\\text{(Reddish}\\\text{brown ppt.)}}}{Fe(OH)_3} \downarrow + 3NH_4Cl$$

$$FeSO_4 + 2NH_4OH \rightarrow \underset{\substack{\text{Ferrous}\\\text{hydroxide}\\\text{(Dirty green ppt.)}}}{Fe(OH)_2} \downarrow + (NH_4)_2SO_4$$

$$Pb(NO_3)_2 + 2NH_4OH \rightarrow \underset{\substack{\text{Lead}\\\text{hydroxide}\\\text{(White ppt.)}}}{Pb(OH)_2} \downarrow + 2NH_4NO_3$$

$$Zn(NO_3)_2 + 2NH_4OH \rightarrow \underset{\substack{\text{Zinc}\\\text{hydroxide}\\\text{(White gelatinous}\\\text{ppt.)}}}{Zn(OH)_2} \downarrow + 2NH_4NO_3$$

$$Zn(OH)_2 + 4NH_4OH \rightarrow \underset{\substack{\text{Tetraammine zinc}\\\text{hydroxide}}}{[Zn(NH_3)_4](OH)_2} + 4H_2O$$

$$Cu(NO_3)_2 + 2NH_4OH \rightarrow \underset{\substack{\text{Copper}\\\text{hydroxide}\\\text{(Pale blue ppt.)}}}{Cu(OH)_2} \downarrow + 2NH_4NO_3$$

$$Cu(OH)_2 + 4NH_4OH \rightarrow \underset{\substack{\text{Tetraammine copper}\\\text{hydroxide}}}{[Cu(NH_3)_4](OH)_2} + 4H_2O$$

Metallic salt solution	Colour of the ppt (In small quantity)	Nature of ppt (Soluble / Insoluble) in excess
(i) Iron (III) chloride	Reddish brown	Insoluble
(ii) Iron (II) sulphate	Dirty green	Insoluble
(iii) Lead nitrate	White	Insoluble
(iv) Zinc nitrate	White	Soluble
(v) Copper sulphate	Pale blue or bluish white	Soluble

Tests for Ammonia:

(A) Ammonia gives dense white fumes with concentrated hydrochloric acid.

$$NH_3 + HCl \rightarrow NH_4Cl$$

(A) Ammonia turns Nessler's reagent brown.

(B) When ammonia is passed through copper sulphate solution first a little, then a bluish white precipitate is formed, which dissolves in excess of ammonia to give deep blue solution, inky blue solution or Prussian blue solution.

(C) As ammonia is basic in nature, therefore it turns moist red litmus blue and phenolphthalein from colourless to pink.

➢ **Uses of Ammonia:** Ammonia is extensively used in the manufacture of nitrogenous fertilisers, nitric acid and other explosives. Generally, it is used as a refrigerant and as a cleansing agent (removes grease).

MNEMONICS

1. Concept: Conditions for Haber's Process

Mnemonics:

CaFé ProMo

Interpretation:

Ca: Catalyst

Fe: Iron

Pro: Promoter

Mo: Molybdenum

2. Concept: Colour change with different indicators in ammonia:

Mnemonics:

Rohan **L**aaya **B**illi **Par P**apa

Mumma **O**utrageously **Y**elled

Interpretation:

R: Red

L: Litmus solution—

B: blue

Par: Phenolphthalein—turns

P: Pink

M: Methyl

O: Orange—

Y: Yellow

Red litmus solution — blue

Phenolphthalein — turns pink

Methyl orange — yellow

3. Concept: Metallic salt solution colours reddish brown only if Iron(III) is taken. It will be dirty green if Iron(II) is taken.

Mnemonics:

I **C**an't **R**ead **B**ible.

Interpretation:

I: Iron

C: Chloride

R: Reddish

B: Brown

Key Terms

➢ **Nessler's Solution:** It is a slightly alkaline solution of potassium mercuric iodide made by adding KOH. Its chemical formula is K_2HgI_4.

Key Facts

➢ Ammonia and ammonium compounds being highly soluble in water do not occur as minerals.

➢ Drying of ammonia gas is done by passing it through a drying tower containing lumps of quicklime (CaO).

➢ Other drying agents like conc. H_2SO_4, P_2O_5 and anhydrous $CaCl_2$ are not used, as ammonia being basic reacts with them.

➢ Ammonia gas is collected in inverted gas jar by the downward displacement of air. Its reason is that it is lighter than air and highly soluble in water, so it can not be collected over water.

- Ammonia is liquefied easily as compared to nitrogen and hydrogen.
- Ammonia burns in oxygen with yellowish green flame and produces water vapours and nitrogen.
- Ammonia reduces heated yellow lead monoxide to greyish metallic lead.

$$\underset{\text{(yellow)}}{3PbO} + 2NH_3 \longrightarrow \underset{\text{(greyish)}}{3Pb} + 3H_2O + N_2$$

- Anhydrous ammonia is a clear, colourless liquid under pressure. It evaporates rapidly and produces cooling effect. This makes ammonia a good refrigerant.
- Aqueous ammonia emulsifies or dissolves fats, grease, etc.
- Ammonia reacts with CO_2 at 150°C and 150 atm. pressure to give urea, which is a valuable nitrogenous fertiliser.

$$2NH_3 + CO_2 \xrightarrow[150 \text{ atm}]{150°C} NH_2CONH_2 + H_2O$$

- Ammonia is the only basic gas.

Example 3

Explain why dry or liquid ammonia is neutral to litmus. **[OEB]**

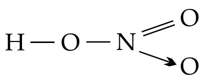
Ans. Ammonia (NH_3) is a covalent compound; therefore, in the absence of ions, dry or liquid ammonia has no effect on litmus.

TOPIC-4: NITRIC ACID

Concepts covered: • Formation of nitric acid in atmosphere • Laboratory preparation of nitric acid • Properties of nitric acid

Revision Notes

- Molecular formula: HNO_3 Molecular mass: 63
- Structure of nitric acid



- Nitric acid was initially called '**aqua-fortis**'.
- Nitric acid occurs in the free state and in the combined state.

Formation of Nitric Acid in Atmosphere

Step 1: Nitrogen present in the atmosphere reacts with oxygen to form nitric oxide.

$$N_2(g) + O_2(g) \rightleftharpoons 2NO(g)$$

Step 2: Nitric oxide is further oxidised to nitrogen dioxide.

$$2NO + O_2 \rightarrow 2NO_2(g)$$

Step 3: This nitrogen dioxide dissolves in rainwater.

$$4NO_2 + 2H_2O + O_2 \rightarrow 4HNO_3$$

Laboratory Preparation of Nitric Acid by Ostwald's Process

Reactants: Distilling a mixture of potassium or sodium nitrate with concentrated sulphuric acid.

$$\underset{}{KNO_3} + \underset{\text{Conc.}}{H_2SO_4} \xrightarrow{<200^\circ C} \underset{\text{Potassium bisulphate}}{KHSO_4} + HNO_3$$

$$NaNO_3 + H_2SO_4 \xrightarrow{<200^\circ C} \underset{\text{sodium bisulphate}}{NaHSO_4} + HNO_3$$

[Board 2024]

Products: Potassium bisulphate or sodium bisulphate and nitric acid vapours.

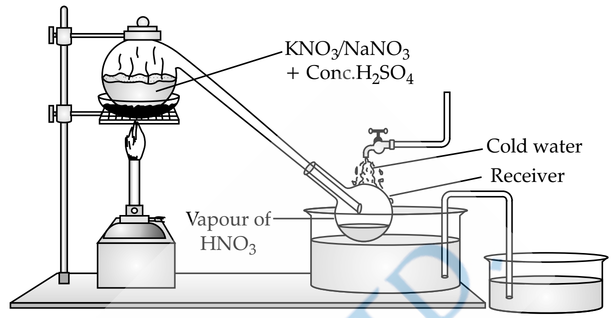


Fig . 8.11. Laboratory preparation of nitric acid

Procedure: A mixture of equal parts by weight of sodium nitrate or potassium nitrate and concentrated sulphuric acid is heated gently in the glass retort.

Laboratory preparation of nitric acid.

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Observation: On heating the mixture in glass retort, the volatile nitric acid is displaced, and the vapours are collected in the receiver, which is cooled from outside with cold water.

Collection: Vapours of nitric acid are condensed to a light yellow liquid. The yellow colour is due to the dissolution of NO_2 gas, as nitric acid is unstable and decomposes easily.

$$4HNO_3 \rightarrow 2H_2O + 4NO_2 + O_2$$

The yellow colour of the acid is removed by passing dry air or CO_2 in it.

Manufacture of Nitric Acid (Ostwald Process)

The process is explained in three steps:

Step 1: Catalytic Oxidation of Ammonia

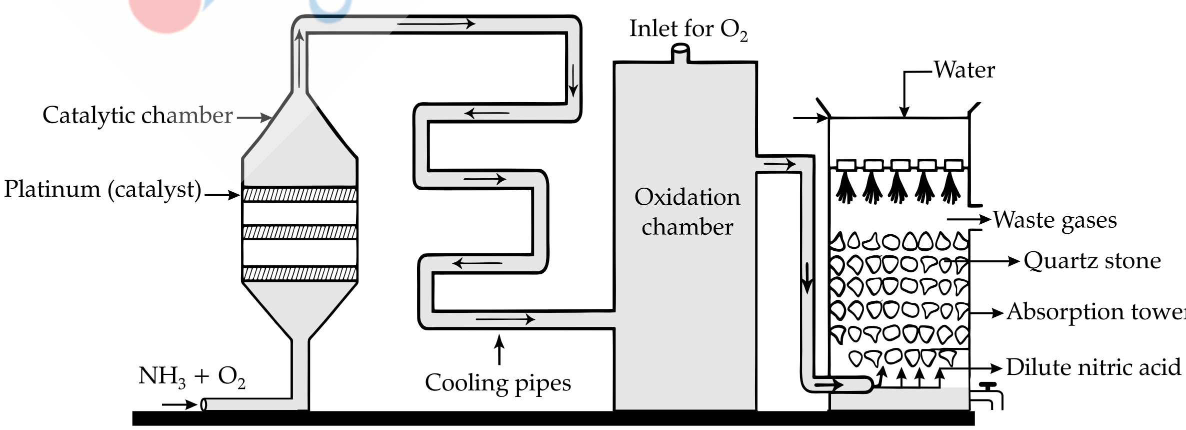

Fig. 8.12: Manufacture of nitric acid by Ostwald process **[Board 2024]**

- There is catalytic oxidation of NH_3 with atmospheric oxygen. Only initial heating is required as reaction is exothermic.

➢ Oxidation of ammonia into nitrogen monoxide takes place.

$$4NH_3 + 5O_2 \xrightarrow[800^\circ C]{Pt} 4NO + 6H_2O + \text{Heat}$$

Step 2: Oxidation of Nitric Oxide in Oxidation Chamber
For complete oxidation of nitric oxide, the gases should be cooled down before entering the oxidation chamber.

$$2NO + O_2(g) \xrightarrow{50^\circ C} 2NO_2(g)$$

Step 3: Absorption of Nitrogen Dioxide in Water: The nitrogen dioxide is passed through the absorption tower, from the top of which warm water trickles.

$$3NO_2(g) + H_2O(l) \rightarrow 3HNO_3(aq) + NO(g)$$

NO formed is recycled and aqueous HNO_3 is concentrated by distillation upto 68% by mass. 98% concentration is achieved by dehydration with concentrated H_2SO_4.

➢ **Chemical Properties of Nitric Acid** **[Board 2024]**

1. Stability

- Pure nitric acid is colourless and **unstable and decomposes** slightly even at ordinary temperature and in the presence of sunlight.

$$4HNO_3 \xrightarrow[\text{Sunlight}]{\Delta} \underset{\text{Brown}}{4NO_2} + 2H_2O + O_2$$

- Thus, the nitric acid stored in a plain glass bottle turns yellow. To avoid the decomposition, nitric acid is generally stored in coloured bottles.

2. Acidic Properties

- It is a strong monobasic acid. It ionises completely in aqueous solution, generating hydronium ions.

$$HNO_3 + H_2O \rightarrow H_3O^+ + NO_3^-$$

- It turns blue litmus to red, methyl orange to pink and phenolphthalein remains colourless.
- Reaction with Alkalis: Neutralises alkalis to form salt and water. Metallic oxides and hydroxides react with dilute HNO_3 to form salt and water.

$$Na_2O + 2HNO_3 \rightarrow 2NaNO_3 + H_2O$$
$$ZnO + 2HNO_3 \rightarrow Zn(NO_3)_2 + H_2O$$
$$NaOH + HNO_3 \rightarrow NaNO_3 + H_2O$$
$$Fe(OH)_3 + 3HNO_3 \rightarrow Fe(NO_3)_3 + 3H_2O$$
$$Mg(OH)_2 + 2HNO_3 \rightarrow Mg(NO_3)_2 + 2H_2O$$

- **Reaction with Carbonates and Bicarbonates:** These react with dil. HNO_3 to give salt, water and carbon dioxide.

$$Na_2CO_3 + 2HNO_3 \rightarrow 2NaNO_3 + H_2O + CO_2\uparrow$$
$$Na_2CO_3 + 2HNO_3 \rightarrow 2NaNO_3 + H_2O + CO_2\uparrow$$
$$CaCO_3 + 2HNO_3 \rightarrow Ca(NO_3)_2 + H_2O + CO_2\uparrow$$

[Board 2024]

Reaction with carbonates and bicarbonates: These react with dil. HNO_3 to form soluble metallic nitrates, water and SO_2 gas. The above reactions prove the acidic nature of nitric acid.

3. Oxidising Properties

- Oxidising properties are due to nascent oxygen, which it gives on decomposition.

$$2HNO_3(\text{Conc.}) \rightarrow 2NO_2 + H_2O + [O]$$
$$2HNO_3(\text{Dil.}) \rightarrow 2NO + H_2O + 3[O]$$

(i) Action on Non-metals: **[Board 2024]**

$$\underset{\text{Carbon}}{C} + 4HNO_3 \rightarrow CO_2 + 2H_2O + 4NO_2$$
$$\underset{\text{Sulphur}}{S} + 6HNO_3 \rightarrow H_2SO_4 + 2H_2O + 6NO_2$$
$$\underset{\text{Phosphorus}}{P} + 5HNO_3 \rightarrow H_3PO_4 + H_2O + 5NO_2$$

(ii) Action on Metals: Nitric acid reacts with all metals except gold and platinum.

- Cold and dilute nitric acid oxidises less active metals such as Cu, Ag, Pb and Hg to their nitrates and liberates nitric oxide.

$$\underset{\text{Copper}}{3Cu} + 8HNO_3 \rightarrow 3Cu(NO_3)_2 + 4H_2O + 2NO$$

- Cold and dilute nitric acid oxidises active metals to their nitrates and liberates nitrous oxide (N_2O).

$$\underset{\text{Zinc}}{4Zn} + 10HNO_3 \rightarrow 4Zn(NO_3)_2 + 5H_2O + N_2O$$
$$\underset{\text{Iron}}{4Fe} + 10HNO_3 \rightarrow 4Fe(NO_3)_2 + 5H_2O + N_2O$$

- Concentrated nitric acid oxidises metals to their nitrates and liberates nitrogen dioxide. **[Board 2024]**

$$Cu + 4HNO_3 \rightarrow Cu(NO_3)_2 + 2H_2O + 2NO_2$$
$$Zn + 4HNO_3 \rightarrow Zn(NO_3)_2 + 2H_2O + 2NO_2$$
$$Fe + 6HNO_3 \rightarrow Fe(NO_3)_2 + 2H_2O + 3NO_2$$

- Metals like Fe, Al, Co and Ni become passive (inert) when treated with pure concentrated nitric acid because of the formation of extremely thin layer of insoluble metallic oxide which stops the reaction.
- Very dilute nitric acid (1% nitric acid) reacts with magnesium and manganese metals.

$$Mg + 2HNO_3 \rightarrow Mg(NO_3)_2 + H_2$$
$$Mn + 2HNO_3 \rightarrow Mn(NO_3)_2 + H_2$$

➢ **Test for Nitric Acid and Nitrates:**

- Concentrated nitric acid gives brown fumes on heating.

$$4HNO_3 \xrightarrow{\Delta} 2H_2O + \underset{\text{Brown fumes}}{4NO_2} + O_2$$

- Nitrates (other than potassium, sodium and ammonium) produce reddish brown fumes of nitrogen dioxide [NO_2] when copper reacts with nitric acid.

$$Cu + \underset{\text{conc.}}{4HNO_3} \xrightarrow{\Delta} Cu(NO_3)_2 + 2H_2O + 2NO_2$$ **[Board 2024]**

$$Cu + 4NaNO_3 + 4H_2SO_4 \xrightarrow{\Delta} 4NaHSO_4 + Cu(NO_3)_2 + 2H_2O + 2NO_2$$

- **Brown Ring Test**

Procedure:

(i) Take an aqueous solution of nitrate or nitric acid in a test tube.

(ii) Add freshly prepared saturated solution of iron [II] sulphate.

(iii) Now add concentrated sulphuric acid carefully from the sides of the test tube, it means that it should not fall dropwise in the test tube.

(iv) Cool the test tube in water.

(v) A brown ring appears at the junction of the two liquids.

Brown Ring Test

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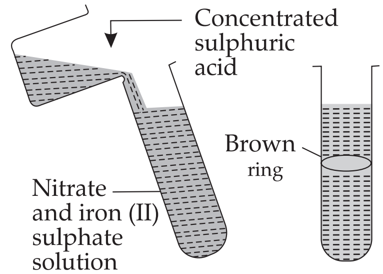


Fig. 8.13. Brown ring test

Reaction:

$$6FeSO_4 + 3H_2SO_4 + 2HNO_3 \rightarrow 3Fe_2(SO_4)_3 + 4H_2O + 2NO$$

$$FeSO_4 + 6H_2O \rightarrow [Fe_2(H_2O)_6]SO_4$$

$$[Fe(H_2O)_6]SO_4 + NO \rightarrow \underset{\text{Pentaaqua-nitroso iron (I) sulphate}}{[Fe(H_2O)_5NO]\,SO_4} + H_2O$$

Key Terms

➢ **Nitrogen Fixation:** The conversion of free atmospheric nitrogen into useful nitrogenous compounds in the soil is called nitrogen fixation.

➢ **Aqua Regia:** It is a mixture of conc. nitric acid (1 part by volume) and conc. hydrochloric acid (3 parts by volume) (1: 3). It is also called as royal water.

Key Facts

➢ Chile saltpetre: $NaNO_3$

➢ During lightning discharge, the nitrogen present in the atmosphere reacts with the oxygen to form nitric oxide.

$$N_2 + O_2 \rightleftharpoons 2NO$$

➢ Nitric acid was formerly known as **aqua fortis**, meaning strong water. It is due to the fact that it reacts with nearly all metals. It can even dissolve silver.

➢ Nitric acid up to 98% (fuming nitric acid) is obtained by distilling 68% HNO_3 over conc. H_2SO_4.

➢ An aqueous solution of nitric acid (68% concentration) forms a constant boiling mixture at 121°C.

➢ Nitric acid is non-poisonous. It has a corrosive action on the skin and causes painful blisters. It stains the skin yellow as it reacts with protein of the skin and forms **xanthoproteic acid**.

➢ Nitric acid stored in a bottle turns yellow. This colour is due to dissolved NO_2 in HNO_3. To avoid the decomposition, nitric acid is normally stored in coloured bottles.

➢ Nitric acid oxidises metals and oxidises hydrogen to water and hydrochloric acid to chlorine. Aqua regia reacts even with noble metals like gold and platinum to give their chlorides. Aqua regia contains nascent chlorine which attacks these metals.

$$Pt + 4[Cl] \longrightarrow \underset{\text{(Soluble)}}{PtCl_4}$$

$$Au + 3[Cl] \longrightarrow \underset{\text{(Soluble)}}{AuCl_3}$$

$$HCl + AuCl_3 \longrightarrow H^+[AuCl_4]^-$$

➢ **In brown ring test**, a freshly prepared ferrous sulphate solution is used because on exposure to the atmosphere, it is oxidised to ferric sulphate which will not give the brown ring. **[Board 2024]**

Example 4

Identify the reactant and write the balanced equation for the following:
Nitric acid reacts with compound Q to give a salt $Ca(NO_3)_2$, water and carbon dioxide.

Ans. Reactant - $CaCO_3$

$CaCO_3 + 2HNO_3 \rightarrow Ca(NO_3)_2 + H_2O + CO_2$

TOPIC-5: SULPHURIC ACID

Concepts covered: • Manufacturing sulphuric acid • Properties of sulphuric acid

Revision Notes

- Sulphuric acid is called the king of chemicals. It is commonly called oil of vitriol.
- It was prepared during the distillation of green vitriol ($FeSO_4.7H_2O$).

$$2FeSO_4.7H_2O \xrightarrow{\Delta} Fe_2O_3 + SO_2 + 13H_2O + H_2SO_4$$

- Sulphuric acid is manufactured by **Contact Process**.

Preparation of sulphuric acid

Scan Me!

(A) Catalytic oxidation of sulphur dioxide in catalytic chamber:
The gases entering the catalytic chamber must be pure, otherwise, it would poison the catalyst.

$$\underset{\text{Sulphur dioxide}}{2SO_2} + O_2 \xrightarrow[450^\circ C]{\text{Vanadium pentaoxide}} \underset{\text{Sulphur trioxide}}{2SO_3}$$

[Board 2024]

The catalyst glows red hot because the reaction is exothermic in nature. So, only initial heating of the catalyst is required.Absorption tower:
SO_3 formed in catalytic chamber is not directly absorbed in water because:

(i) The reaction is highly exothermic in nature.

(ii) It forms a homogeneous mixture. A dense mist of minute particles of sulphuric acid is formed which is not easily condensed. Instead, sulphur trioxide formed is absorbed in concentrated sulphuric acid to form pyrosulphuric acid or oleum ($H_2S_2O_7$).

$$SO_3 + \underset{\text{(conc.)}}{H_2SO_4} \rightarrow \underset{\substack{\text{Pyrosulphuric acid} \\ \text{or} \\ \text{Oleum}}}{H_2S_2O_7}$$

(B) Dilution of Oleum: It is diluted by adding calculated amount of water to obtain concentrated sulphuric acid of desired concentration.

$$H_2S_2O_7 + \underset{\substack{\text{Calculated} \\ \text{amount}}}{H_2O} \rightarrow \underset{\text{Concentrated}}{2H_2SO_4}$$

- Concentrated sulphuric acid is colourless, odourless, viscous, hygroscopic liquid which is soluble in water in all proportions.

➢ It is a strong dibasic acid. The salts of sulphuric acid are called sulphates and bisulphates. Dilute sulphuric acid shows the typical properties of dilute acids.

Reactions of Sulphuric Acid
Scan Me!

(A) Reaction with bases:

It forms salt and water. It undergoes neutralisation reaction.

$$\underset{\text{Potassium hydroxide}}{2KOH} + \underset{\text{(dil.)}}{H_2SO_4} \rightarrow \underset{\text{Potassium sulphate}}{K_2SO_4} + 2H_2O$$

$$\underset{\text{Sodium hydroxide}}{2NaOH} + \underset{\text{(dil.)}}{H_2SO_4} \rightarrow \underset{\text{Sodium sulphate}}{Na_2SO_4} + 2H_2O$$

$$\underset{\text{Copper hydroxide}}{Cu(OH)_2} + \underset{\text{(dil.)}}{H_2SO_4} \rightarrow \underset{\text{Copper sulphate}}{CuSO_4} + 2H_2O$$

$$\underset{\text{Ammonium hydroxide}}{2NH_4OH} + \underset{\text{(dil.)}}{H_2SO_4} \rightarrow \underset{\text{Ammonium sulphate}}{(NH_4)_2SO_4} + 2H_2O$$

(B) Reaction with metallic carbonates and bicarbonates:

It liberates carbon dioxide, producing effervescence.

$$\underset{\text{Potassium carbonate}}{K_2CO_3} + \underset{\text{(dil.)}}{H_2SO_4} \rightarrow \underset{\text{Potassium sulphate}}{K_2SO_4} + H_2O + CO_2\uparrow$$

$$\underset{\text{Potassium bicarbonate}}{KHCO_3} + \underset{\text{(dil.)}}{H_2SO_4} \rightarrow \underset{\text{Potassium bisulphate}}{KHSO_4} + H_2O + CO_2\uparrow$$

$$\underset{\text{Sodium carbonate}}{Na_2CO_3} + \underset{\text{(dil.)}}{H_2SO_4} \rightarrow \underset{\text{Sodium sulphate}}{Na_2SO_4} + H_2O + CO_2\uparrow$$

$$\underset{\text{Sodium bicarbonate}}{NaHCO_3} + \underset{\text{(dil.)}}{H_2SO_4} \rightarrow \underset{\text{sodium bisulphate}}{NaHSO_4} + H_2O + CO_2\uparrow$$

$$\underset{\text{Sodium bicarbonate}}{2NaHCO_3} + \underset{\text{(dil.)}}{H_2SO_4} \rightarrow \underset{\text{sodium sulphate}}{Na_2SO_4} + 2H_2O + 2CO_2\uparrow$$

(C) Reaction with Active metals: On reaction of active metals with dilute acids, hydrogen gas is liberated.

$$Zn + \underset{\text{(dil.)}}{H_2SO_4} \rightarrow \underset{\text{Zinc Sulphate}}{ZnSO_4} + H_2\uparrow$$

$$Mg + \underset{\text{(dil.)}}{H_2SO_4} \rightarrow \underset{\text{Magnesium Sulphate}}{MgSO_4} + H_2\uparrow$$

$$Fe + \underset{\text{(dil.)}}{H_2SO_4} \rightarrow \underset{\text{Ferrous Sulphate}}{FeSO_4} + H_2\uparrow$$

(D) Reaction with metallic sulphites and bisulphites: On reaction of active metals with dilute acids, choking, odorous sulphur dioxide gas is liberated.

$$\underset{\text{Potassium Sulphite}}{K_2SO_3} + \underset{\text{(dil.)}}{H_2SO_4} \rightarrow \underset{\text{Potassium Sulphate}}{K_2SO_4} + H_2O + \underset{\text{Sulphur dioxide}}{SO_2}\uparrow$$

$$\underset{\text{Potassium Bisulphite}}{KHSO_3} + \underset{\text{(dil.)}}{H_2SO_4} \rightarrow \underset{\text{Potassium Bisulphate}}{KHSO_4} + H_2O + \underset{\text{Sulphur Dioxide}}{SO_2}\uparrow$$

$$\underset{\text{Sodium Sulphite}}{Na_2SO_3} + \underset{\text{(dil.)}}{H_2SO_4} \rightarrow \underset{\text{Sodium Sulphate}}{Na_2SO_4} + H_2O + \underset{\text{Sulphur Dioxide}}{SO_2}\uparrow$$

$$\underset{\text{Sodium Bisulphite}}{2NaHSO_3} + \underset{\text{(dil.)}}{H_2SO_4} \rightarrow \underset{\text{Sodium Sulphate}}{Na_2SO_4} + 2H_2O + \underset{\text{Sulphur Dioxide}}{2SO_2}\uparrow$$

(E) Reaction with metallic sulphides: On reaction of metallic sulphides with dilute sulphuric acid, hydrogen sulphide gas having rotten eggs smell is liberated.

$$\underset{\text{Zinc Sulphide}}{ZnS} + \underset{\text{(dil.)}}{H_2SO_4} \rightarrow \underset{\text{Zinc Sulphate}}{ZnSO_4} + \underset{\text{Hydrogen Sulphide}}{H_2S}$$

$$\underset{\text{Iron(II) Sulphide}}{FeS} + \underset{\text{(dil.)}}{H_2SO_4} \rightarrow \underset{\text{Iron(II) Sulphate}}{FeSO_4} + \underset{\text{Hydrogen Sulphide}}{H_2S}$$

(A) Reaction with metallic oxide:

It forms salt and water.

$$Al_2O_3 + \underset{\text{(dil.)}}{3H_2SO_4} \rightarrow Al_2(SO_4)_3 + 3H_2O$$

$$CuO + \underset{\text{(dil.)}}{H_2SO_4} \rightarrow CuSO_4 + H_2O$$

$$ZnO + \underset{\text{(dil.)}}{H_2SO_4} \rightarrow ZnSO_4 + H_2O$$

$$MgO + \underset{\text{(dil.)}}{H_2SO_4} \rightarrow MgSO_4 + H_2O$$

(B) It shows characteristic colour change with indicators:

Indicator	Colour change
Blue litmus	Red
Phenolphthalein	Remains colourless
Methyl orange	Red or Pink

Example 5

1. Which property of sulphuric acid is shown by the reaction of concentrated sulphuric acid with:

(i) Ethanol?

(ii) Carbon?

Ans. (i) Dehydrating agent/dehydration

(ii) Oxidising agent/oxidation

➢ Concentrated sulphuric acid acts as

(A) Least volatile acid: Nitrates and chlorides on reaction with concentrated sulphuric acid form their corresponding acids, which are more volatile than sulphuric acid. It acts as a non-volatile acid.

$$NaCl + H_2SO_4(conc.) \xrightarrow[200°C]{\text{below}} NaHSO_4 + HCl$$

$$KNO_3 + H_2SO_4 \xrightarrow[200°C]{\text{below}} KHSO_4 + HNO_3$$

(B) Dehydrating agent: Concentrated sulphuric acid acts as dehydrating agent, as it has strong affinity for water. Dehydrating agent is a chemical compound which removes chemically combined elements of water, hydrogen and oxygen in the ratio of 2: 1.

(i) Dehydration of formic acid:

$$\underset{\text{Formic acid}}{HCOOH} + H_2SO_4(conc.) \rightarrow \underset{\text{Carbon monoxide}}{CO} + [H_2O.H_2SO_4]$$

[Board 2024]

(ii) Dehydration of Oxalic acid:

$$\underset{\text{Oxalic acid}}{\begin{matrix} COOH \\ | \\ COOH \end{matrix}} + H_2SO_4(conc.) \rightarrow \underset{\text{Carbon monoxide}}{CO} + CO_2 + [H_2O.H_2SO_4]$$

(iii) Dehydration of ethanol:

$$\underset{\text{Ethanol}}{C_2H_5OH} + H_2SO_4(conc.) \xrightarrow{170°C} \underset{\text{Ethene}}{C_2H_4} + [H_2O.H_2SO_4]$$

(iv) Dehydration of blue vitriol ($CuSO_4.5H_2O$):
Blue coloured crystalline copper sulphate, on coming in contact with concentrated H_2SO_4, becomes white and crumbles down to form powder.

$$\underset{\substack{\text{Hydrated}\\ \text{copper sulphate}\\ \text{(Blue)}}}{CuSO_4.5H_2O} + H_2SO_4(\text{conc.}) \rightarrow \underset{\substack{\text{Anhydrous}\\ \text{Copper sulphate}\\ \text{(White)}}}{CuSO_4} + [5H_2O.H_2SO_4]$$

(v) Dehydration of sugar:
On adding concentrated sulphuric acid to sugar, initially the crystals become brown, steam is released, causing a lot of frothing, and finally a **black porous mass** is left behind.

$$\underset{\text{Sugar}}{C_{12}H_{22}O_{11}} + H_2SO_4(\text{conc.}) \rightarrow \underset{\substack{\text{Black}\\ \text{mass}}}{12C} + [11H_2O.H_2SO_4]$$

(A) Oxidising agent: It oxidises
(i) Metals

$$\underset{\text{Zinc}}{Zn} + \underset{\text{(conc.)}}{2H_2SO_4} \rightarrow \underset{\text{Zinc sulphate}}{ZnSO_4} + 2H_2O + SO_2$$

$$\underset{\text{Copper}}{Cu} + \underset{\text{(conc.)}}{2H_2SO_4} \rightarrow \underset{\substack{\text{Copper}\\ \text{sulphate}}}{CuSO_4} + 2H_2O + SO_2$$

(ii) Non-metals

$$C + \underset{\text{(conc.)}}{2H_2SO_4} \rightarrow CO_2 + 2H_2O + 2SO_2$$

$$S + \underset{\text{(conc.)}}{2H_2SO_4} \rightarrow 2H_2O + 3SO_2$$

(iii) Inorganic compounds

$$H_2S + \underset{\text{(conc.)}}{H_2SO_4} \rightarrow S + 2H_2O + SO_2\uparrow$$

(C) Drying agent: Concentrated sulphuric acid acts as a drying agent, i.e., it is used to remove moisture from the gases and other compounds without undergoing any chemical reaction.

MNEMONICS

Concept: Role of Conc. Sulphuric Acid
Mnemonics:
Li**V**e**D** **O**n **D**iet
Interpretation:
L: Least
V: Volatile
D: Dehydrating
O: Oxidising
D: Drying

Key Terms

- **King of Chemicals:** Sulphuric acid is called the king of chemicals.
- **Oil of vitriol:** Sulphuric acid was obtained as an oily viscous liquid by heating crystals of green vitriol. So, it is known as oil of vitriol.

Key Facts

- **Barytes:** $BaSO_4$ (Barium sulphate)
- **Gypsum:** $CaSO_4.2H_2O$
- **Kieserite:** $MgSO_4.H_2O$
- Water is never poured on acid to dilute it, as large amount of heat is evolved which changes water to steam. The steam so formed causes spurting of acid, which can cause burn injuries, so dilution is done by pouring acid on a given amount of water in a controlled manner by continuous stirring. The evolved heat is dissipated in water itself, and so the spurting of the acid is minimised.

Course of Action
Maximum Time = 3:00 Hrs
Maximum Questions = 86

9 ORGANIC CHEMISTRY

Learning Objectives

The students will be able to:
- *Explain the tetravalency and catenation property of carbon.*
- *Discuss structure and isomerism in organic compounds.*
- *Describe the homologous series.*
- *Simple Nomenclature of hydrocarbons with simple functional groups.*
- *Discuss the preparation of ethane, ethene and acetylene.*
- *Differentiates between alkanes, alkenes and alkynes based on their properties.*
- *Discuss the preparation and properties of ethanol and carboxylic acid.*

TOPIC-1: ORGANIC COMPOUNDS

Concepts covered: • Unique nature of carbon atoms, • Characteristics of organic compounds, • Cycloalkanes, • Hydrocarbons

Revision Notes

Organic Compounds
Scan Me!

➢ The word "**organic**" means pertaining to life. Earlier, it was regarded that organic compounds can only be produced by nature under the influence of living force called **vital force**.

➢ The vital force theory was soon discarded when a German scientist, Friedrich Wohler, synthesise an organic compound urea in the laboratory, by heating ammonium cyanate.

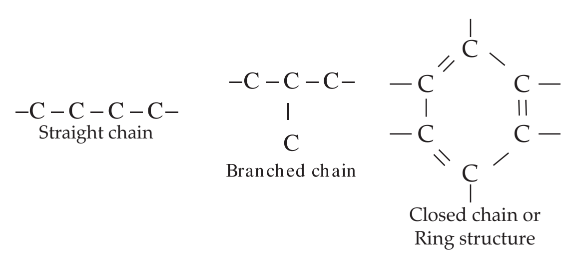
$$\underset{\text{Ammonium cyanate}}{NH_4CNO} \xrightarrow{\text{Heat}} \underset{\text{Urea}}{CO(NH_2)_2}$$

Later on, Kolbe prepared acetic acid (CH_3COOH) and Berthelot synthesised methane (CH_4) gas in the laboratory.

➢ All organic compounds essentially contain carbon atom. Carbon has four **valence electrons**. Therefore, to satisfy its valency carbon shares its electrons with other carbon atoms or with the atoms of other elements. As a result of sharing, it leads to the formation of straight chain, branched chain, closed chain or the ring structure.

➢ **Unique nature of carbon atoms:**

(i) Carbon has four valence electrons. It can neither



lose nor gain electrons to attain octet. It forms covalent bonds by sharing its four electrons with other atoms. It is known as **tetravalency** of the carbon atoms.

(ii) Carbon atom possesses a unique property to link with one another by means of covalent bonds to form long chains (or rings) of carbon atoms. This property of forming bonds with atoms of the same elements is called as **catenation**. **[Board, 2021]**

➢ **Characteristics of organic compounds are:**

(i) All organic compounds are **covalent** in nature.

(ii) Almost all the organic compounds are insoluble in water but soluble in organic solvents like benzene, ether, carbon tetrachloride.

(iii) All have relatively low melting point and boiling point.

(iv) All organic compounds are combustible in nature.

➢ Earlier chemists basically classified hydrocarbons as either **aliphatic or aromatic**. The classification was done on the basis of their source and their properties.

➢ **Cycloalkanes:** These hydrocarbons possess one or multiple carbon rings. The hydrogen atom is attached to the carbon ring.

➢ **Aromatic Hydrocarbons:** These are also called as **arenes**. Arenes are compounds which consist of at least one **aromatic ring**.

➢ **Hydrocarbons:** A compound made up of hydrogen and carbon only is called **hydrocarbon**. For example, CH_4, C_2H_2, C_2H_6, etc. The most important natural source of hydrocarbon is petroleum or crude oil. Hydrocarbons are further divided into two main groups:

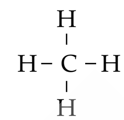
(i) Aliphatic or Open Chain Compounds

(ii) Cyclic (closed) chain compounds.

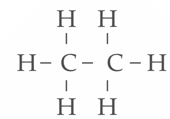
The aliphatic compounds are further divided into saturated and unsaturated hydrocarbons.

➢ **Saturated hydrocarbons:** A hydrocarbon in which the carbon atoms are connected by only single bonds is called **saturated hydrocarbon**. They are named as alkanes. It is represented by the general formula C_nH_{2n+2} where n is the number of carbon atoms.

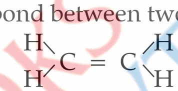
Methane, CH_4



Ethane, C_2H_6



➢ **Unsaturated hydrocarbons:** A hydrocarbon in which two carbon atoms are connected by a double bond or a triple bond is called as an **unsaturated hydrocarbon**. Alkenes are the hydrocarbons with double bond between two carbon atoms. Alkynes are the hydrocarbons with triple bond between two carbon atoms.



Ethene

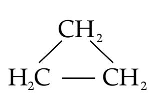
$H-C \equiv C-H$

Ethyne

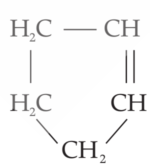
➢ Difference between Saturated and Unsaturated Hydrocarbons: **[Board, 2024]**

Saturated Hydrocarbons	Unsaturated Hydrocarbons
i. All the four valencies of each carbon atom are satisfied by forming single covalent bonds with carbon and with hydrogen atoms.	**i.** The valencies of at least two carbon atoms are not fully satisfied by the hydrogen atoms.
ii. Carbon atoms are joined only by a single covalent bond.	**ii.** Carbon atoms are joined by double covalent bonds, or by triple covalent bonds.
iii. They are less reactive due to the non-availability of electrons in the single covalent bond, and therefore they undergo substitution reaction.	**iii.** They are more reactive due to the presence of electrons in the double or the triple bond, and therefore, undergo addition reaction.

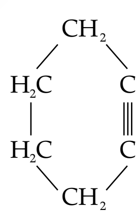
➢ **Carbocyclic compounds:** Cyclic or **closed** chain hydrocarbons contain three or more carbon atoms in their molecule. Cyclic compounds containing single, double and triple bonds are called **cycloalkanes, cycloalkenes and cycloalkynes** respectively.



Cyclopropane (C_3H_6)

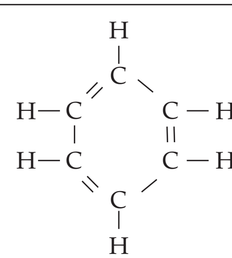


Cyclopentene C_5H_8



Cyclohexyne (C_6H_8)

Some hydrocarbon contains at least one benzene ring in their molecules. It is a ring of six carbon atom having C — C single and C = C double bond in alternate positions.



Benzene

➢ **Alkyl Group:** Alkyl group is represented by 'R'. The general formula of alkyl group is C_nH_{2n+1} (where n = number of carbon atoms). The group formed by the removal of one hydrogen atom from alkane molecules is called an alkyl group. An alkyl group is named by replacing the suffix 'ane' of the alkane with the suffix – yl. **[Board, 2024]**

$\underset{\text{Methane}}{CH_4} \xrightarrow{-H} \underset{\text{Methyl}}{-CH_3}$

$\underset{\text{Ethane}}{C_2H_6} \xrightarrow{-H} \underset{\text{Ethyl}}{-C_2H_5}$

➢ **Homologous Series:** When the organic compounds having similar structural formula, same functional group are arranged in order of increasing molecular weights, they form a homologous series.

➢ **Characteristics of a Homologous Series**

(i) They have similar general and structural formula and same chemical properties.
(ii) The two adjacent members of homologous series differ by CH_2 unit.
(iii) The molecular mass of two adjacent homologous differ by 14 amu.
(iv) The members of a homologous series can be prepared by similar methods of preparation.

➢ **Homologous Series of Alkanes:** (General formula C_nH_{2n+2})

Alkane	Formula
1. Methane	CH_4
2. Ethane	C_2H_6
3. Propane	C_3H_8
4. Butane	C_4H_{10}
5. Pentane	C_5H_{12}

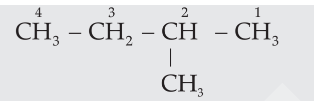
➢ **Importance of Homologous Series:**

(i) It helps in systematic study of organic compounds.
(ii) By knowing the properties of any member of a homologous series, we can predict the properties of other members of the series.

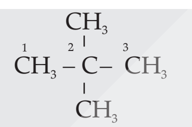
➢ **Isomerism:** Two or more compounds having the same molecular formula but different physical and chemical properties are called isomers and this phenomenon is known as **isomerism**. Example of butane. **[ICSE 2024]**

Example - **(i)** $\overset{5}{C}H_3 - \overset{4}{C}H_2 - \overset{3}{C}H_2\ \overset{2}{C}H_2 - \overset{1}{C}H_3$

Pentane (n – pentane)

(ii) 

2-methyl butane (*iso* -pentane)

(iii) 

2,2 dimethyl propane
(*neo* -pentane) **[Board, 2020]**

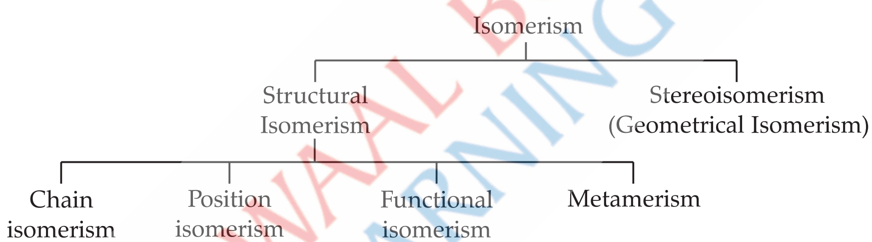
Isomerism
Scan Me!

Structural Isomerism
Scan Me!

Stereoisomerism
Scan Me!

➢ **There are two types of isomerism:**

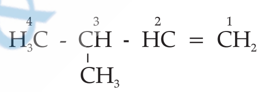
(1) Structural isomerism: Chain isomerism, position isomerism, functional isomerism, metamerism and tautomerism.
(2) Stereoisomerism: Optical and geometrical isomerism.



➢ **Chain Isomerism:** When **the same** molecular formula represents two or more compounds which differ in the length of carbon chain without altering the position of double or triple bond then the isomerism is said to be chain isomerism. For example: C_5H_{10}

$\overset{5}{H_3C} - \overset{4}{H_2C} - \overset{3}{H_2C} - \overset{2}{C}H = \overset{1}{C}H_2$

Pent–1–ene



3-Methyl but - 1- ene

➢ **Position Isomerism:** The isomerism in this case differ with respect to the position of the double bond and triple bond. For example: Butyne C_4H_6.

$\overset{4}{H_3C} - \overset{3}{C}H_2 - \overset{2}{C} \equiv \overset{1}{C}H$

(But - 1 - yne)

$\overset{4}{H_3C} - \overset{3}{C} \equiv \overset{2}{C} - \overset{1}{C}H_3$

(But - 2 - yne)

[Board, 2023]

➢ **Functional Isomerism:** A **functional group** is an atom or a group of atoms which imparts characteristics properties to the organic compound. In functional isomerism the compounds have same number molecular formula but have different functional groups.

Series	Functional Group	General Formula	Example
Alkane	Hydrocarbon chain	C_nH_{2n+2}	Methane, CH_4 Ethane, C_2H_6
Alkene	Double bond $>C=C<$	C_nH_{2n}	Ethene, C_2H_4 Propene, C_3H_6
Alkyne	Triple bond $-C \equiv C-$	C_nH_{2n-2}	Ethyne, C_2H_2 Propyne, C_3H_4
Alcohol	– OH group	$C_nH_{2n+1}OH$	Methanol, CH_3OH Ethanol, C_2H_5OH

Aldehyde	– CHO group	$C_nH_{2n+1}CHO$	Ethanal, CH_3CHO Propanal, C_2H_5CHO
Carboxylic acid	– COOH group	$C_nH_{2n+1}COOH$	Ethanoic acid, CH_3COOH Propanoic acid, C_2H_5COOH **[Board, 2024]**
Halides -			
Chloride	– Cl	$C_nH_{2n+1}Cl$	Chloromethane CH_3Cl Chloroethane C_2H_5Cl
Bromide	– Br	$C_nH_{2n+1}Br$	Bromomethane CH_3Br Bromoethane C_2H_5Br
Iodide	– I	$C_nH_{2n+1}I$	Iodomethane CH_3I Iodoethane C_2H_5I

➢ **Nomenclature of Organic Compounds:** Nomenclature is the system of assignment of names to organic compounds. Organic compounds have two names, common name or trivial system and IUPAC name.

Trivial System: The basic of naming organic compounds by the trivial system is its (a) Source **(b)** Properties **(c)** Latin or Greek origin.

IUPAC System: (International Union of Pure and Applied Chemistry)

IUPAC name takes up only one molecular structure of the compound and assign only one name to the compound.

According to this system, the name of an organic compound consists of three parts:

(i) Word Root **(ii)** Suffix

(iii) Prefix

IUPAC

Scan Me!

(i) Word Root: The Word root indicates the total number of carbon atoms present in the **longest** carbon chain belonging to the compound. For example, 'Meth' refers to a chain with 1 carbon atom and 'Hept' refers to a chain with 7 carbon atoms.

Number of carbon atoms	Word Root
C_1	Meth
C_2	Eth
C_3	Prop
C_4	But
C_5	Pent
C_6	Hex
C_7	Hept
C_8	Oct
C_9	Non
C_{10}	Dec

(ii) Suffix: The part of the name which appears after the word root.

● **Primary Suffix:** It tells us about the nature of carbon chain i.e., whether saturated or unsaturated.

Name	Group	Suffix
Alkane	$>C-C<$	ane
Alkene	$>C=C<$	ene
Alkyne	$-C\equiv C-$	yne

● **Secondary Suffix:** It is written after the primary suffix for functional group present in the compound.

Name	Group	Suffix
Alcohol	– OH	ol
Aldehyde	– CHO	al
Carboxylic acid	– COOH	oic acid

(iii) Prefix: There are many groups which are not regarded as functional groups in the IUPAC system. These are regarded as side chains and represent as prefix. A prefix is placed **before** the word root while naming a particular compound.

$$-\overset{5}{C}-\overset{4}{C}-\underset{\displaystyle CH_3}{\overset{3}{C}}-\overset{2}{C}-\overset{1}{C}-$$

3-methyl

➢ **Rules for naming an organic compound:**

(i) Selection of carbon chain: The longest chain of carbon atoms in the structure of the compound is found first. The compound is then named as a derivative of the alkane hydrocarbon.

$$\overset{1}{C}-\overset{2}{C}-\overset{3}{C}-\overset{4}{C}-\overset{5}{C}$$

Longest chain of 5 carbon atoms, so word root is 'pent'.

(ii) The alkyl group present as side chains (branches) are considered as substituent and named separately as methyl (– CH_3) or ethyl (– C_2H_5) group.

$$\overset{1}{C}-\underset{\displaystyle C_2H_5}{\overset{2}{C}}-\overset{3}{C}-\overset{4}{C}-\overset{5}{C}$$

2-ethyl

(iii) The carbon atoms of the longest carbon chain are numbered in such a way that the alkyl groups (substitutes) get the lowest possible number.

$$\overset{1}{C} - \overset{2}{C}(CH_3) - \overset{3}{C} - \overset{4}{C} - \overset{5}{C}$$

2-methyl

(iv) If the functional group is also present in the chain, then the carbon atoms are numbered in such a way that the functional group gets the **smallest** possible number.

$$\overset{5}{CH_3} - \overset{4}{CH}(CH_3) - \overset{3}{CH_2} - \overset{2}{CH}(OH) - \overset{1}{CH_3}$$

4-methylpentan - 2- ol

(v) If different types of substituents are attached in the chain, they are named alphabetically.

$$\overset{1}{CH_3} - \overset{2}{CH}(Br) - \overset{3}{CH_2} - \overset{4}{CH}(Cl) - \overset{5}{CH_2} - \overset{6}{CH_3}$$

2-bromo-4-chlorohexane

$$\overset{1}{CH_3} - \overset{2}{CH}(CH_3) - \overset{3}{CH_2} - \overset{4}{CH}(C_2H_5) - \overset{5}{CH_2} - \overset{6}{CH_2} - \overset{7}{CH_3}$$

4-ethyl 2-methylheptane

(vi) The IUPAC name of the compound is obtained by writing the position and name of alkyl group just before the name of parent 'hydrocarbon'.

$$CH_3 - \overset{3}{CH} - \overset{2}{CH_2} - \overset{1}{CH_3}$$
$$\overset{4}{CH_2} - \overset{5}{CH_2} - \overset{6}{CH_3}$$

3-methylhexane

(vii) Multiple alkyl groups are named as di, tri or tetra for two, three or four respectively.

$$\overset{1}{CH_3} - \overset{2}{CH_2} - \overset{3}{C}(CH_3)_2 - \overset{4}{CH_2} - \overset{5}{CH_2} - \overset{6}{CH_3}$$

3,3-dimethylhexane

MNEMONICS

Concept: Homologous series of alkanes.

Mnemonics: **M**any **e**lephants **p**refer **b**lue **p**ineapples

Interpretation:

M – methane

E – ethane

P – propane

B – butane

P – pentane

Key Terms

➢ **Organic Chemistry:** It is the study of hydrocarbons and their derivatives.

➢ **Catenation:** It is the property by virtue of which atoms of the same element get linked together through covalent bonds so as to form long straight, branched or closed chain or rings.

➢ **Tetravalency:** Carbon atom shows tetravalency because it has four electrons in its valence shell.

➢ **Functional Group:** An atom or group of atoms or some other characteristics structural feature which gives special properties to a compound.

➢ **Homologous Series:** It is a series of compound having similar structural formulae, same functional group and hence similar chemical properties. **[Board, 2023]**

➢ **Structural Formula:** It gives up the relative arrangements of bonded atoms in a molecule.

Key Facts

➢ Carbon is a necessary element in every organic compound.

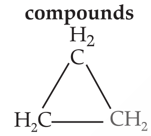
➢ The characteristic of the carbon atom by virtue of which it forms four covalent bonds, is known as tetravalency of carbon.

➢ The unique nature of carbon atom, *i.e.*, catenation and tetravalency gives rise to the formation of a large number of compounds.

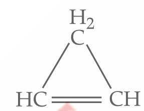
➢ Classification of Hydrocarbons

Hydrocarbons
(Compounds of carbon and hydrogen)

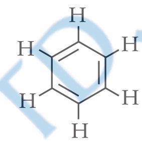
- **Open chain**
  (Acyclic compounds/ Aliphatic compounds)
  - **Saturated**
    (Single bonds between all carbon atoms)
    - **Alkanes**
      (Paraffins)
      ($CH_3-CH_2-CH_2-CH_3$)
      butane
  - **Unsaturated**
    (At least one double or triple bond)
    - **Alkenes**
      (Olefins)
      Double bond
      $\left(\begin{matrix}H\\H\end{matrix}\!>C=C<\!\begin{matrix}H\\H\end{matrix}\right)$
    - **Alkynes**
      (Acetylenes
      Triple bond
      ($H-C\equiv C-H$)
- **Closed chain**
  (Cyclic compounds)
  - **Alicyclic compounds**

    Cyclopropane

    Cyclopropene
  - **Aromatic compounds**
    (Special types of compounds, include benzene ring compounds)

    Benzene
    [ring containing six carbon atoms having carbon-carbon single and carbon-carbon double bonds in alternate position]

Example 1

1. Write the IUPAC name of the following compounds:

(i) $H-\overset{\overset{\displaystyle O}{\|}}{C}-H$

(ii) $H_3C-\overset{\overset{\displaystyle O}{\|}}{C}-CH_3$

(iii) $\begin{array}{c} \quad H \quad OH \quad H \quad H \\ \quad | \quad\;\; | \quad\;\; | \quad\; | \\ H-C-C-C-C-H \\ \quad | \quad\;\; | \quad\;\; | \quad\; | \\ \quad H \quad H \quad H \quad H \end{array}$

(iv) $C_2H_5-O-C_2H_5$

U A I

Ans. (i) Methanal
(ii) Propanone
(iii) Butan-2-ol
(iv) Ethoxy ethane

2. Give reasons for the following:
(i) Hydrocarbons are excellent fuels.
(ii) Almost 90% of all known compounds are organic in nature.

A A I

Ans. (i) Hydrocarbons are excellent fuels because they ignite easily at low temperature and liberate large amount of heat without producing harmful products.

(ii) Because 90% of all known compounds contain C and H and are obtained from or are present in the living world. (Plants and animals).

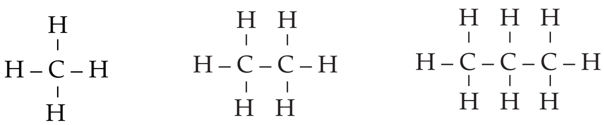
TOPIC-2: HYDROCARBONS (ALKANES, ALKENES AND ALKYNES), ALCOHOLS AND CARBOXYLIC ACIDS

Concepts covered: • Alkanes: • Structural formula, • Laboratory preparation of methane and ethane, • Properties of methane and ethane, • Alkenes, • Preparation of ethene, • Properties of alkenes, • Alkynes, • Laboratory preparation of ethyne, • Properties of alkynes, • Preparation, properties and Uses of Ethanol, • Properties and Uses of Acetic Acid.

Revision Notes

➢ **Alkanes** are the **saturated** hydrocarbons which contain only single covalent bonds between two carbon atoms. They posses the general formula C_nH_{2n+2}, Where n = is the number of the first member of carbon atoms. The first member of series (n=1) is methane CH_4 and second member (n=2) is ethane, C_2H_6.

➢ **Structural formula of Alkanes**:



Methane Ethane Propane

➢ **Laboratory preparation of methane and ethane:** Methane is prepared by anhydrous sodium acetate (sodium ethanoate) with sodalime, NaOH + CaO.

$$\underset{\text{Sodium acetate}}{CH_3COONa(s)} + NaOH(s) \xrightarrow[\Delta]{CaO} Na_2CO_3(s) + \underset{\text{Methane}}{CH_4(g)}\uparrow$$

[BOARD 2017, 2019]

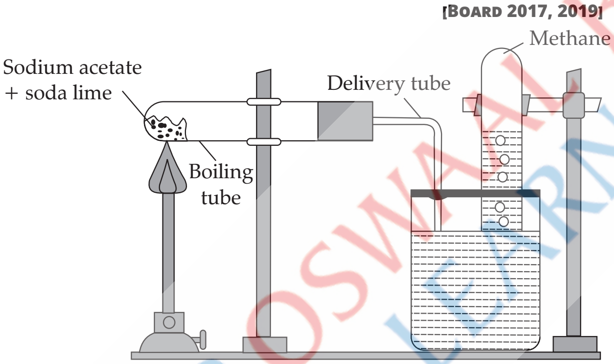


Fig. 9.1 Preparaton of methane (CH_4)

Ethane is prepared by heating anhydrous sodium propionate with sodalime.

$$\underset{\text{Sodium propionate}}{C_2H_5COONa(s)} + NaOH(s) \xrightarrow[\Delta]{CaO} Na_2CO_3(s) + \underset{\text{Ethane}}{C_2H_6}\uparrow$$

➢ **Other methods of preparations of Methane:**

(i) Methane can also be prepared by the action of water on aluminium carbide.

$$\underset{\text{Aluminium carbide}}{Al_4C_3(s)} + 12H_2O(l) \longrightarrow 4Al(OH)_3(s) + \underset{\text{Methane}}{3CH_4(g)}$$

(ii) Methane can also be prepared by the reduction of methyl iodide (iodo methane) with Zn-Cu couple Zn/HCl.

$$\underset{\text{Iodomethane}}{CH_3I} + 2[H] \xrightarrow{\text{Zn–Cu Couple}} \underset{\text{Methane}}{CH_4} + HI$$

[Board 2017]

➢ **Other methods of Preparation of Ethane:**

(i) Ethane can also be prepared by the reduction of bromoethane with Zn-Cu couple or Zn/HCl.

$$\underset{\text{Bromoethane}}{C_2H_5Br} + 2[H] \xrightarrow{\text{Zn–Cu Couple}} \underset{\text{Ethane}}{C_2H_6} + HBr$$

[BOARD 2020]

(ii) Ethane from alkyl halides - Methyl iodide or methyl bromide reacts with sodium metal in the presence of dry ether, than ethane gas is produced.

$$2CH_3I + 2Na \xrightarrow{\text{Dry ether}} \underset{\text{Ethane}}{H_3C-CH_3} + 2NaI$$

➢ **Physical Properties:**

1. Methane is colourless and odourless gas. Its melting point is -183ºC and boiling point is -162ºC. It is insoluble in water but soluble in organic solvents.

2. Ethane is colourless and odourless gas. Its boiling point is -89ºC and melting point is -172ºC. It is insoluble in water but soluble in organic solvents.

➢ **Chemical Properties:**

(i) Combustion of Alkanes: Methane and ethane burn in air to form carbon dioxide and water with the evolution of large amount of heat.

$$CH_4 + 2O_2 \rightarrow CO_2 + 2H_2O + \text{Energy}$$

$$2C_2H_6 + 7O_2 \rightarrow 4CO_2 + 6H_2O + \text{Energy}$$

[BOARD 2020]

(ii) Substitution Reactions:

Methane reacts with chlorine in the presence of diffused sunlight to form mono-chloromethane, CH_3Cl.

$$CH_4(g) + Cl_2(g) \rightarrow \underset{\text{Monochloromethane}}{CH_3Cl(g)} + HCl(g)$$

With excess of chlorine, the remaining three H atoms are successively replaced by Cl atoms to form dichloromethane, trichloromethane and tetrachloro-methane.

$$CH_3Cl + Cl_2 \longrightarrow \underset{\text{Dichloromethane}}{CH_2Cl_2} + HCl$$

$$CH_2Cl_2 + Cl_2 \longrightarrow \underset{\text{Trichloromethane (chloroform)}}{CHCl_3} + HCl$$

$$CHCl_3 + Cl_2 \longrightarrow \underset{\text{tetrachlormethane}}{CCl_4} + HCl$$

These reactions are called **substitution rections** because chlorine atom successively replaces hydrogen atoms in the methane molecule.

Similarly it will happen with C_2H_6.

Alkenes: Alkenes are also called **olefins** because the lower member of alkenes form oily products when they were treated with chlorine or bromine. **Alkenes** form a homologous series having the general formula C_nH_{2n}. Ethene is the first member of the alkene series.

➢ **Preparation of Ethene**

i. Dehydration of ethyl alcohol.

Reactants: Ethanol and conc. sulphuric acid.

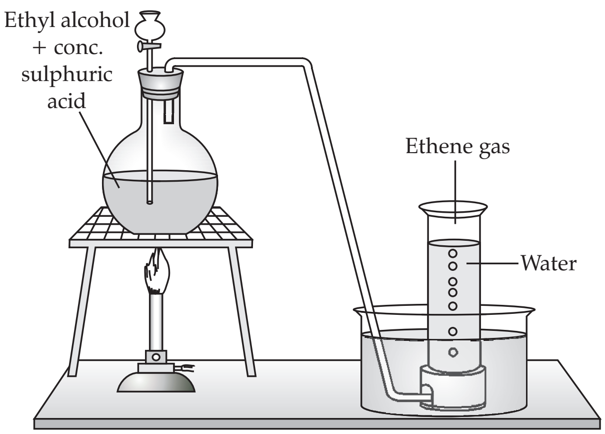


Fig. 9.2 Laboratory Preparation of ethene

➢ **Chemical Reaction:**

$$\underset{\text{Ethyl Alcohol}}{CH_3CH_2OH} + H_2SO_4 \xrightarrow{100°C} \underset{\text{Ethyl hydrogen sulphate}}{CH_3CH_2HSO_4} + H_2O$$

$$CH_3CH_2HSO_4 \xrightarrow[170°C]{\text{excess } H_2SO_4} \underset{\text{Ethene}}{CH_2 = CH_2} + H_2O$$

[Board 2023]

➢ **Collection:** The gas is collected by downward displacement of water because:

(A) It is an inflammable gas.

(B) It is insoluble in water.

(ii) By dehydration: Industrial preparation:

By passing ethanol vapours through a tube containing alumina (Al_2O_3) at 300°C.

$$C_2H_5OH \xrightarrow[300°C]{Al_2O_3} C_2H_4 + H_2O$$

(iii) Dehydrogenations:

When an alkyl halide is heated gently in an alcoholic medium, it forms corresponding alkene.

$$\underset{\text{Ethyl chloride}}{C_2H_5Cl} + \underset{\text{(Alc. hot and conc.)}}{KOH} \rightarrow C_2H_4 + KCl + H_2O$$

$$\underset{\text{Ethyl bromide}}{C_2H_5Br} + \underset{\text{(Alc. hot and conc.)}}{KOH} \rightarrow C_2H_4 + KBr + H_2O$$

➢ **Physical properties**

Ethene is a colourless and **inflammable gas** with a peculiar odour. Its boiling point is - 102°C and melting point is –169°C. It is sparingly soluble in water but highly soluble in organic solvents.

➢ **Chemical properties**

(i) Addition of hydrogen (hydrogenation) in the presence of catalyst such as platinum or palladium at ordinary temperature or nickel at 200°C.

$$\underset{\text{Ethene}}{C_2H_4} + H_2 \xrightarrow[200°C]{Ni} \underset{\text{Ethane}}{C_2H_6}$$

(ii) Addition of halogens (halogenation): Ethene reacts with halogens in presence of carbon tetrachloride as a solvent. The order of the reaction with halogens is $F_2 > Cl_2 > Br_2 > I_2$.

$$\underset{\text{Ethene}}{CH_2{=}CH_2} + Cl_2 \xrightarrow{CCl_4} \underset{\text{(1, 2-dichloroethane)}}{\underset{Cl}{\overset{|}{CH_2}} - \underset{Cl}{\overset{|}{CH_2}}}$$

$$\underset{\text{Ethene}}{CH_2{=}CH_2} + Br_2 \xrightarrow{CCl_4} \underset{\text{(1, 2-dibromoethane)}}{\underset{Br}{\overset{|}{CH_2}} - \underset{Br}{\overset{|}{CH_2}}}$$

[Board 2019]

(iii) Addition of water (hydration)

$$CH_2 = CH_2 + H_2O \xrightarrow{H^+} \underset{\text{Ethyl alcohol}}{C_2H_5OH}$$

(iv) Addition of HCl: Ethene reacts with HCl at room temperature. The order of reaction of hydrogen halides with alkanes is HI > HBr > HCl > HF.

$$\underset{\text{Ethene}}{CH_2 = CH_2} + HCl_{(aq)} \longrightarrow \underset{\left(\begin{array}{c}\text{Ethyl chloride}\\ \text{Chloroethane}\end{array}\right)}{CH_3CH_2Cl}$$

(v) Polymerisation: Ethene polymerises to produce polyethene.

$$n\underset{\text{Ethene}}{H_2C = CH_2} \xrightarrow{\text{Polymerization}} \underset{\text{Polyethene}}{[H_2C - CH_2]_n}$$

(vi) Combustion of ethene: Ethene burns in air with a sooty flame producing a large amount of heat.

$$C_2H_4 + 3O_2 \longrightarrow 2CO_2 + 2H_2O + \text{Heat}$$

➢ **Uses of Ethene:**

(i) It is used for ripening of fruits.

(ii) It is used in making polythene.

(iii) It is used in making epoxy ethane.

Alkynes

• The aliphatic hydrocarbons, containing a triple bond (-C ≡ C-) between two carbon atoms are called **alkynes**. General formula for Alkynes is C_nH_{2n-2}. Ethyne is the first member of alkyne series.

• Molecular formula of Ethyne is C_2H_2.

(i) Laboratory Preparation of Ethyne (Acetylene)

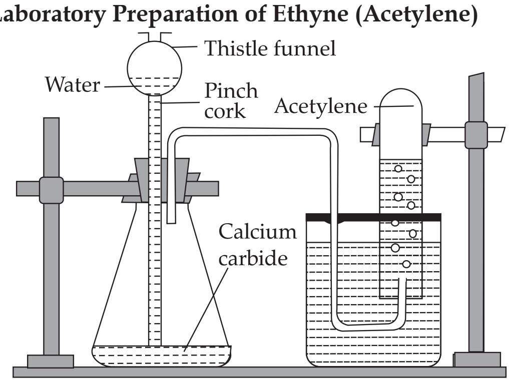


Fig. 9.3 Laboratory preparation of ethyne

➢ Reactants: Water and Calcium carbide

[Board 2019, 2023, 2024]

$$\underset{\text{Calcium carbide}}{CaC_2} + 2H_2O \rightarrow \underset{\text{Calcium hydroxide}}{Ca(OH)_2} + \underset{\text{Ethyne}}{C_2H_2\uparrow}$$

➢ **Collection:** The gas collected by downward displacement of water, since it is insoluble in water.

(ii) Preparation from 1,2-dibromoethane:

$$CH_2Br\text{-}CH_2Br \xrightarrow{\text{alc.KOH}} CH_2{=}CHBr \xrightarrow{NaNH_2} \underset{\text{Ethylene}}{CH \equiv CH}$$

1,2-dibromoethane

➢ **Physical Properties**

It is a colourless gas with an ether like odour. Its boiling point is - 75°C. It liquefies at - 84°C. It is negligibly soluble in water but highly soluble in organic solvents.

➢ **Chemical Properties**

(i) Oxidation of ethyne: Ethyne burns in excess air with a brilliant white flame to produce carbon dioxide, water vapours and a large amount of heat.

$$\underset{\text{Ethene}}{2CH \equiv CH} + 5O_2 \rightarrow 4CO_2\uparrow + 2H_2O + \text{heat}$$

(ii) Addition of hydrogen (catalytic hydrogenation): In the presence of nickel, platinum or palladium.

$$\underset{\text{Ethyne}}{C_2H_2} + H_2 \xrightarrow{Ni} \underset{\text{Ethylene}}{C_2H_4}$$

$$\underset{\text{Ethylene}}{C_2H_4} + H_2 \xrightarrow{Ni} \underset{\text{Ethane}}{C_2H_6}$$

(iii) Reaction with chlorine: Acetylene in an inert solvent reacts with chlorine to give first dichloroethene and then tetrachloroethane.

$$\underset{\text{Acetylene}}{HC \equiv CH} + Cl_2 \xrightarrow{Ni} \underset{\substack{\text{Acetylenedichloride} \\ \text{(1,2-dichloroethene)}}}{CHCl{=}CHCl} + Cl_2 \rightarrow \underset{\substack{\text{Acetylenetetrachloride} \\ \text{(1,1,2,2-tetrachloroethane)}}}{CHCl_2{-}CHCl_2}$$

Acetylene reacts vigorously with chlorine gas in the presence of sunlight to give out flames.

$$C_2H_2 + Cl_2 \rightarrow 2C + 2HCl$$

(iv) Reaction with Bromine: Ethyne reacts with bromine in carbon tetrachloride to first form dibromoethene and then tetrabromoethane. During addition, the red - brown colour of bromine gets decolourized.

$$\underset{\text{Acetylene}}{C_2H_2} \xrightarrow{+Br_2} \underset{\text{Acetylene dibromide}}{C_2H_2Br_2} \xrightarrow{+Br_2} \underset{\text{Acetylene tetrabromide}}{C_2H_2Br_4}$$

(v)Reaction with HCl:

$$\underset{\text{Ethyne}}{CH \equiv CH} + HCl \rightarrow \underset{\text{Chloroethene}}{CH_2{=}CHCl} + HCl \rightarrow \underset{\text{1,1-dichloroethane}}{CH_3{-}CHCl_2}$$

➢ **Uses**

(i) As an illuminant in oxy-acetylene lamp.

(ii) For artificial ripening and preservation of fruits.

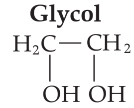
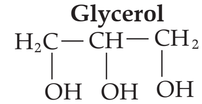
(iii) For oxy-acetylene welding at very high temperature.

Ethanol And Acetic Acid

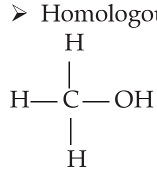
➢ The hydroxyl derivatives of alkanes are called alcohol. Alcohols are formed by replacing one or more hydrogen of alkane by hydroxyl group (–OH).

➢ General formula of alcohol is R-OH.

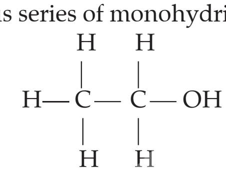
➢ Alcohols are classified based on the number of hydrogens replaced by -OH group in alkane:

Monohydric Alcohol	Dihydric Alcohol	Trihydric Alcohol
Ethanol	Glycol	Glycerol
H_3C-CH_2-OH		

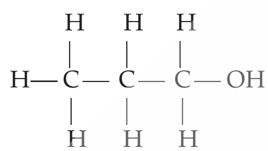
➢ Homologous series of monohydric alcohols are:



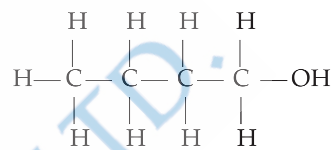
Methanol



Ethanol



Propanol



Butanol

Ethanol – Ethanol is a alcohol with the formula C_2H_5OH.

➢ **Preparation**: Alcohol is prepared by the replacement of one or more hydrogen from alkanes.

Ethyl alcohol or ethanol is prepared by the alkane.

Preparation of ethanol by hydrolysis of alkyl halide: In laboratory ethanol is prepared by the hydrolysis of alkyl halide by hot dilute alkali.

Chemical reaction **[Board 2024]**

$$C_2H_5Cl + KOH\,(aq.) \xrightarrow{\text{Boil}} C_2H_5OH + KCl$$

➢ **Physical Properties** – It is inflammable volatile liquid soluble in water and also in organic solvents. Its boiling point is 78.3°C. Its density is 0.79 at 298 K.

➢ **Chemical Properties:**

• **Combustion**: Ethanol burns in presence of air and form carbon dioxide and water.

$$C_2H_5OH \rightarrow CO_2 + H_2O$$

• **Action with sodium**: Ethanol reacts with sodium at room temperature and forms sodium ethoxide.

$$2C_2H_5OH + 2Na \rightarrow \underset{\text{Sodium ethoxide}}{2C_2H_5ONa} + H_2$$

• **Reaction with acetic acid**: Ethanol reacts with acetic acid in presence of sulphuric acid and produce ester. So, the reaction is known as esterification.

$$C_2H_5OH + CH_3COOH \xrightarrow{\text{conc. } H_2SO_4} \underset{\text{(Ethyl acetate)}}{CH_3COOC_2H_5} + H_2O$$

• On dehydration with conc. Sulphuric acid, ethyl alcohol gives ethene. **[Board 2024]**

➢ **Uses of alcohol**:

• Used as a solvent and for the manufacturing of chemicals such as ether, iodoform and acetic acid.

• Used in thermometers and as a preservative for biological specimen.

• Used in alcoholic beverages.

• **Denatured alcohol**: It a mixture of ethyl alcohol with

5 % methyl alcohol, copper sulphate and pyridine. It is used in chemical industries but can not be used in alcoholic beverages.

• **Spurious alcohol**: It is a alcohol made by improper distillation. It is a mixture of alcohol with large proportion of methanol. It is used as a solvent for paints and varnishes.

Carboxylic Acid: **Acetic acid:** An organic compound with a carboxyl functional group is known as carboxylic acid.

Acetic acid or ethanoic acid is represented as:

$$CH_3COOH \quad \text{or} \quad CH_3-\overset{\overset{\displaystyle O}{\|}}{C}-OH$$

➢ In laboratory it is prepared by oxidation of ethyl alcohol or acetaldehyde using acidified potassium dichromate.

➢ **Physical Properties of Acetic Acid**: It is a colourless liquid with a specific pungent smell.

➢ **Glacial acetic acid**: On cooling the anhydrous acetic acid form a solid crystalline mass which resemble to ice. And that's why it is called glacial acetic acid.

➢ **Chemical properties**

• It is a weak acid and turns blue litmus red.

• It reacts with alkalis and produces salts and water.

MNEMONICS

Concept: Preparation of methane.

Mnemonics: SAW SaiL GuM

Interpretation:

S – sodium
A – acetate
W – with
S – soda
L – lime
G – gives
M – methane

Key Terms

➢ **Alkanes:** These are the hydrocarbons in which all the linkages between the carbon atom are single covalent bonds.

➢ **Fire-damp:** It is called 90% methane found in cavities in coal.

➢ **Pyrolysis:** The decomposition of a compound by heat in the absence of air is known as Pyrolysis.

➢ **Cracking:** It is defined as a process in which pyrolysis occurs in alkanes.

Key Facts

➢ Alkanes are relatively unreactive under ordinary conditions so, they are called paraffins.

➢ Alkanes with more than three carbon atoms form isomers.

➢ Methane is considered as a green house gas. It is 20 times more effective in trapping heat in comparison to CO_2.

➢ Methane can not be prepared by Wurtz reaction as this reaction is not suitable for the preparation of alkanes with odd number of carbon atoms.

➢ Soot is used in the manufacturing of printing inks and tyres.

➢ All alkanes react with chlorine, bromine and iodine in a similar manner, producing the corresponding substituted products.

➢ Alkenes are also known as olefins (oil-forming) because the lower members of alkenes form oily products on reacting with chlorine or bromine.

➢ Ethyne is used for oxy-acetylene welding at very high temperature. It is also used for artificial ripening and preservation of fruits.

Example 3

Define following

(A) Denatured alcohol

(B) Spurious alcohol R

Ans. (A) Denatured alcohol: It a mixture of ethyl alcohol with 5 % methyl alcohol, copper sulphate and pyridine.

(B) Spurious alcohol: It is a alcohol made by improper distillation. It is a mixture of alcohol with large proportion of methanol.